Tutorial

A step-by-step walkthrough of QuerySurge™



Built by RTTS THE SOFTWARE QUALITY EXPERTS

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Welcome to QuerySurge

QuerySurge is the smart Data Testing solution that automates the data validation and ETL testing of Big Data, Data Warehouses, Business Intelligence Reports and Enterprise Applications with full DevOps functionality for continuous testing. By analyzing and pinpointing any differences in the data, QuerySurge ensures that the data extracted from source systems remains intact in the target and complies with transformation requirements. QuerySurge is an essential asset to every data testing process.

Challenges QuerySurge Can Solve

- Your need for data quality at speed. Validate up to 100% of all data up to 1,000 x faster than traditional testing. more »
- Your test automation issues. Now you can automate all your data testing, from the kickoff of tests to performing the validation to automated emailing of the results and updating your test management system. more »
- Your ability to test across platforms**,** whether you use a Big Data Lake, Data Warehouse, traditional database, NoSQL document store, BI reports, flat files, JSON files, SOAP or restful web services, xml, mainframe files, or any other data store. more »
- Your need to integrate for Continuous Delivery. QuerySurge integrates with most Data Integration/ETL solutions, Build/Configuration solutions, and QA/ Test Management solutions. more »
- Your ability to analyze your data**,** with our Data Analytics Dashboard & Data Intelligence Reports. more »

Additional Features & Benefits

- **Integration with leading Test Management Solutions**. QuerySurge offers integration with Micro Focus ALM, IBM RQM, and Microsoft DevOps (formerly TFS & VSTS). Now you can store all of your data tests and receive all of your results in your favorite test management solution. more »
- **Query Wizards — fast and easy with no SQL coding needed.** The smart Wizards provide both novices and experienced team members with the ability to easily and quickly validate their data with no SQL coding required. more »
- **DevOps and Continuous Testing.** Dynamically generate, execute, and update tests and data stores utilizing API calls. more »
- **Proven ROI**. Making a business case for QuerySurge is easy. QuerySurge returns 1,287%. more »

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Resources

Knowledge Base and Community Forums provide a hub for extensive Query*Surge* information and the answers to the most frequently asked questions about Query*Surge*. To access the QuerySurge Knowledge Base, click here >>

- **Video Library:** The Video & Slide Library provides tutorials, demos and webinars on QuerySurge. To access the Video Library, click here: http://www.querysurge.com/product-tour/video-library
- **Contact Us:** The Contact Us page provides a form where you can submit comments, questions, or suggestions. http://www.querysurge.com/company/contact-us



Direct Link to Knowledge Base is provided within Querysurge. Select the help icon located on the bottom panel.

Note: This tutorial requires the installation of 'Tutorial + Sample Data' in the QuerySurge installer. This component can be selected during the initial *QuerySurge* installation. If you are using the Trial in the Cloud, the Sample Data is already installed. If you are using the Trial Download, please select this option during installation.

Installation Requirements

QuerySurge is an enterprise data testing solution that will typically be used to validate large data sets. Since you are testing either a Big Data store or Data Warehouse with potentially massive quantities of data, QuerySurge needs the right hardware to perform at speeds that will meet your team's needs.

Therefore, the more CPU, RAM, and hard drive space you have, the better and faster QuerySurge performs.

For a trial setup, QuerySurge can be run on a desktop with modest resources, but that will limit QuerySurge's performance and ability to scale. For additional information on system requirements, click here: QuerySurge System Requirements.

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Important Terms



QueryPair

A pair of SQL queries where one query retrieves data from a source file or database and another SQL query retrieves data from a target database, data warehouse or data mart.



Agent

Performs the query tasks. Agents execute queries against source and target data stores and return the results to the QuerySurge database.



Design-Time Run

Allows you to execute a *QueryPair* test to make sure that it behaves as you expect. This is not intended for actual test execution purposes.



Query Snippet

Reusable piece of SQL code that can be embedded in one or more queries. The purpose of a Snippet is to minimize the number of places you need to make changes on the same code in different queries.



Test Suite

A collection of QueryPairs used for execution purposes. This feature allows you to group QueryPairs for execution that is independent of your 'Design Library' organization



Scenarios

A collection of *Suites* that are scheduled for execution.



Query Wizard

A tool that allows you to generate *QueryPairs* automatically, requiring no manual SQL coding. The *Query Wizard* generates tests that can cover about 80% of all data in a data warehouse automatically.[1]



Widgets

Project *Widgets* give you a real-time view into your project progress at all levels, from *QueryPair* development to execution and results.



Command Line Integration

Provides the ability to schedule *Test Suites* to run using Windows Task Scheduler or integrate with both Data Integration (ETL) and Continuous Build systems.

[1] A recent poll conducted by RTTS on targeted LinkedIn groups found that 80% of columns in data warehouse tables have no transformations.

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Query*Surge*

Tutorial



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Consider This...

ZCity is a popular electronics corporation that is in the market for a new data warehouse so that corporate business personnel can assess the activities that are occurring throughout their sales regions. ZCity recently merged with one of its competitors, XMart, and now has two sales teams. The sales system environment for ZCity and XMART both reside on a MySQL database.

The business has decided that a data warehouse that runs nightly incremental ETL loads will help them make more effective business decisions. The analysts created a mapping document that maps out each source field to its corresponding target field from each source system. This includes any data transformations that the business deems necessary to be entered into the warehouse. The corporate IT team has completed development of a MySQL data warehouse (the target database) and an ETL process that includes transformation logic (from the mapping document) to load the source data from each source into the data warehouse.

The business' Quality Assurance team has been tasked with verifying the transformation logic and has purchased QuerySurge to validate 100% of the data being transformed from source to target. The Quality Assurance Manager has planned for the Quality Assurance team to review the mapping documents and build test cases, from the same logic that the developers used when creating the ETL code. The Query*Surge* term for test cases is "QueryPairs". The team is confident that this approach will not only cover testing 100% of the data but will also prove the mapping documents are comprehensive.

Their idea is that if two separate teams, the Development Team and the Quality Assurance Team, are building queries and logic from the same mapping documents separately and can attain the same results, then there is a very good chance there will be no defects in the production environment.

That's the plan; Now, it's time to get testing!

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Overview

You will perform the following tasks in the tutorial. It is recommended that you use two monitors. It also might be easier to print the tutorial.

- Generate QueryPairs using the "Query Wizards"
- Review existing and create new QueryPairs
- Review existing and create new Reusable Query *Snippets*
- Execute "Design-Time Runs"
- Investigate "Design-Time Run" failures
- Schedule and execute a *Scenario*
- Generate and review reports
- Review *Widgets*

The tutorial is comprised of the following assets:

1. ZCITY Source

- MySQL sales database (3 tables)
- 27 *QueryPairs*
- 1 *Test Suite* including all ZCITY *QueryPairs*
- 1 Reusable Query *Snippet*

2. XMART Source

- MySQL sales database (4 tables)
- 31 *QueryPairs*

3. Data Warehouse Target

- MySQL data warehouse (Dimensional Model)
  - 5 dimension tables
  - 2 fact tables

4. Mapping and Model Document (printing and reviewing this document is recommended)

The document is linked within the tool from the setup wizard and can also be downloaded from here: Tutorial Models and Mapping Document.

- ZCITY mappings and data model
- XMART mapping and file data model
- Data warehouse data model

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First and Subsequent Logins to QuerySurge

Upon QuerySurge installation, a project administrative user has been created for you, or an Administrator has already provided you with your own login credentials.

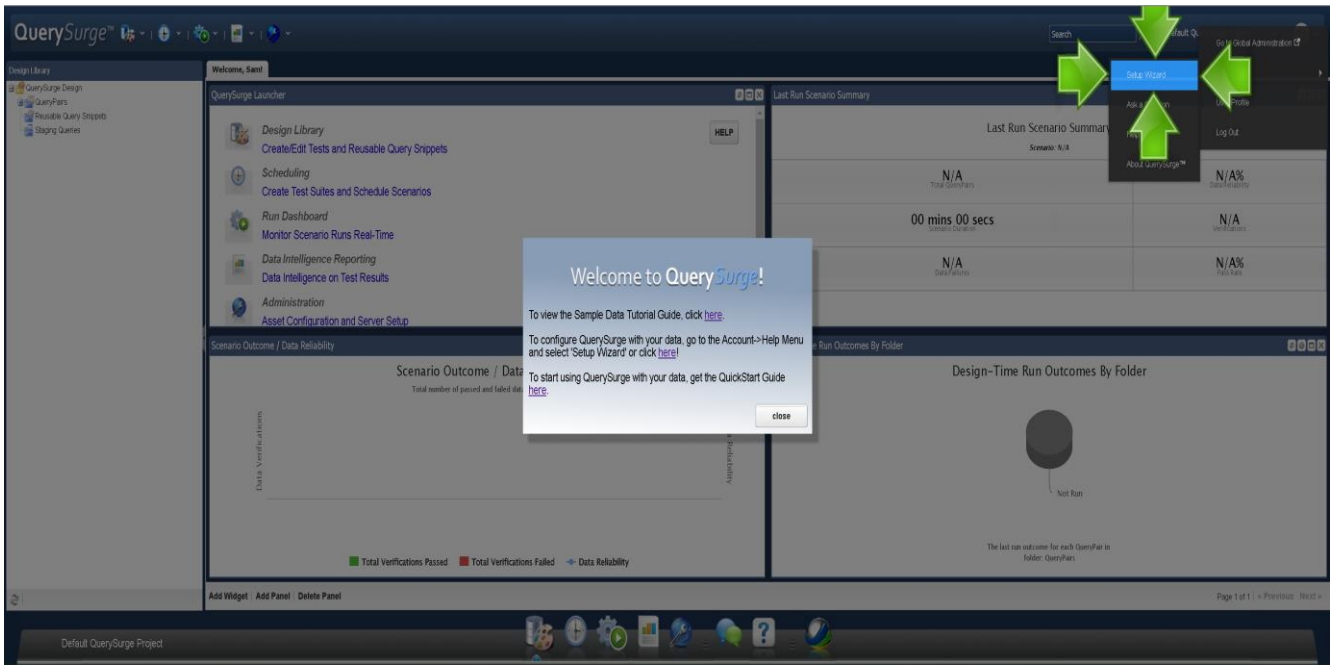
Note: If you have already logged into QuerySurge, this section will provide a review

1. On the installation machine, click **Start.**
2. Under the **Start** menu select All Programs > QuerySurge > Launch QuerySurge.
3. Query*Surge* will launch in your default browser.
4. If your Query*Surge* install is local, enter your credentials or use the credentials admin/admin. If you are using a Cloud trial, use the credentials clouduser/clouduser. Click **Login**.

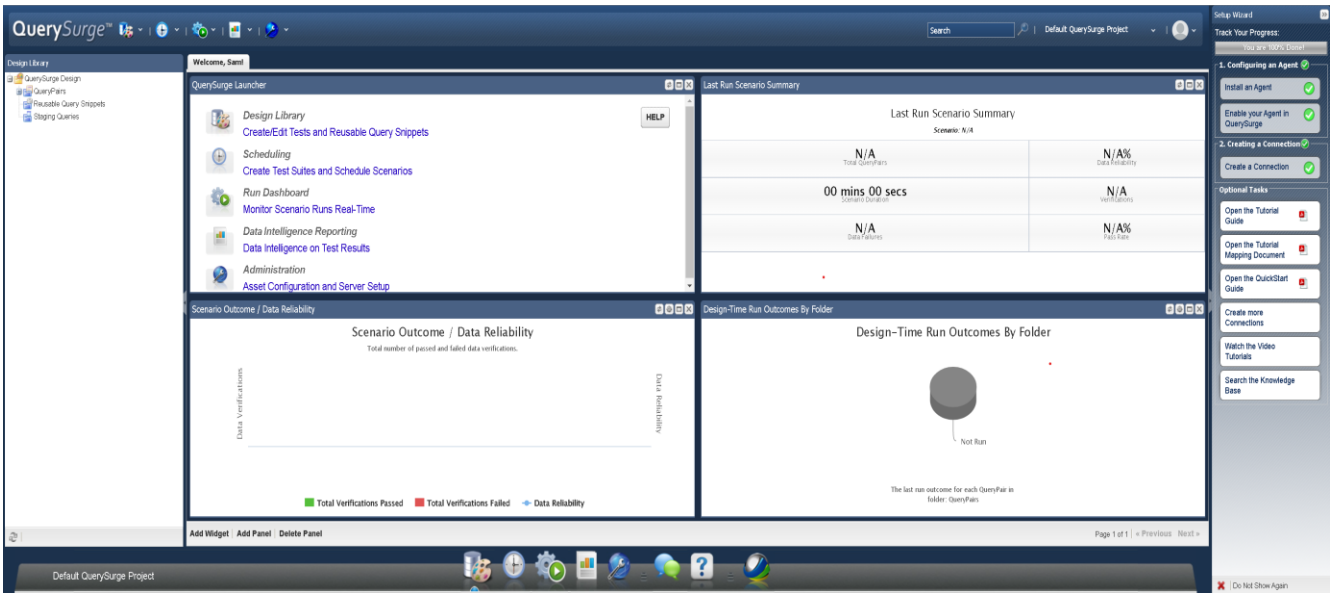


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5. If this is your first time launching Query*Surge*, a splash screen will display. If this is not the first time launching Query*Surge*, the "Setup Wizard" is always available to you on the top toolbar.



6. Click **"Setup Wizard"** on the top toolbar if not already displayed.



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Points of interest:

- *Setup Wizard* panel on the right
  - The panel is divided into 3 sections:
    - Configuring an Agent
    - Creating a Connection
    - Optional Tasks
  - The panel shows that an *Agent* has been installed and enabled (The green circle with the check mark next to task 'Install an Agent' and 'Enable your Agent in QuerySurge' signifies a completed task) 
  - The 'Create a Connection' link brings you to the connection wizard
  - The 'Optional Tasks' section allows you to easily navigate to documentation and videos or to create additional connections
- *Data Analytics Dashboard* panel in the middle
  - The center panel displays QuerySurge's *Data Analytics Dashboard* and *Data Widgets*, which is a configurable dashboard that provides a view into project information (more on this later)

7. Collapse the 'Setup Wizard' by clicking the button located at the top right corner of the Querysurge window. 

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Create *QueryPairs* Using the *Query Wizard* for *Column-Level Comparison*

A *QueryPair* is what QuerySurge refers to as a test case – two queries with the following characteristics:

- A SQL query that retrieves data from a source file or database and
- Another SQL query that retrieves data from a target big data store, data warehouse, data mart or database

The *Query Wizard* is a feature that allows you to generate *QueryPairs* automatically requiring no SQL coding. It is a fast and easy way to create *QueryPairs* for both skilled team members and beginners with less experience who want to increase testing speed.

Based on a LinkedIn poll of Data Warehouse experts, on average 80% of columns in a typical data warehouse have no transformations at all. The *Query Wizards* were developed to generate tests that can cover approximately 80% of all data in a data warehouse quickly and without writing SQL code. The *Wizards* are user friendly and provide amazing results for both beginners and experienced testers.

The *Query Wizard* can generate *QueryPairs* for:

- **Column-Level Comparison:** this is great for Big Data stores and Data Warehouses where tables will have some columns with transformations (20% on average) and some columns with no transformations (80% on average). The *Column-Level Comparison* is for the specific columns with no transformation. Columns with transformation will require SQL coding.

- **Table-Level Comparison:** this is great for data migrations and database upgrades with no transformation at all. Many tables can be compared simultaneously and quickly.

- **Row Count Comparison:** this works with all - Big Data stores, Data Warehouses, Data Migrations and Database Upgrades. Many tables can be compared simultaneously and quickly.

A folder can be created that contains all the *QueryPairs* generated. These *QueryPairs* can also be added to a new *Test Suite* so they can be easily run together (more on this later).

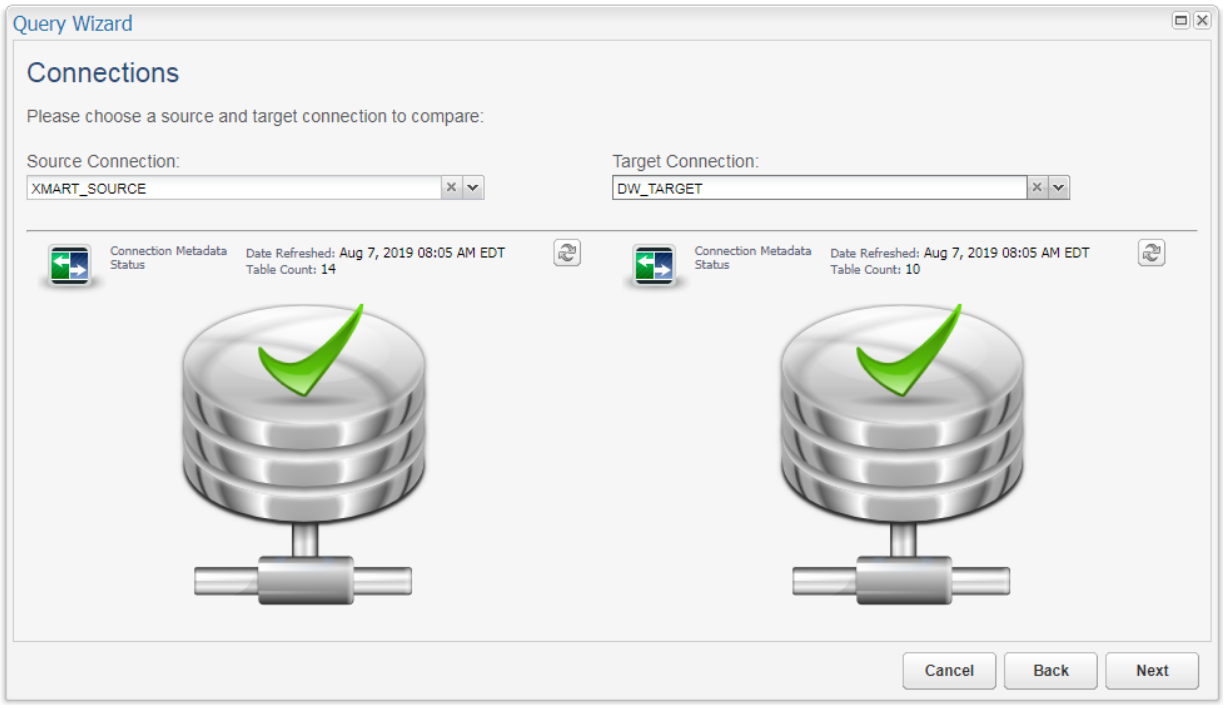
Note: This part of the tutorial illustrates the use of the *Query Wizard* for creating a *QueryPair* with *Column-Level Comparisons*

1. Click the dropdown menu on the **"Design Menu"** icon in the top toolbar and select **"Launch Query Wizard"**



2. Click the **Next** button.
3. Select "XMART_SOURCE" from the **"Source Connection"** drop down menu
4. Select "DW_TARGET" from the **"Target Connection"** drop down menu.

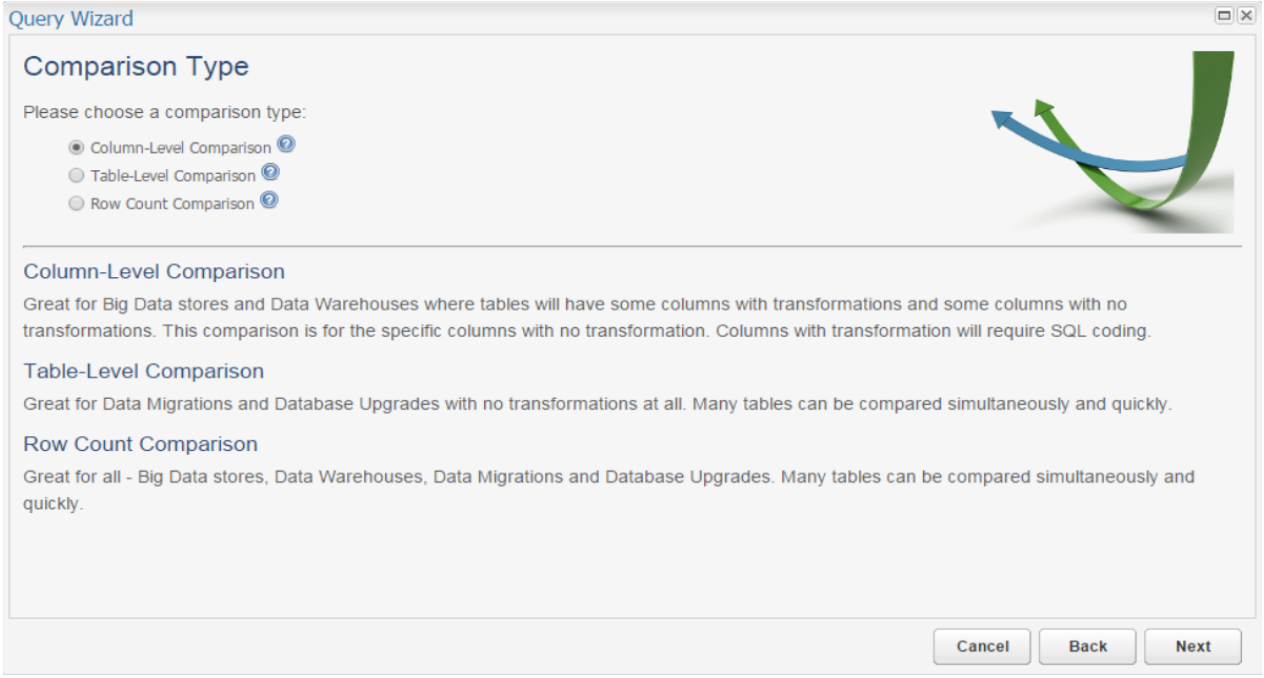
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Points of interest:

- Metadata (or data about our data) pre-populates for the selected databases

5. Click the **Next** button**.**
6. Keep the default **"Column-Level Comparison"** selected on the **"Comparison Type"** dialog window.

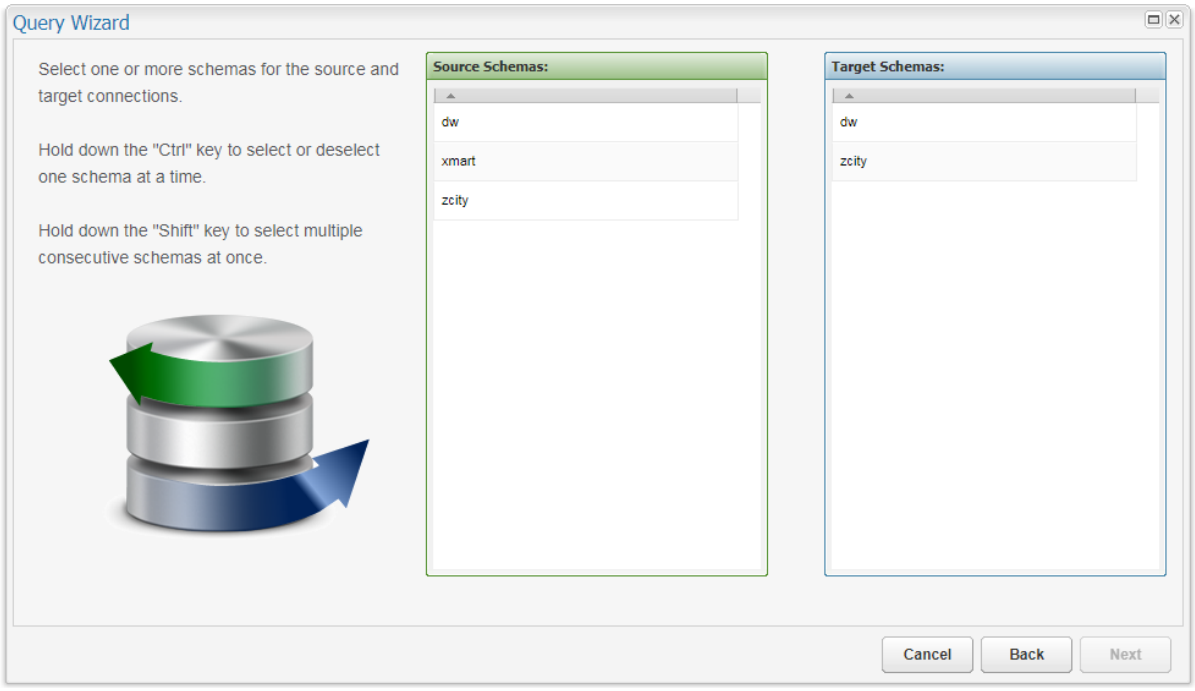


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Points of interest:

- *Column-Level Comparison* validates data points in the selected columns and allows for the addition of Filtering (WHERE Clause) and Sorting (ORDER BY)
- *Table-Level Comparison* validates each data point in the source table to its corresponding data point in the target table
- *Row Count Comparison* validates the number of rows in each of the selected tables

7. Click the **Next** button



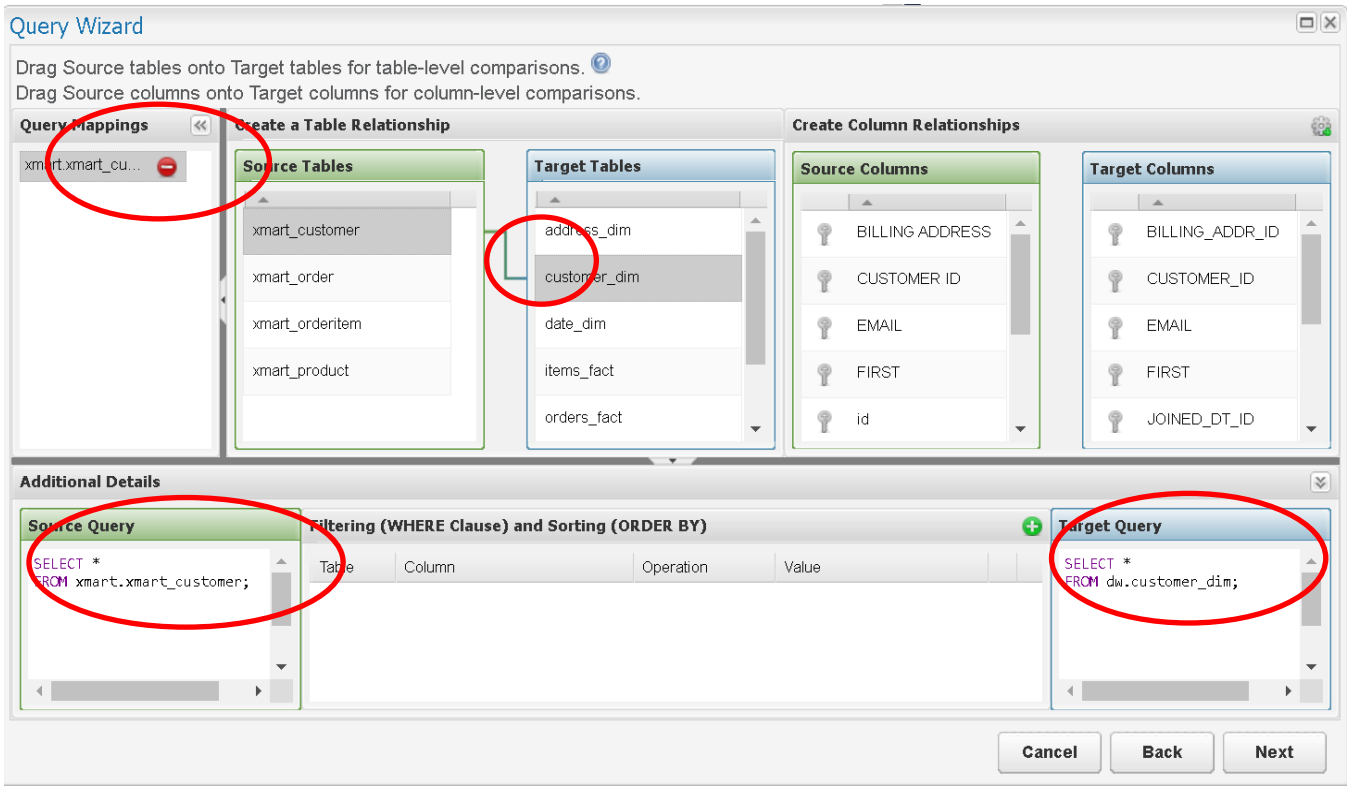
Points of interest:

- Schemas available for selection are retrieved from the source and target databases that you previously selected
- One, many, or all schemas can be selected

8. Select 'xmart' on the **"Source Schemas"** pane and 'dw' on the **"Target Schemas"** pane and click **Next**

9. Select and drag 'xmart_customer' under **"Source Tables"** and drop it into it's corresponding **"Target Tables"** 'dw.customer_dim'

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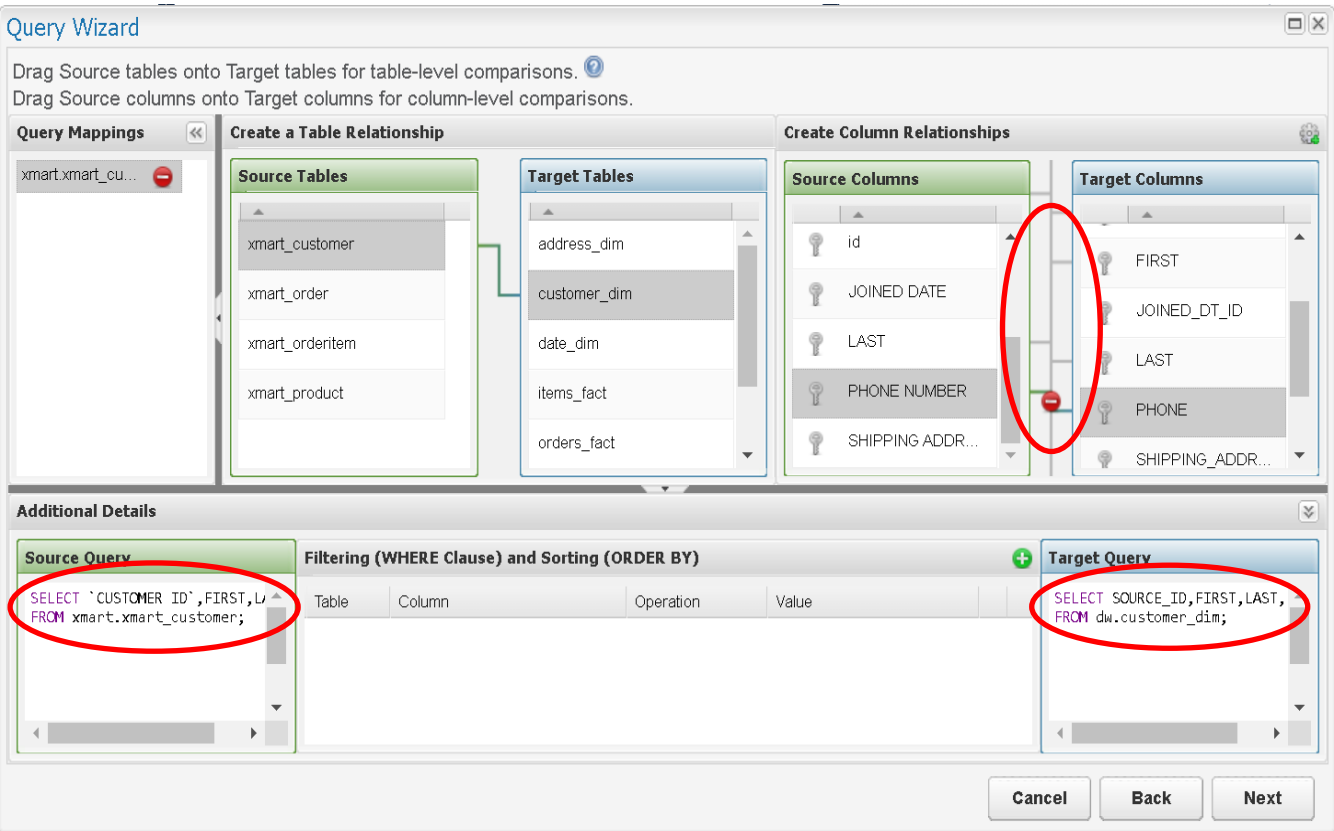
Points of interest:

- As a source table is dropped onto its corresponding target table, its 'Query Mapping' appears in the left-hand column (if desired, you can double click to rename it)
- Mapped tables show a line connecting the source and target table with the selected 'Query Mappings' line highlighted green
- Auto generated SQL appears in the 'Source Query' and 'Target Query' panes at the bottom of the *Query Wizard* window

10. Select and drag 'CUSTOMER ID' under **"Source Columns"** and drop it into its corresponding **"Target Columns"** 'SOURCE_ID'.
11. Select and drag "FIRST" under **"Source Columns"** and drop it onto "FIRST" in its corresponding **"Target Columns"**.
12. Select and drag "LAST" under **"Source Columns"** and drop it onto "LAST" in the corresponding **"Target Columns"**.
13. Select and drag "EMAIL" under **"Source Columns"** and drop it into "EMAIL" in its corresponding **"Target Columns"**.

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14. Select and drag “PHONE NUMBER” under **“Source Columns”** and drop it into “PHONE” in its corresponding **“Target Columns”**.



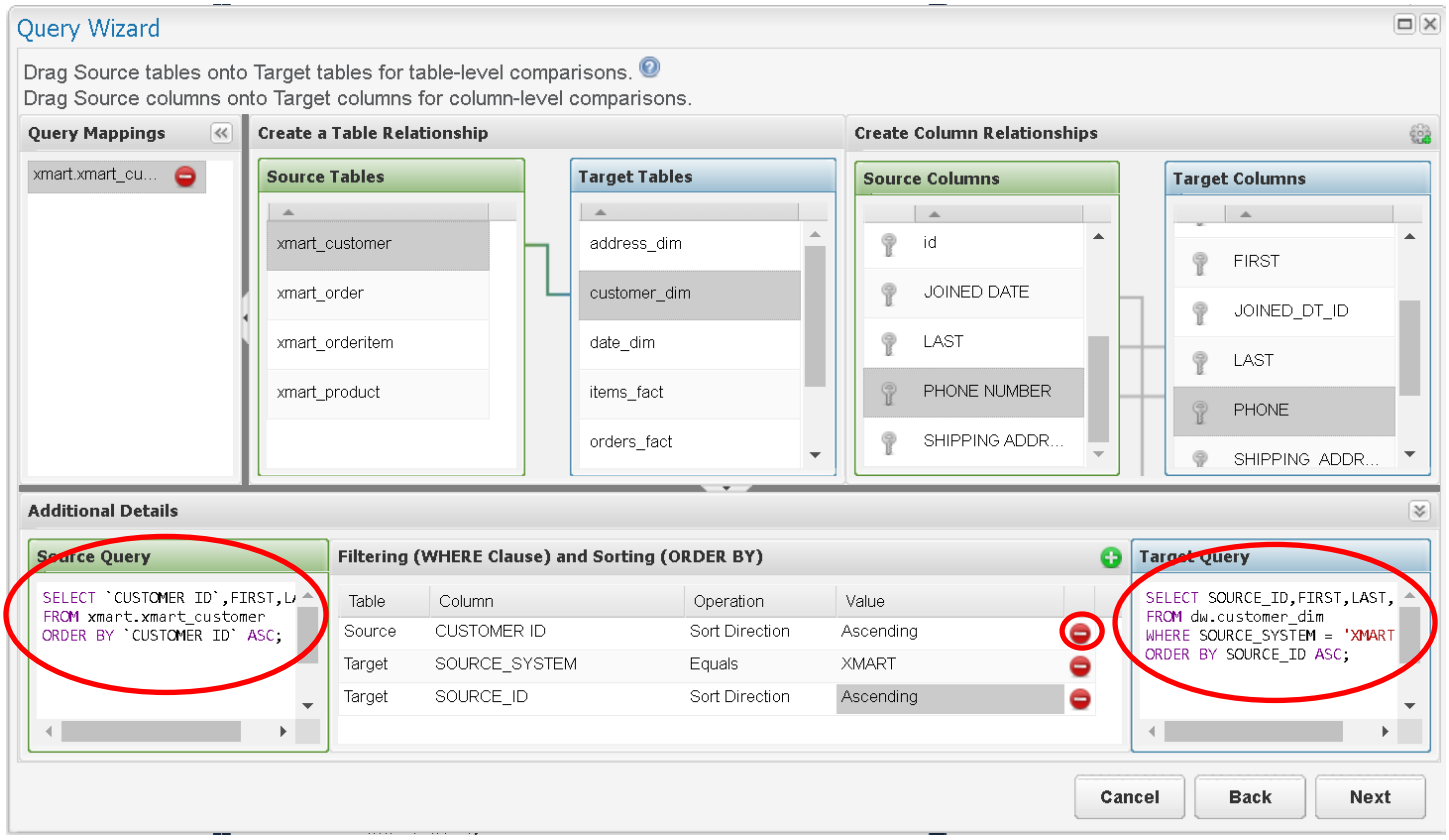
Points of interest:

- As a source column is dropped onto its corresponding target column the auto-generated SQL will change to reflect your selections
- Column relationships show lines connecting the source and target columns with the currently selected relationships highlighted in green
- Relationships can be removed by clicking the ‘Remove Relationship’ icon

15. In the lower middle of the screen under **“Filtering (WHERE Clause) and Sorting (ORDER BY)”** click the **“Add New Criteria”** icon
16. Under **“Table”** in the newly created criteria field click the dropdown menu and select ‘Source’.
17. Under **“Column”** click the dropdown menu and select ‘CUSTOMER ID’.
18. Under **“Operation”** click the dropdown menu and select ‘Sort Direction’.
19. Under **“Value”** click the dropdown menu and select ‘Ascending’.
20. Click the **“Add New Criteria”** icon.

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21. Under **"Table"** click the dropdown menu and select 'Target'.
22. Under **"Column"** click the dropdown menu and select 'SOURCE_SYSTEM'
23. Under **"Operation"** click the dropdown menu and select 'Equals'.
24. Under **"Value"** type 'XMART'.
25. Click the **"Add New Criteria"** icon .
26. Under **"Table"** click the dropdown menu and select 'Target'.
27. Under **"Column"** click the dropdown menu and select 'SOURCE_ID'.
28. Under **"Operation"** click the dropdown menu and select 'Sort Direction'.
29. Under **"Value"** click the dropdown menu and select 'Ascending'.

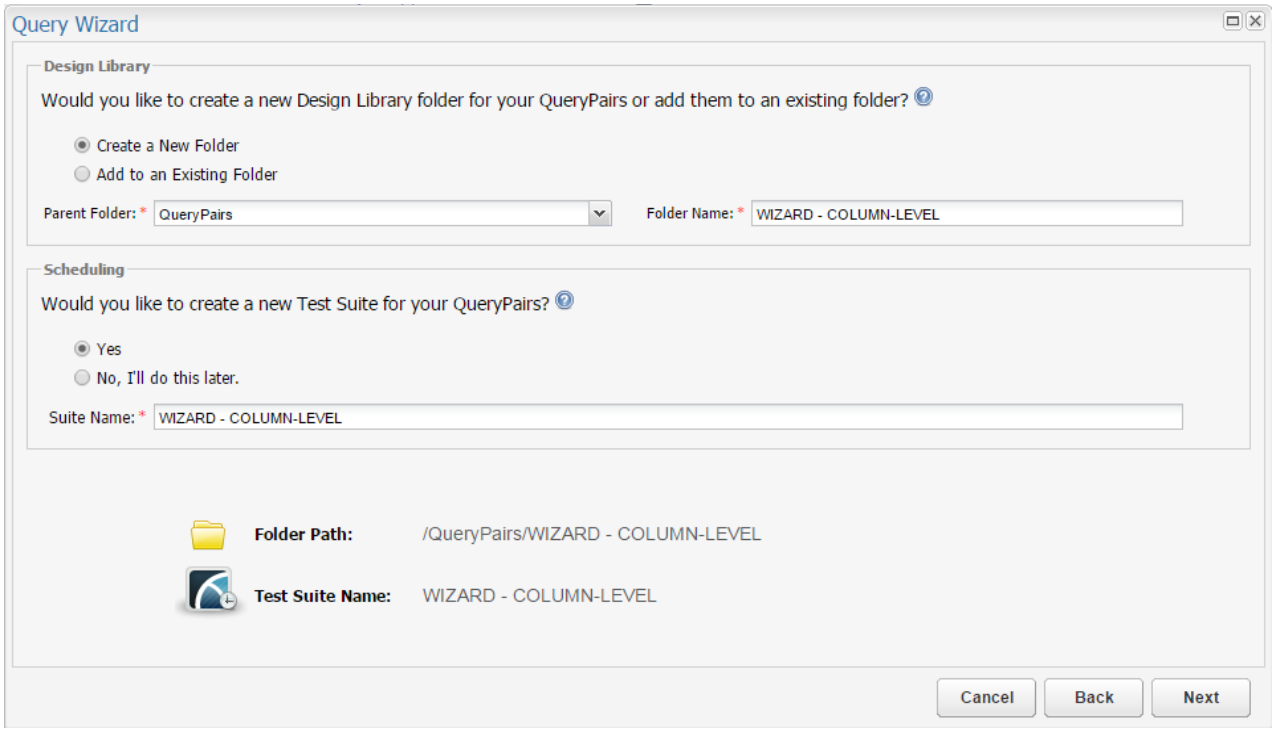


Points of interest:

- Added criteria appear in the 'Filtering (WHERE Clause) and Sorting (ORDER BY)' section
- Auto generated SQL based on added filter criteria appear in the query panes
- Criteria can be removed by clicking the 'Remove Criteria' icon

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30. Click the **Next** button.
31. Under **"Design Library"** select **"Create New Folder"** if not already selected.
32. Under **"Parent Folder"** click the dropdown menu and select 'QueryPairs'.
33. Type 'WIZARD – COLUMN-LEVEL' in the **"Folder Name"** field.
34. Under **"Scheduling"** select **"Yes"**.
35. Type 'WIZARD – COLUMN-LEVEL' in the **"Suite Name"** field.

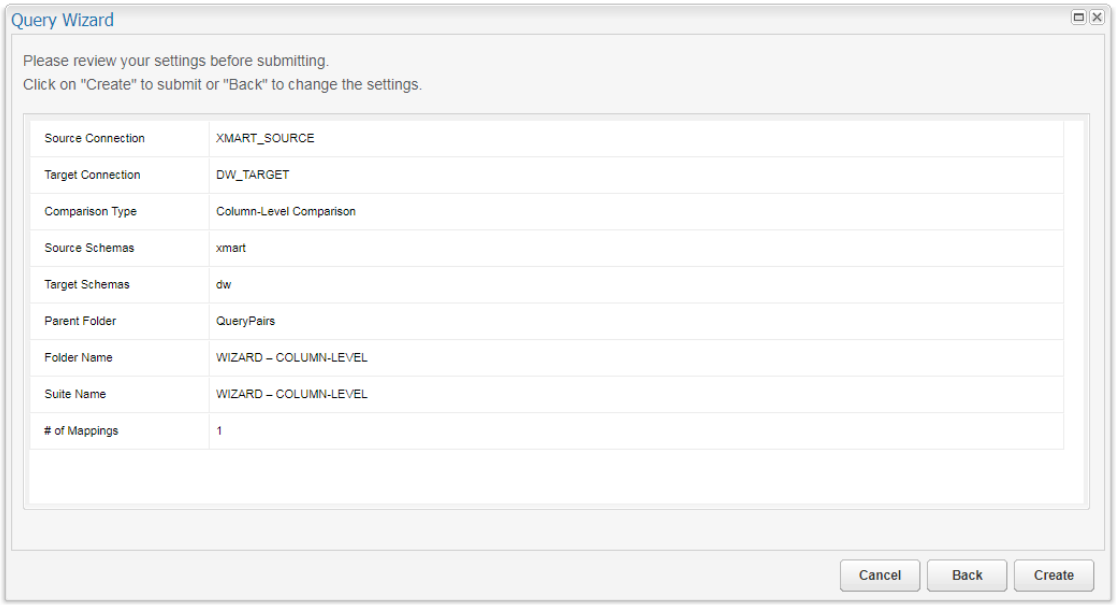


Points of interest:

- The *Query Wizard* will automatically create the following test assets:
    - A new folder in the 'Design Library' with the name 'WIZARD – COLUMN-LEVEL'
    - All *QueryPair(s)* for testing the tables selected
    - A new *Test Suite* in the 'Scheduling' module with the generated *QueryPair(s)*

36. Click the **Next** button.

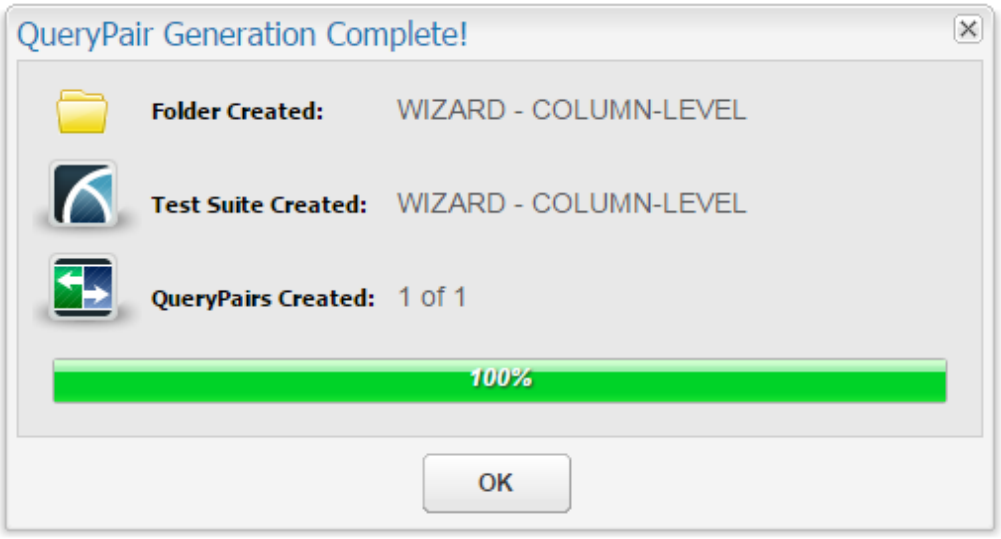
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Points of interest:

- A review screen appears with the details for your *Query Wizard* execution

37. Click the **Create** button.



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Points of interest:

- A folder named 'WIZARD – COLUMN-LEVEL' was created in the 'Design Library' module
- A *Test Suite* named 'WIZARD – COLUMN-LEVEL' was created in the 'Scheduling' module
- One *QueryPair* was created

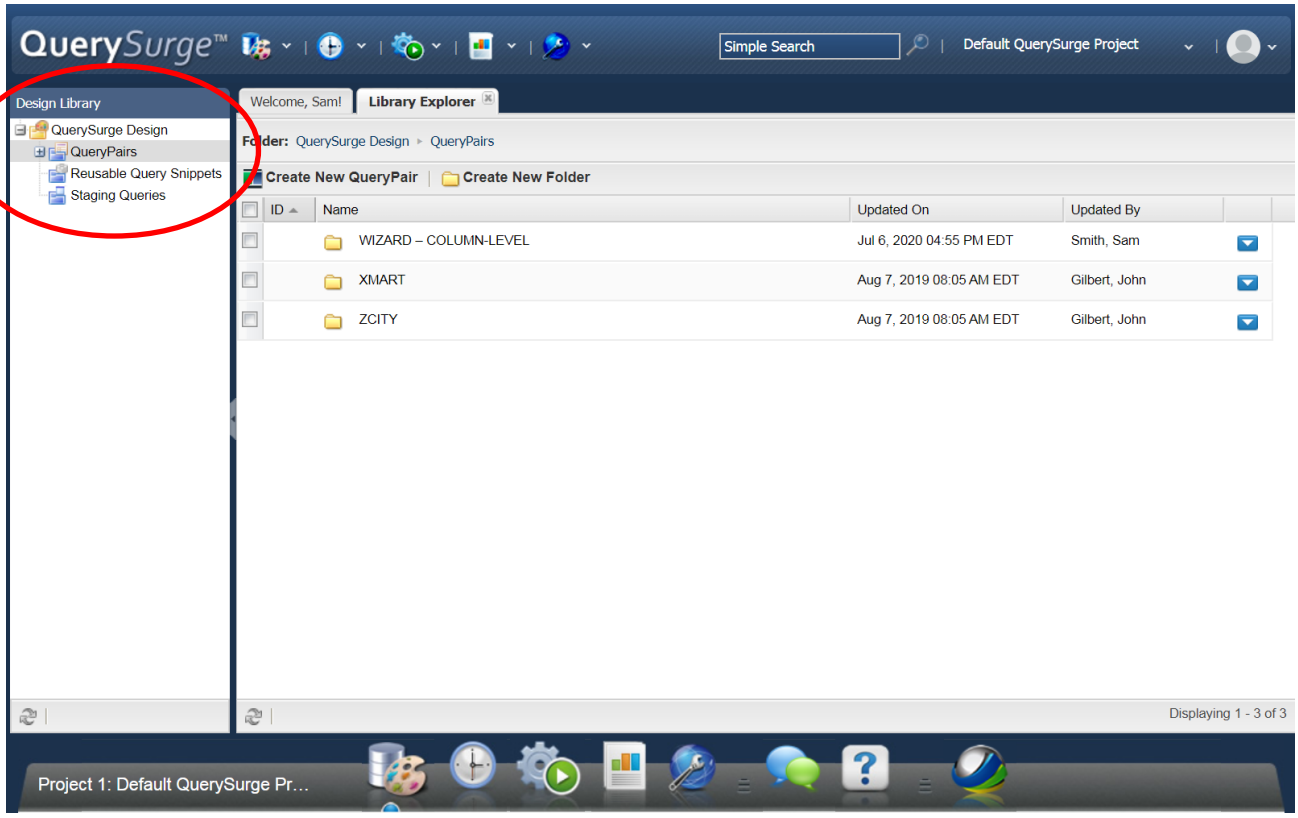
38. Click the **OK** button.

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Execute a *QueryPair* that passes

The QueryPair created in the previous chapter will be utilized for this section of the tutorial

1. In the **"Design Library"** panel on the left, select "QueryPairs".

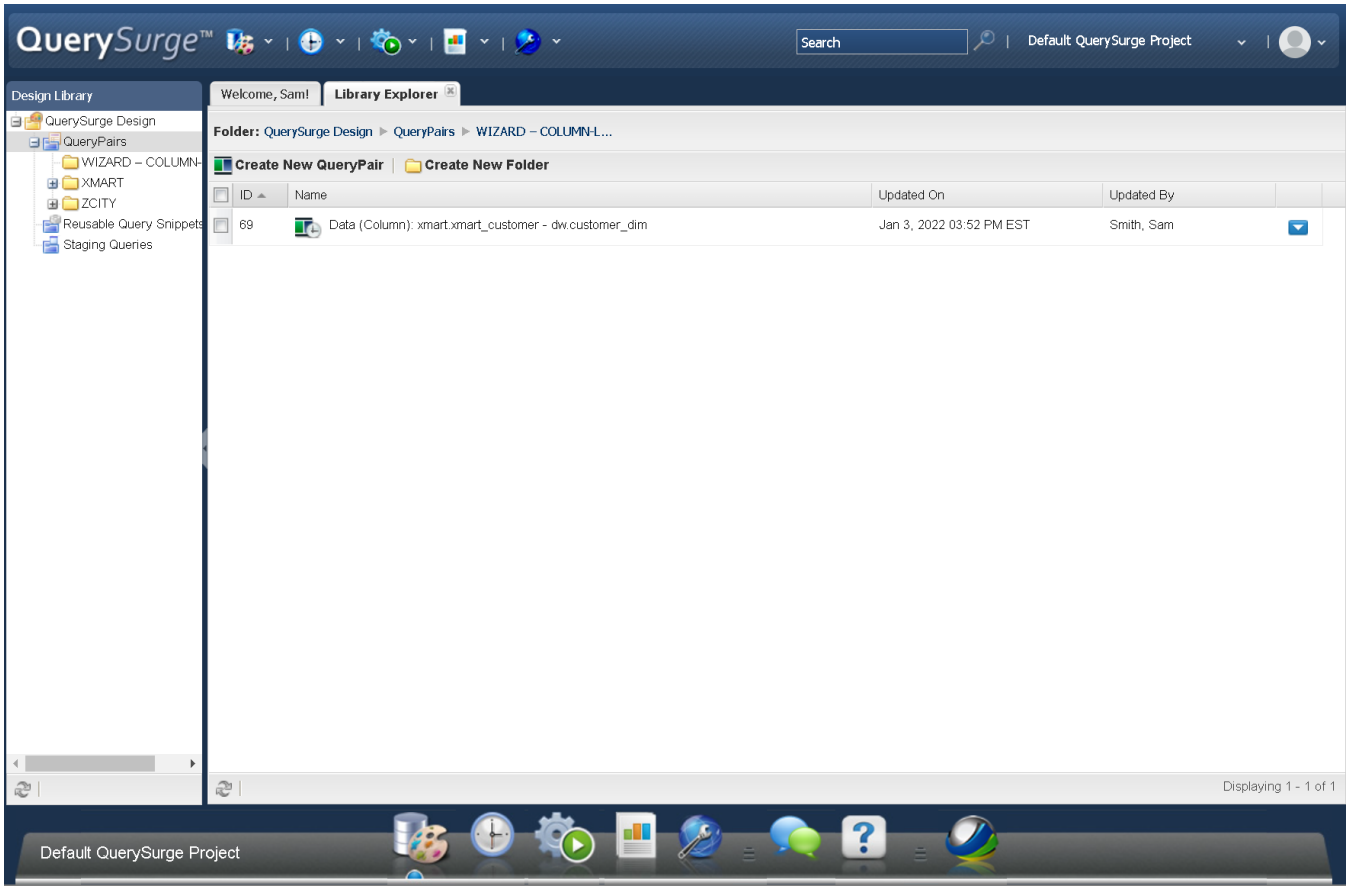


Points of interest:

- Query*Surge* utilizes a folder structure similar to Microsoft Windows Explorer
- Multiple sub-folders can be contained within folders
- The left panel allows for simple navigation
- The tutorial has two main folders, one for each source system, and the folder you created while running the *Query Wizard*

2. Double click the 'WIZARD - COLUMN-LEVEL' folder in the center panel.

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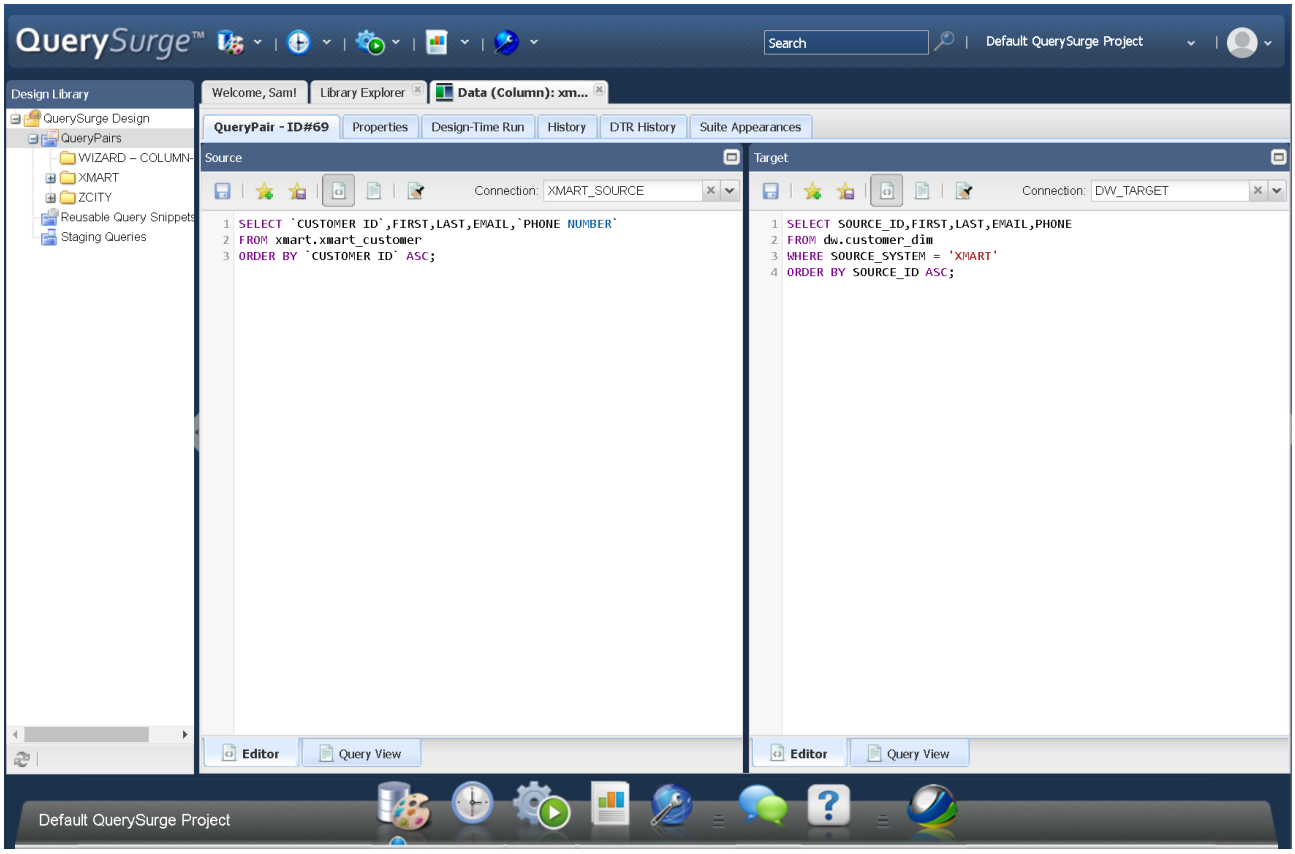


Points of interest:

- The Query*Pair* contained within the 'WIZARD - COLUMN-LEVEL' folder is displayed on the center panel
- In this case, there is only one *QueryPair* that was created during the previous chapter

3. Double click the 'Data (Column): xmart.xmart_customer - dw.customer_dim' *QueryPair*

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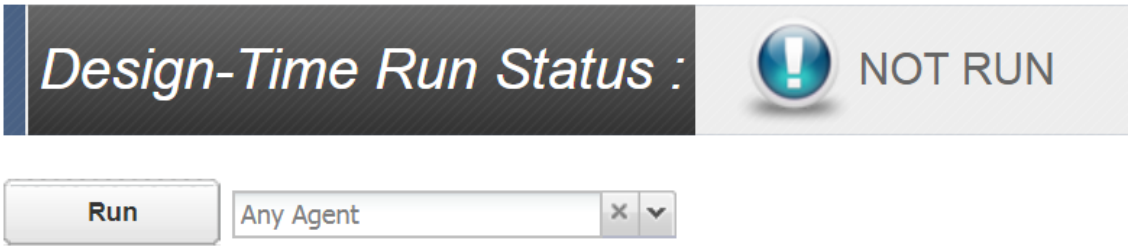
Points of interest:

- The *QueryPair* was created to validate that the source customer data was loaded into the data warehouse. (See mapping 1.01 – 1.05 of the tutorial mapping document - a link to the document is located on the 'Setup Wizard')
    - This *QueryPair* is validating the direct map (no logical transformation) of the data from the Source, XMART, to the target data warehouse
- The 'QueryPair' tab contains
    - 'Source Query' and 'Connection' – for storing the query with a specific database connection
    - 'Target Query' and 'Connection' – for storing the query with a specific database connection
    - 'Save' icons – for saving the source or target query 
- The 'Properties' tab contains
    - *QueryPair* name, description, and mapping fields
    - Data Type Checking – Ability to broadly check data types (e.g., text vs numeric) or convert all types to varchar (String) before comparison
    - Row Count Options – to set reporting for row count differences

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  - Shared Key Column Definition – to set a column from either the source or target query as key
  - Column Threshold Options – Ability to allow fields to pass if within a user-defined threshold
  - Ignorable Columns – Ability to ignore columns during analysis
  - Duplicate Row Options – to set whether QuerySurge uses a comparison algorithm to manage duplicate rows in your data
- The 'Design-Time Run' tab contains
  - Detailed results for each *QueryPair*. This tab allows you to execute a *QueryPair* test run to make sure that it behaves as you expect
- The 'History' tab shows the history of changes to the *QueryPair*
- The 'DTR History' tab shows pass/fail information from previous 'Design-Time Runs' (currently empty as no runs have been executed yet)

4. Click the **"Design-Time Run"** tab.

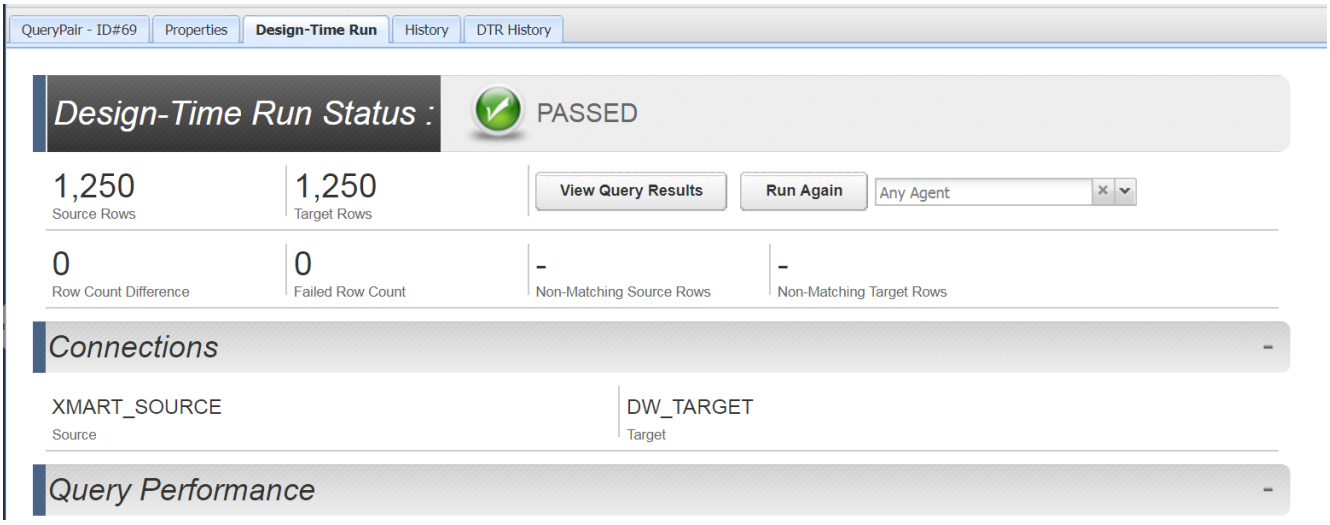


Points of interest:

- The first time clicking on the 'Design-Time Run' for a *QueryPair* will indicate that a run has not been previously executed
- A specific *Agent*, if desired, can be selected in the drop-down menu (not required)

5. Click the **"Run"** button to execute the *QueryPair.*

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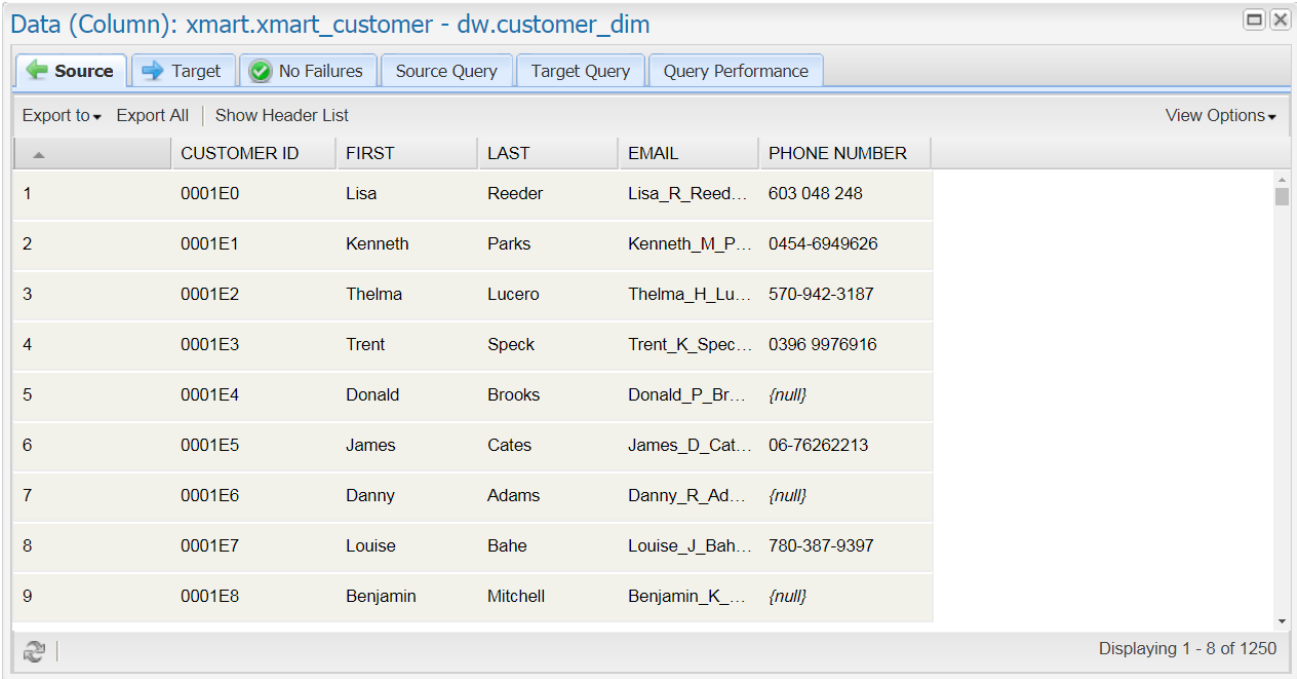


Points of interest:

- When executing a 'Design-Time Run', QuerySurge goes through the following phases:
  - Loading – QuerySurge is loading the *QueryPair* on the *Agent* for execution
  - Running – The *Agent* is running the target and source queries
  - Analyzing – QuerySurge is comparing the results
- The *QueryPair* passes
- Both source and target queries returned 1250 rows, and all rows matched from source to target
- QuerySurge displays the following information about the last 'Design-Time Run'
  - Source Rows – number of source rows returned
  - Target Rows – number of target rows returned
  - Row Count Difference – the difference between the number of source and target rows returned
  - Failed Row Count – number of rows with data differences
  - Non-Matching Source Rows – number of source rows not found in the target (based on key column(s))
  - Non-Matching Target Rows – number of target rows not found in the source (based on key column(s))
- 'Connections' and 'Query Performance' metrics

6. Click **"View Query Results"**

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Points of interest:

- The "QueryPair Results" window contains six tabs
  - Source Tab – displays the source result rows
  - Target Tab – displays the target result rows
  - Failures Tab – displays data failures (not populated in this example as no rows failed)
  - Source Query Tab – displays the source query
  - Target Query Tab – displays the target query
  - Query Performance – displays metrics
- Since this is a passing *QueryPair*, the source and target tabs display the same data (note: the column headers have been aliased)

7. Close the query results window.

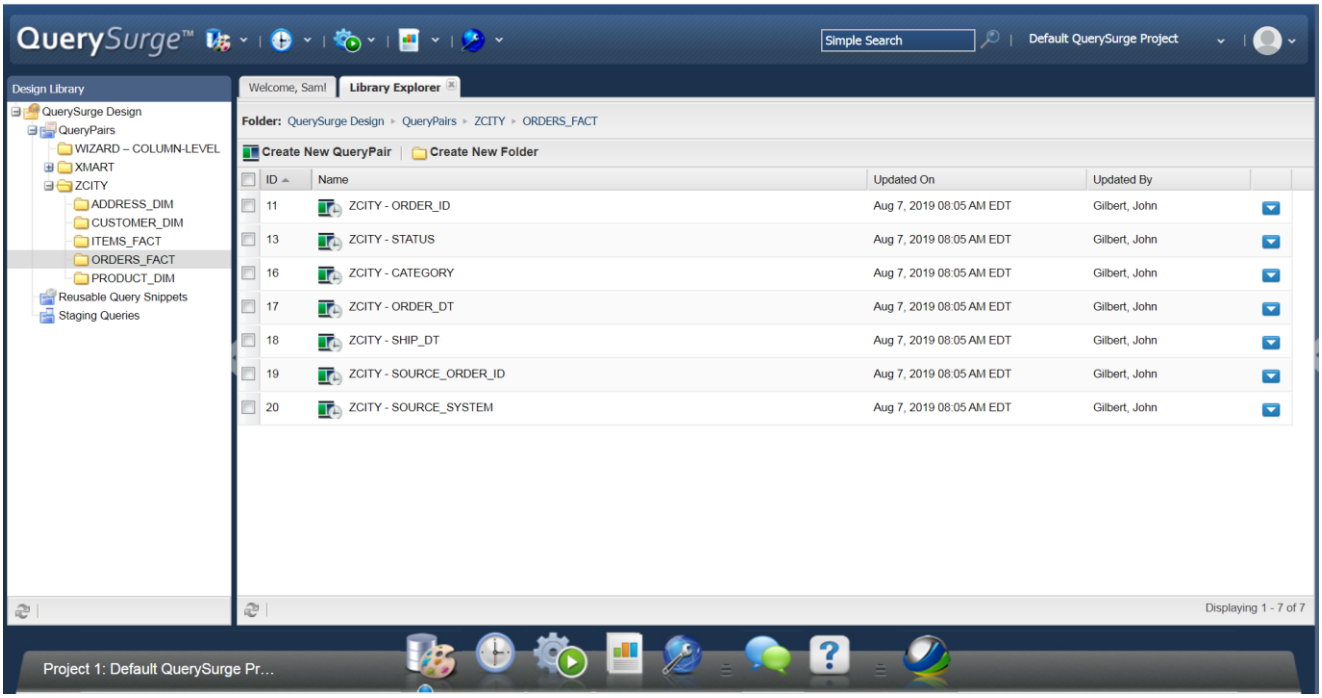
For more information on *QueryPairs*, please visit *Design Library*→*Working with QueryPairs* in the Knowledge Base

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Execute a QueryPair That Fails

Use QuerySurge to execute a QueryPair that fails.

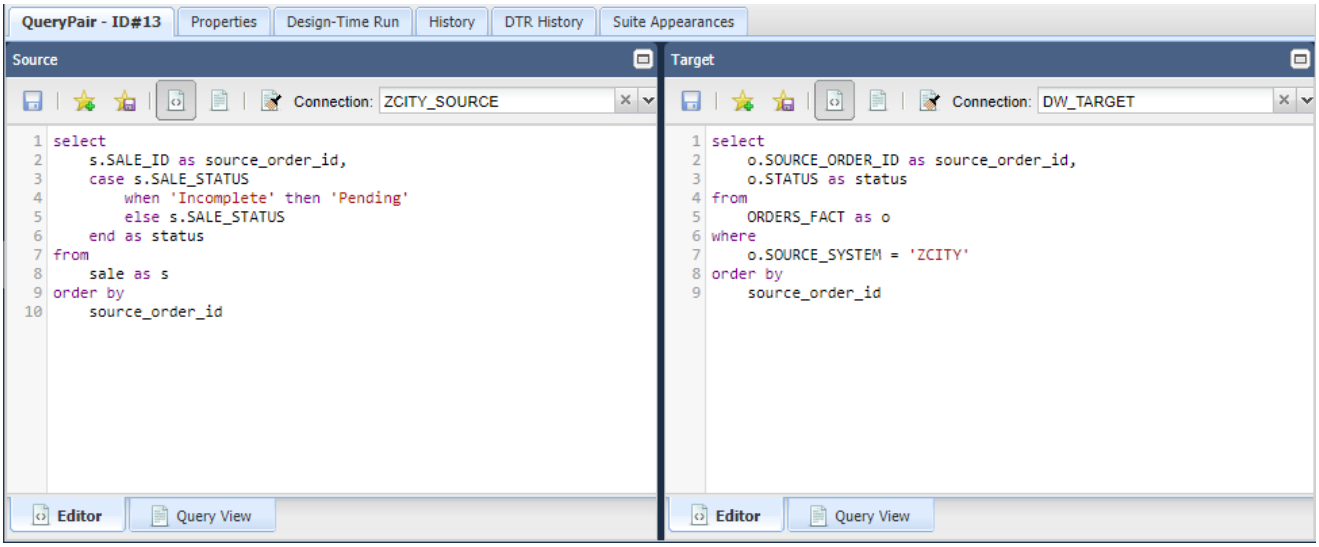
1. In the **"Design Library"** tree (left panel), navigate to the folders 'QueryPairs -> ZCITY -> ORDERS_FACT'



Points of interest:

- The *QueryPairs* contained within 'ORDERS_FACT' folder is displayed

2. Double click the 'ZCITY – STATUS' *QueryPair*.

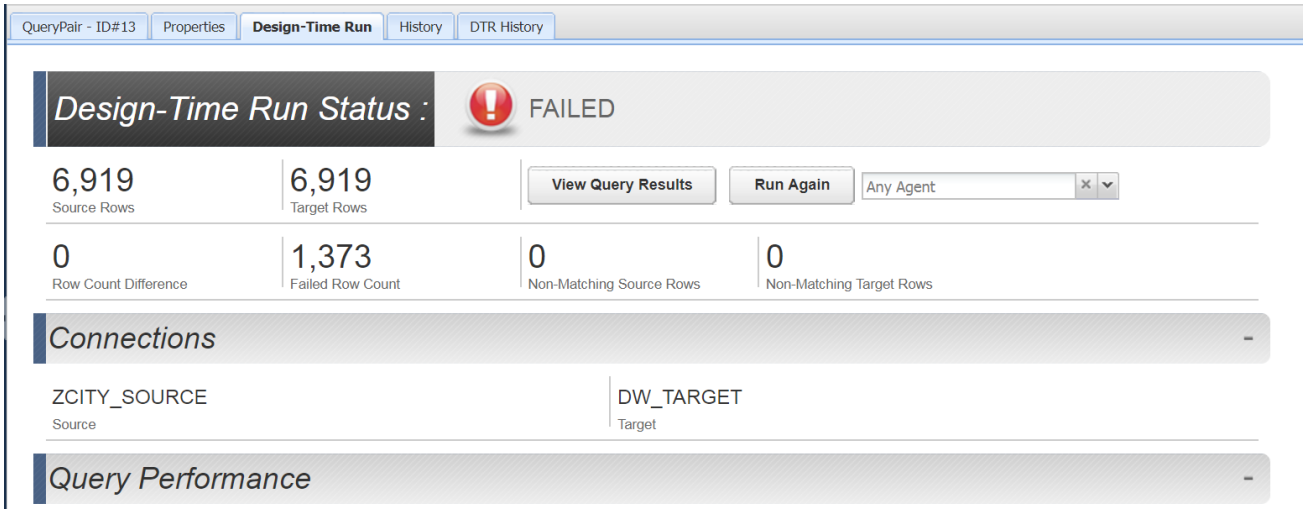


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Points of interest:

- The 'ZCITY – STATUS' *QueryPair* was created to validate the source SALE_STATUS column was loaded into the data warehouse correctly (See tutorial mapping document on the 'Setup Wizard').
  - This *QueryPair* is validating that the SALE_STATUS from source ZCITY and target data warehouse are loaded with the following logic:
    - Populate target ORDERS_FACT.STATUS using the following logic: if the SALE_STATUS value in the Sale table is 'Incomplete', then change to 'Pending'. Otherwise, leave as is.

3. Click the **"Design-Time Run"** tab.
4. Click the **Run** button to execute the *QueryPair.*

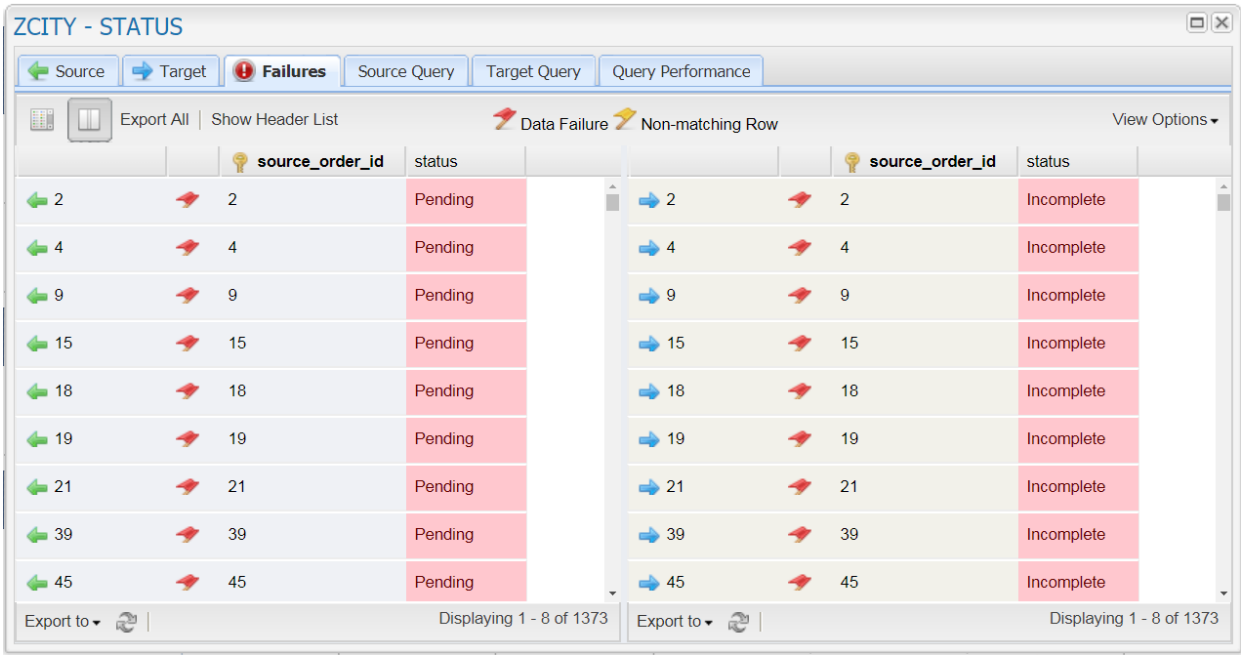


Points of interest:

- The 'ZCITY – STATUS' *QueryPair* fails. (If one or more fields do not match, then QuerySurge fails the *QueryPair*)
- Both source and target returned 6,919 rows
- 1,373 rows failed

5. Click the **"View Query Results"** button.
6. Click the **"Failures"** tab.

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Points of interest:

- The rows with left-pointing green arrows are source rows
- The rows with right-pointing blue arrows are target rows
- Notice the source rows contain the value 'Pending' and the target rows contain the value 'Incomplete'
- All failures were 'Data Failures', as indicated by a red flag
- If there were any 'Non-matching Rows', they would appear with a yellow flag
- Failures are shown with a red background

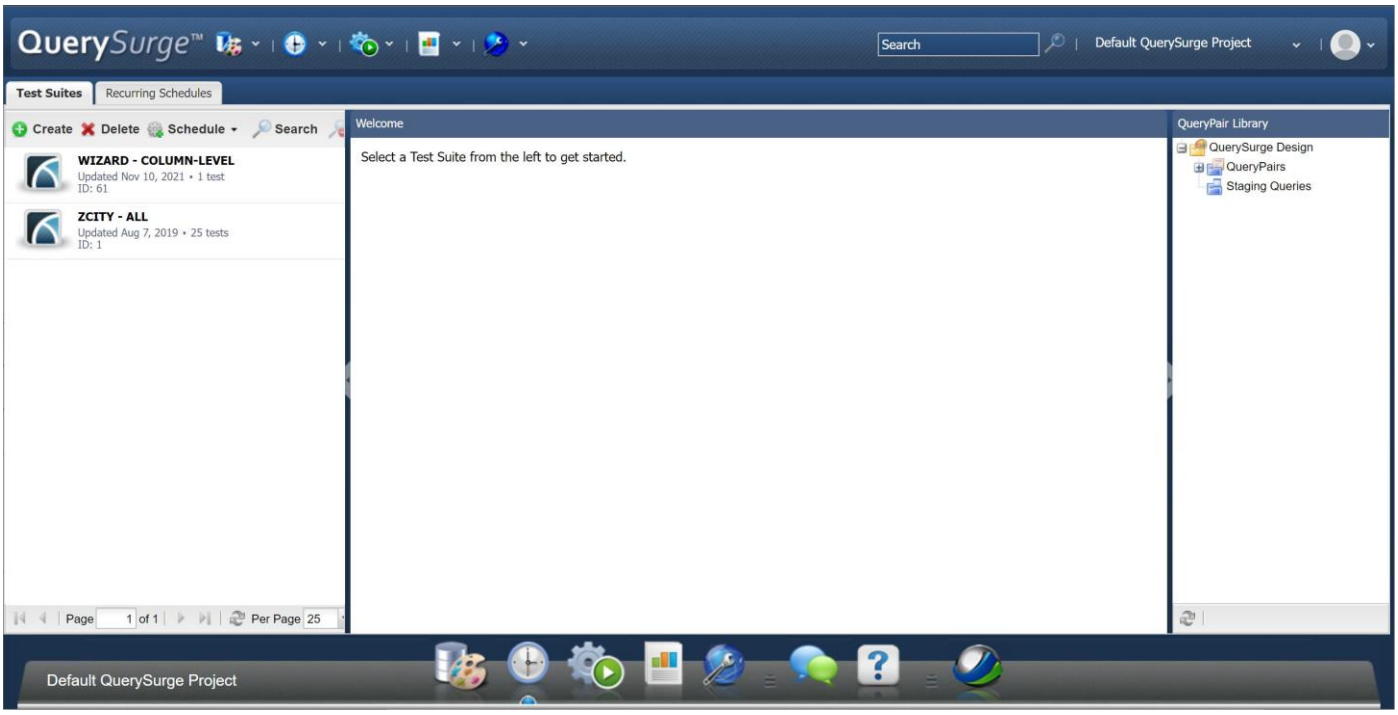
7. Close the 'ZCITY – STATUS' query results window.

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Create and Execute a Test Suite

The *Test Suites* feature of QuerySurge lets you organize your *QueryPairs* in *Test Suites* - collections of *QueryPairs* for execution. This feature allows you a different level of organization for execution purposes that is independent of your 'Design Library' organization.

1. Click the **"Scheduling"** icon located on the bottom panel. 

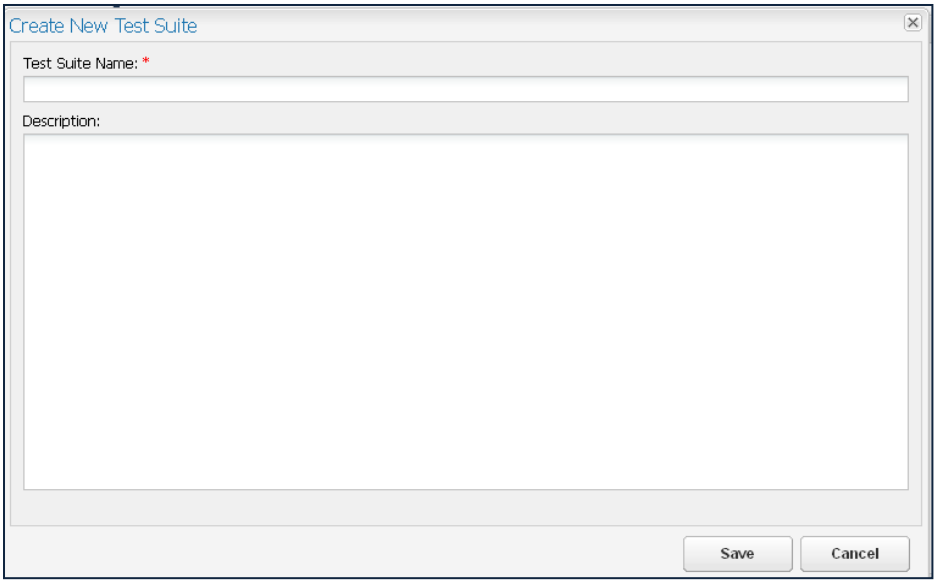


Points of interest:

- A *Test Suite* has been created for you called 'ZCITY – ALL'
  - 'ZCITY – ALL' contains all the *QueryPairs* associate with the ZCITY source
- The 'WIZARD – COLUMN-LEVEL' suite was created in a previous section of this tutorial
- Left Panel – displays all the available *Test Suites* that can be run
- Middle Panel – displays tests associated with a *Test Suite* (currently no *Test Suite* is selected)
- Right Panel – displays your 'QueryPair Library' tree

2. Click **"Create Suite"** on the left panel.

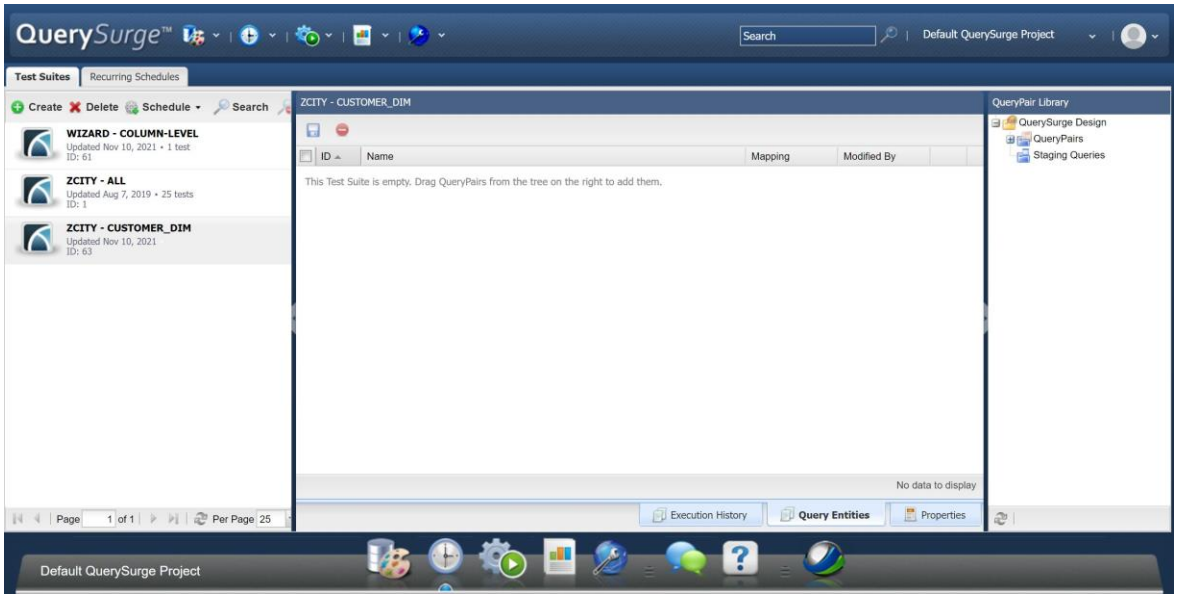
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Points of interest:

- The "Create New Test Suite" dialog contains:
  - Test Suite Name – a name for your suite
  - Description – a description for your suite

3. Enter 'ZCITY – CUSTOMER_DIM' in the **"Test Suite Name"** field
4. Enter 'To test the customer dimension fields' in the **"Description"** field
5. Click the **Save** button.

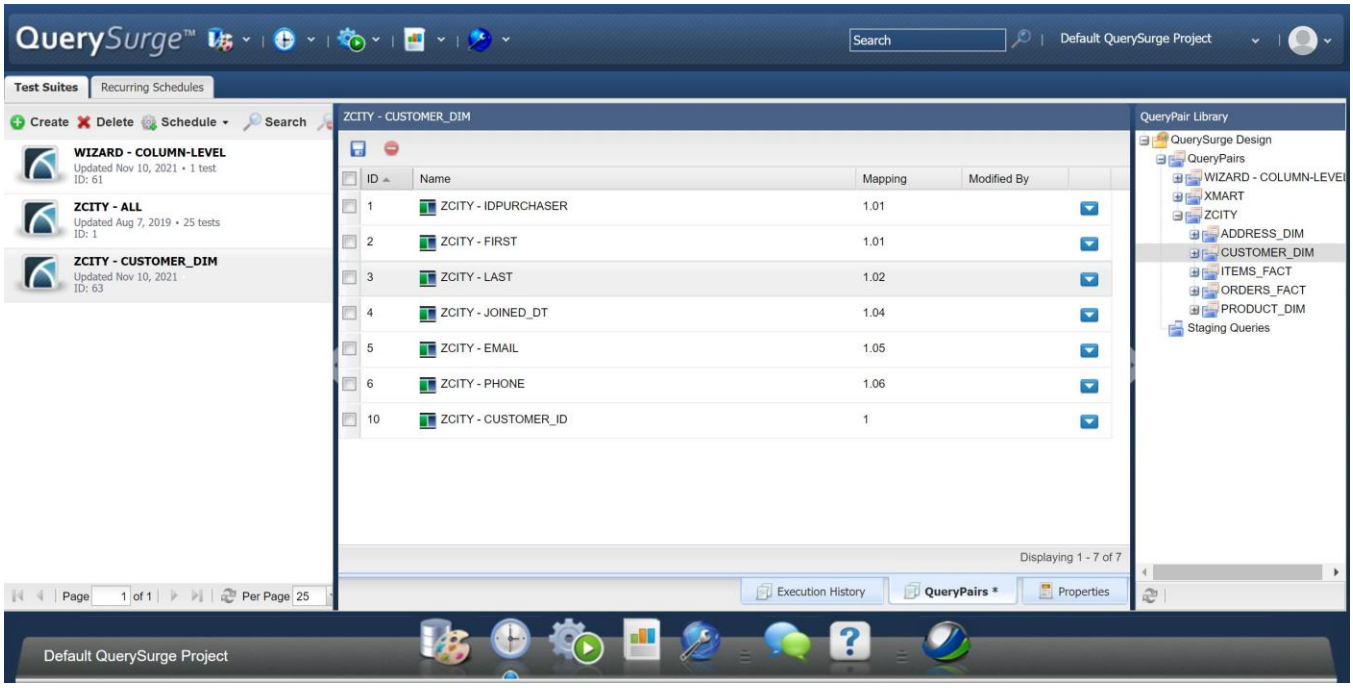


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Points of interest:

- Your new *Test Suite* is currently empty

6. Click "+" next to 'QueryPairs' in 'QueryPair Library' panel on the right and double click 'ZCITY'.
7. Drag and drop the 'CUSTOMER_DIM' folder to the 'ZCITY – CUSTOMER_DIM' test suite panel in the middle.

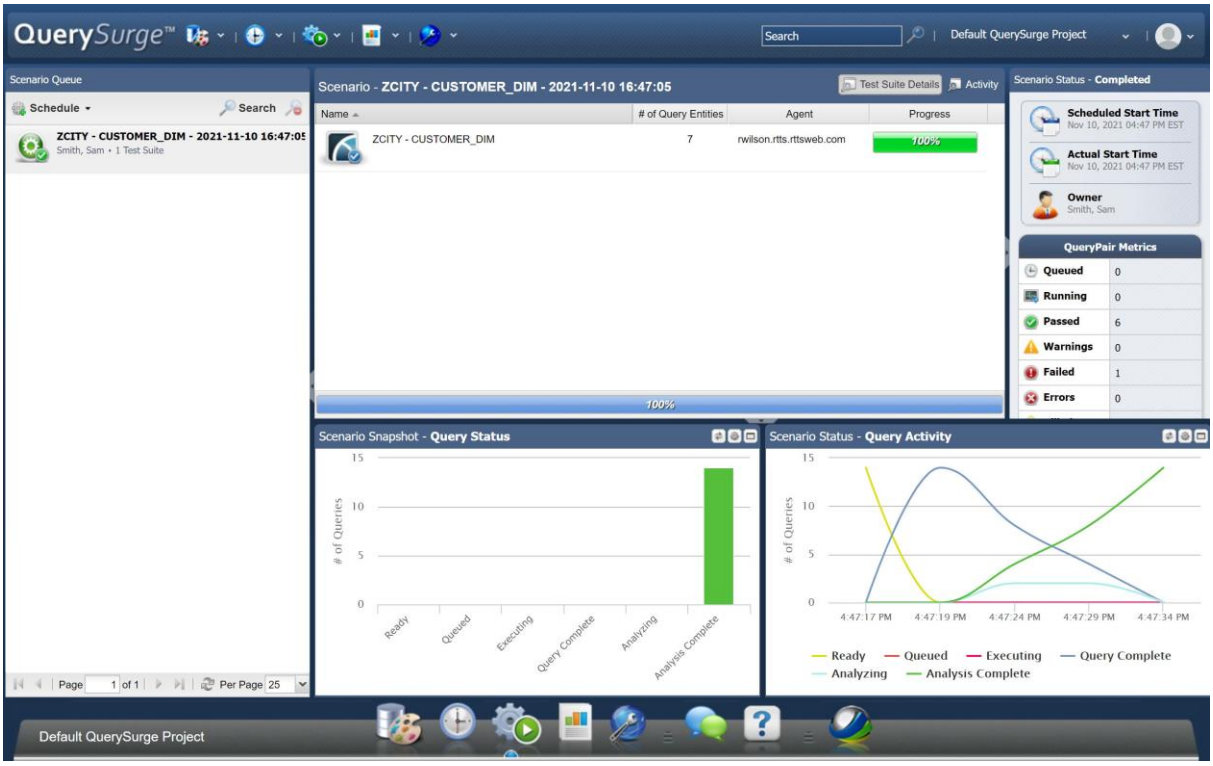


Points of interest:

- The 'ZCITY - CUSTOMER_DIM' *Test Suite* now contains all *QueryPairs* from the 'CUSTOMER_DIM' folder

8. Click the **Save** button in the upper left corner of the middle panel to save your Test Suite.
9. Right click on the "ZCITY – CUSTOMER_DIM" on the Test Suite panel (Left panel) and select **"Run Now"**.
10. Accept all defaults and click the **OK** button on the "Run Scenario" dialog.
11. QuerySurge will automatically toggle you over to the "Run Dashboard" and the Test Suite will execute.

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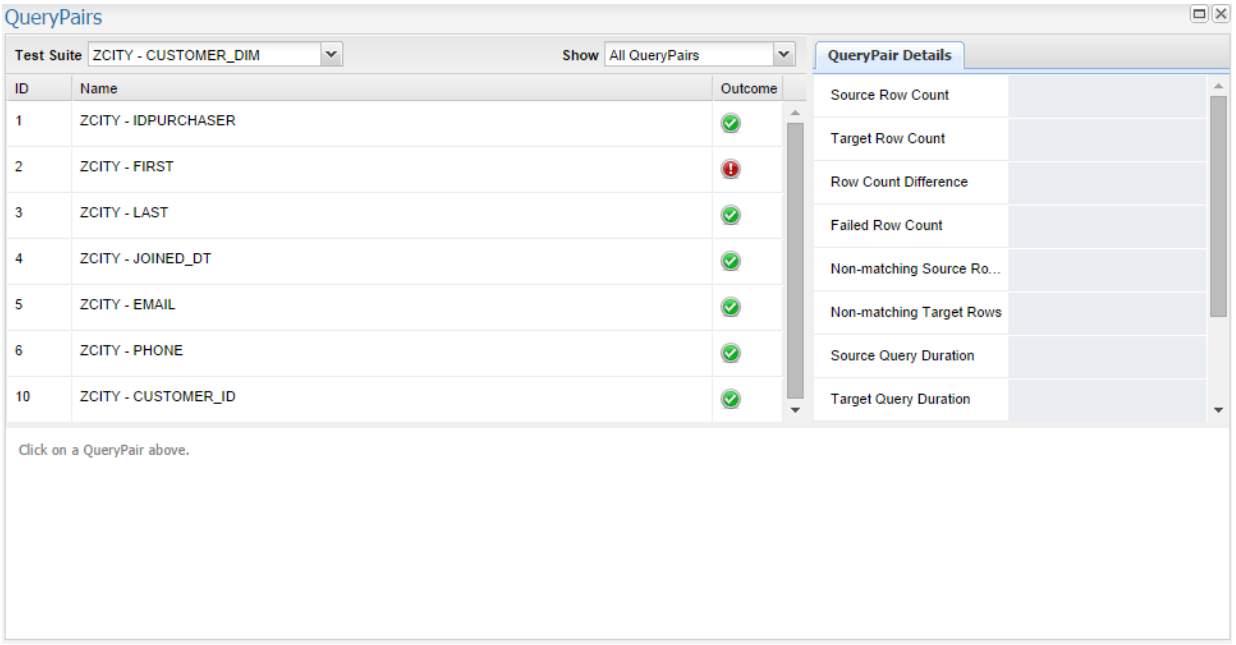


Points of interest:

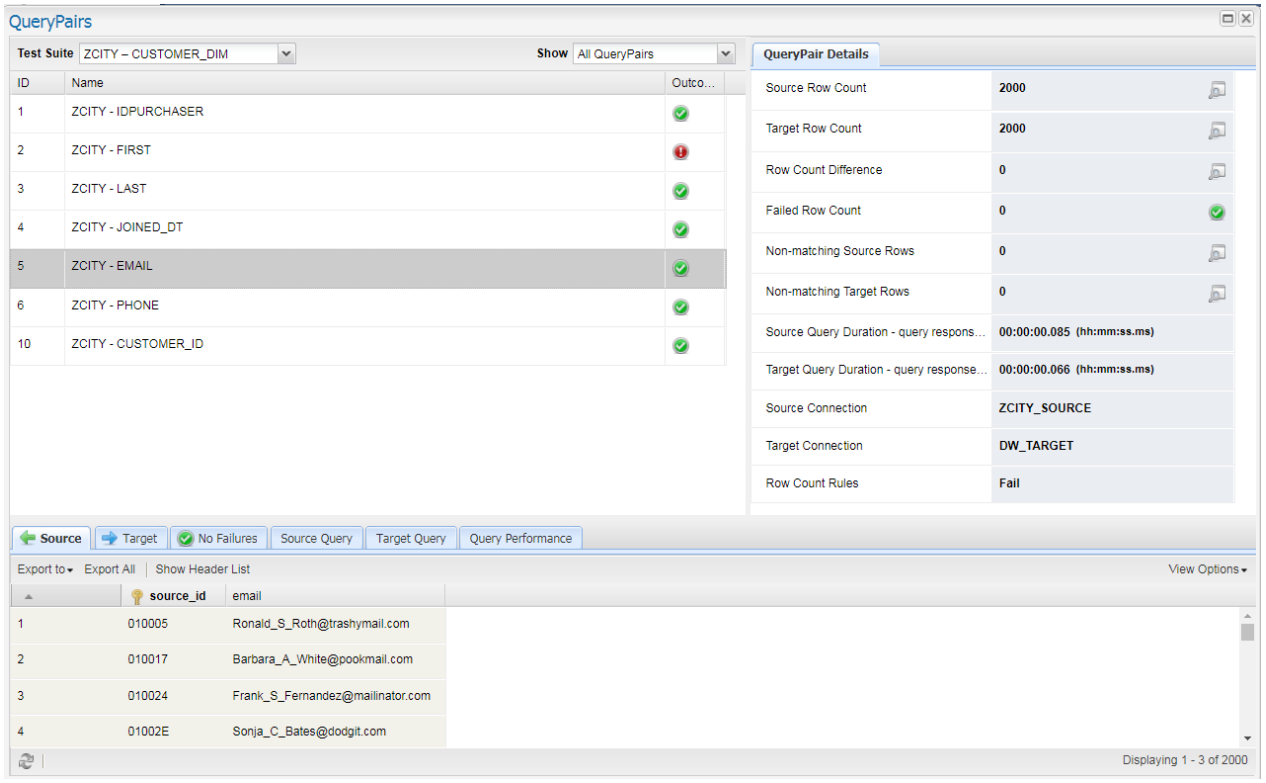
- Left Panel – 'Scenario Queue' contains all Scenarios, including those previously executed
  - The executed Scenario was created for you with a single test suite named 'ZCITY – CUSTOMER_DIM'
  - Notice QuerySurge appended a date/time stamp to the end of the Scenario name
- Upper Middle Panel – the currently selected Scenario's Test Suites and their progress
- Upper Right Panel – additional information and metrics
- Lower Middle/Right Panel – graphs of progress

12. Double click the Test Suite 'ZCITY – CUSTOMER_DIM' to see detailed data about the Test Suite execution results displayed in the middle panel.

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13. Click on the "ZCITY – EMAIL" *QueryPair*



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Points of interest:

- Notice the lower panel has a layout similar to the "Design-Time Run" execution results
- Lower panel contains
  - Source Tab – displays the source result rows
  - Target Tab – displays the target result rows
  - Failures Tab – displays data failures (not populated in this example as no rows failed)
  - Source Query Tab – displays the source query
  - Target Query Tab – displays the target query
  - Query Performance – displays metrics
- Right panel contains
  - Source Row Count – displays number of source rows
  - Target Row Count – displays number of target rows
  - Row Count Difference – displays number of extra rows in source or target
  - Failed row count – displays number of data failures
  - Non-Matching Source Rows – displays number of source rows that do not match target rows
  - Non-Matching Target Rows – displays number of target rows that do not match source rows
- You can export results to Excel, CSV file, or an XML document using the 'Export to' dropdown menu located below the source tab on the lower panel

14. Close the "QueryPairs" Test Suite results window.

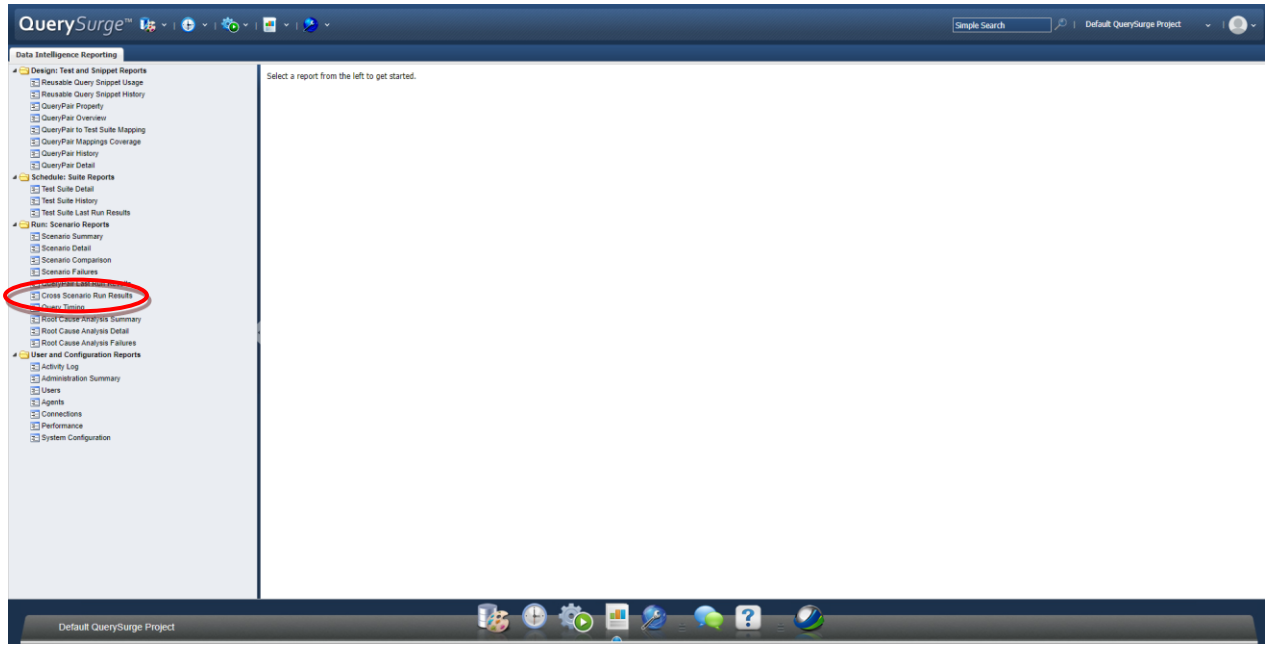
For more information on Execution, please visit *Scheduling → Running a Test Suite* in the Knowledge Base

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Running a Detailed Scenario Report

The “Reporting Module” shows the formal reports offered by QuerySurge. Some reports have configurable options which are found once the report type has been selected and opened. Once executed, reports can be saved as a PDF or exported to Excel.

1. Click the **“Reporting”** icon in the bottom panel. 

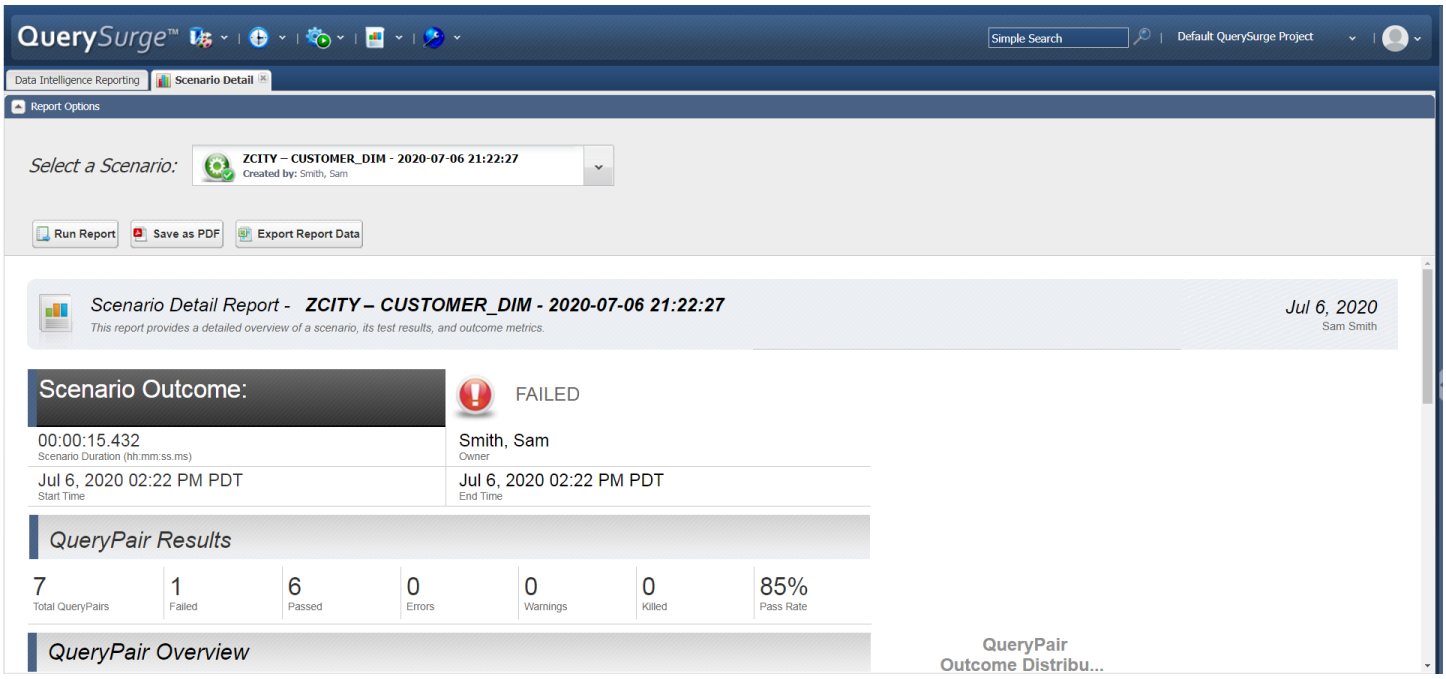


Points of interest:

- A list of reports is shown in the ‘Reporting Center’ (left panel)

2. Select the “Scenario Detail” report under “Run: Scenario Reports”.
3. Click **Open Report**.
4. Under **“Select a Scenario”** select ‘ZCITY – CUSTOMER_DIM’ from the dropdown list. (a date time stamp was appended to the end of the Suite name from the previous exercise)
5. Click the **Run Report** button.

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Points of interest:

- In the screenshot above the "Scenario Outcome" has an overall status of "FAILED" since one of the *QueryPair*s failed
- The "Test Results" displays the number of passed and failed *QueryPairs*
- Additional information is displayed in the "Overview" section

6. Close the **"Scenario Detail"** report tab.

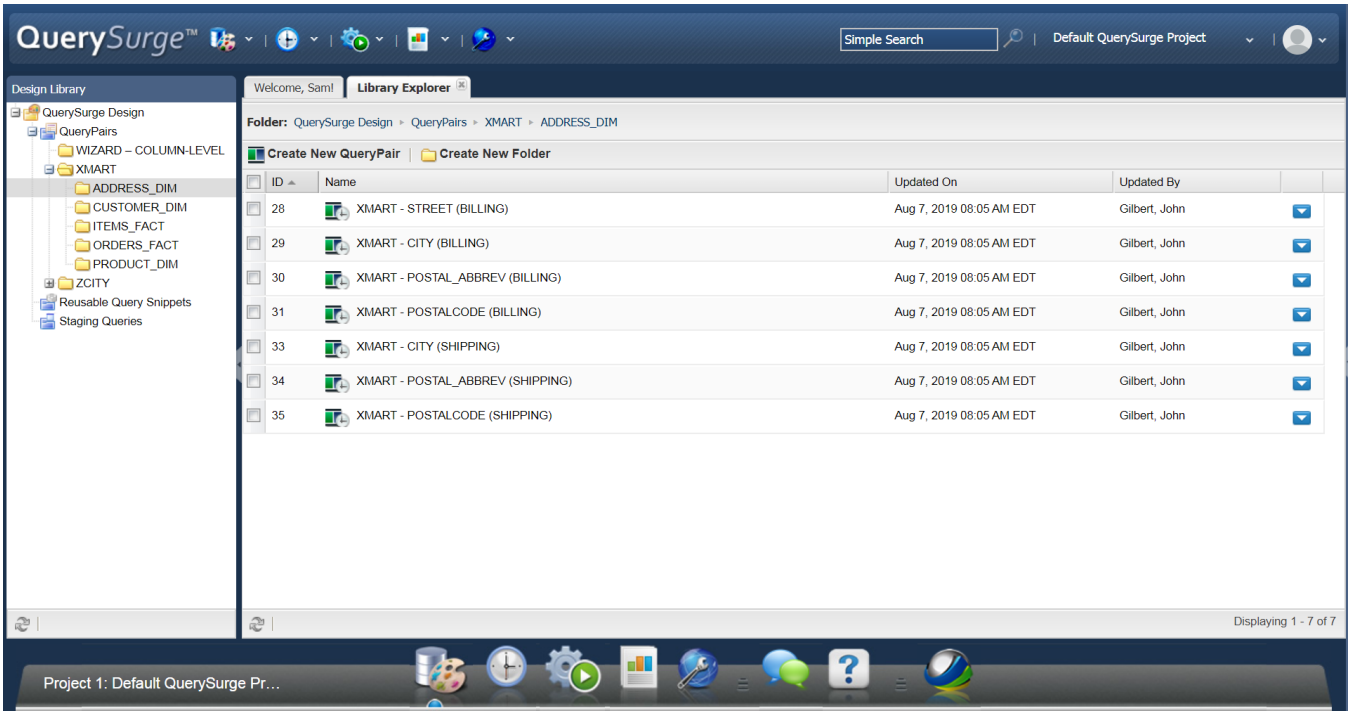
For more information on Reports, please visit *Reporting* in the Knowledge Base

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Create a *QueryPair*

Use QuerySurge to create a new *QueryPair*.

1. Click the **"Design Library"** icon located on the bottom panel 
2. In the left panel under **"Design Library"** select QueryPairs and select the "XMART" folder and then the "ADDRESS DIM" sub-folder.



3. Click the "**Create New QueryPair**" button.

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Points of interest:

- The "Create a New QueryPair" dialog contains:
  - QueryPair Name field – a name for your *QueryPair*
  - Description field – a description for your *QueryPair*
  - Mapping field – for reference back to a mapping document if applicable. (Data mapping is a process used in data projects by which different data models are linked to each other. Mappings are typically used for transformations in an ETL process or for consolidation of multiple databases and/or redundant columns.)
    - See the Tutorial Mappings document, which can be found from the Setup Wizard

4. Type "XMART – STREET (SHIPPING)" in the **"QueryPair Name"** field.
5. Type "Extract the Street portion of XMART_CUSTOMER.SHIPPING ADDRESS by pulling all text before the first comma" in the **"Description"** field. (This description was taken from the tutorial mapping document for the XMART mapping 5.05)
6. Enter "5.05" in the **"Mapping**" field.
7. Click the **Save** button.

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8. Type or copy/paste the following under the **"Source"** editor panel:

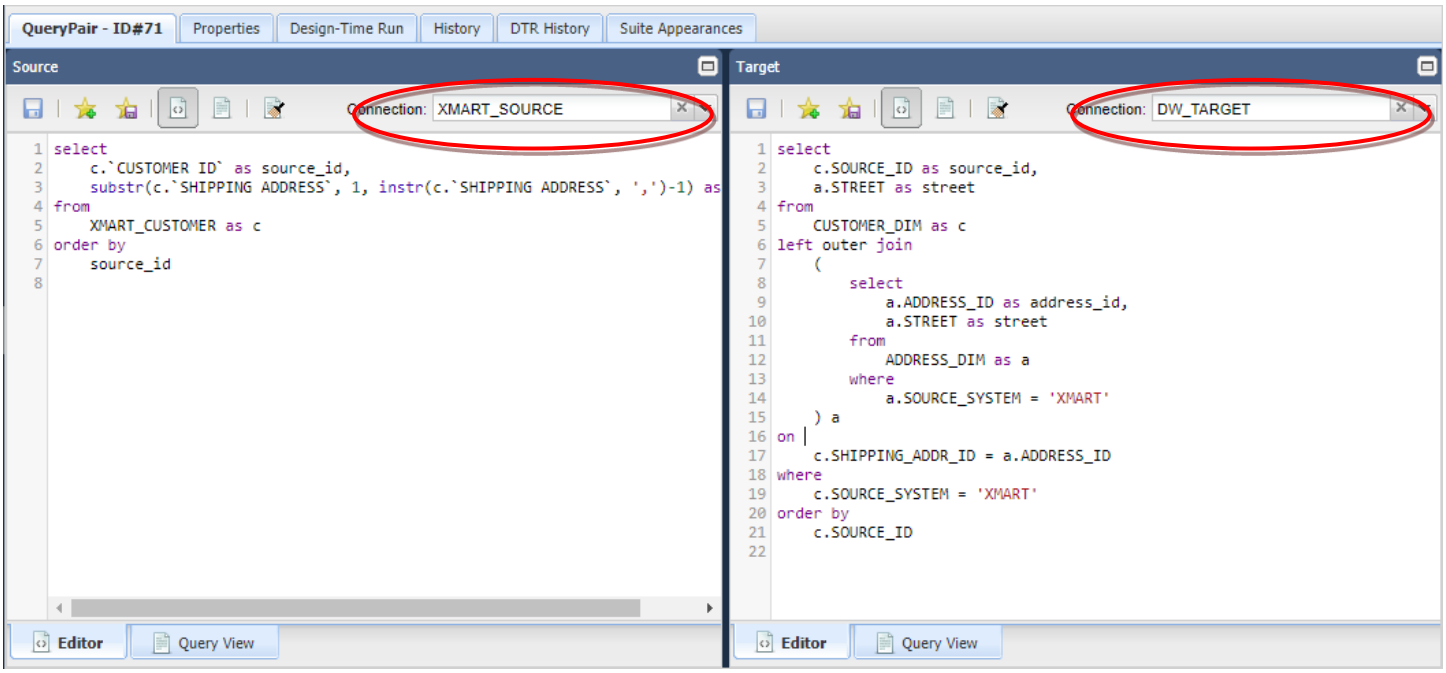
```
select
	c.`CUSTOMER ID` as source_id,
	substr(c.`SHIPPING ADDRESS`, 1, instr(c.`SHIPPING ADDRESS`, ',')-1) as street
from
	XMART_CUSTOMER as c
order by
	source_id
```

9. Type or copy/paste the following under the **"Target"** editor panel:

```
select
	c.SOURCE_ID as source_id,
	a.STREET as street
from
	CUSTOMER_DIM as c
left outer join
	(
		select
			a.ADDRESS_ID as address_id,
			a.STREET as street
		from
			ADDRESS_DIM as a
		where
			a.SOURCE_SYSTEM = 'XMART'
	) a
on
	c.SHIPPING_ADDR_ID = a.ADDRESS_ID
where
	c.SOURCE_SYSTEM = 'XMART'
order by
	c.SOURCE_ID
```

10. Under **"Connection"** select "XMART_SOURCE" from the dropdown menu in the **"Source"** panel.
11. Under **"Connection"** select "DW_TARGET" from the dropdown menu in the **"Target"** panel.
12. Click the **"Save"** icon in both the source and target editor panels. 

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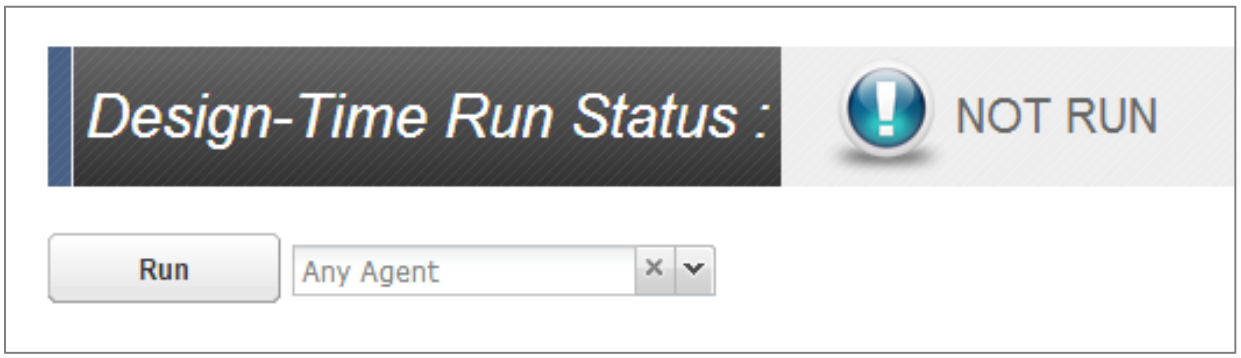


Points of interest:

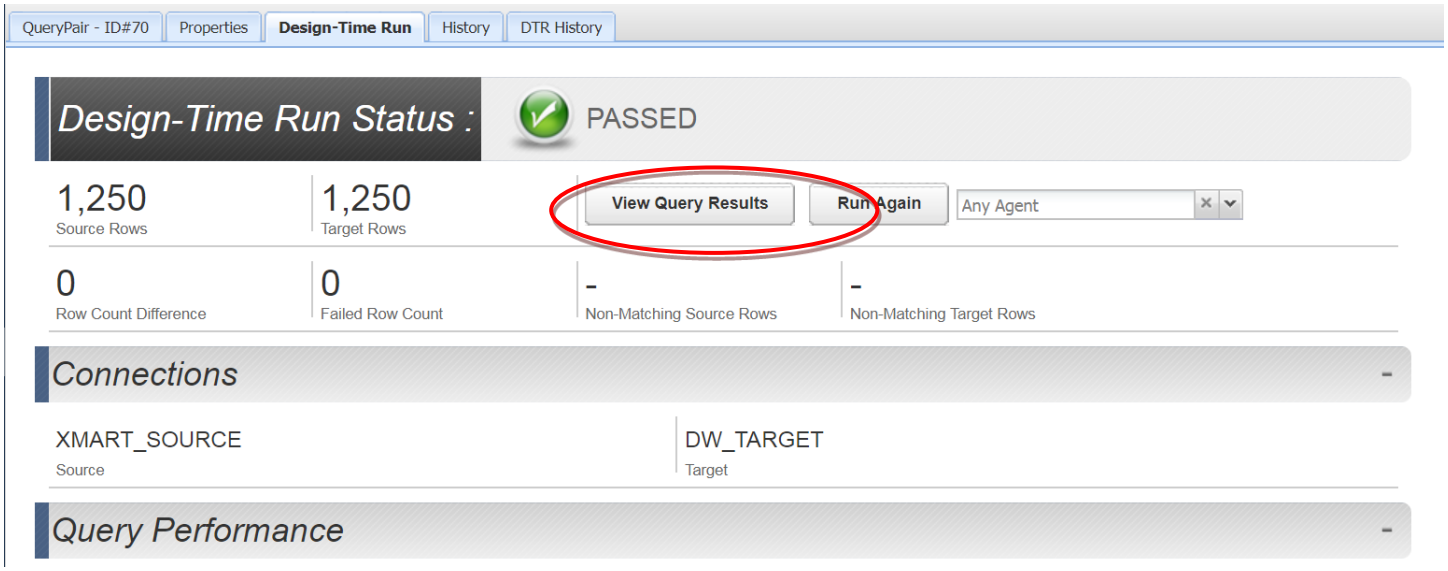
- The 'XMART – STREET (SHIPPING)' *QueryPair* is implemented to validate that the source SHIPPING_ADDRESS column was loaded into the data warehouse correctly (see Tutorial Mapping document from the 'Setup Wizard')
    - This *QueryPair* is validating that the SHIPPING_ADDRESS column from source XMART, to the target data warehouse is loaded with the following logic:
        - Extract the street name from "XMART_CUSTOMER.SHIPPING ADDRESS" by retrieving all the text before the first comma

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13. Click the "**Design Run Time"** tab.

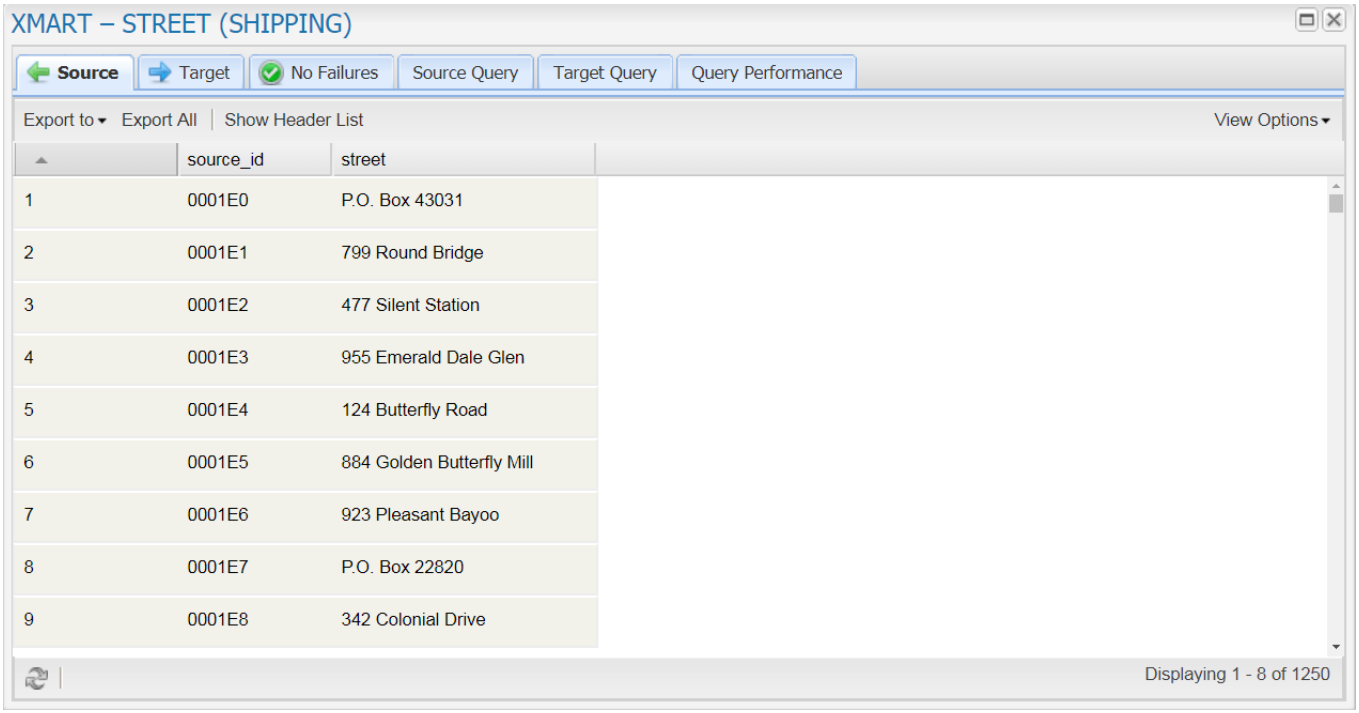


14. Click the "**Run"** button.



15. Click the "**View Query Results"** button.

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16. Close the “XMART – STREET (SHIPPING)” query results window

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Create a *QueryPair* Using a Reusable *Snippet*

A query *Snippet* is a reusable piece of SQL code that can be embedded in one or more queries. The purpose of a S*nippet* is to make common SQL fragments reusable across multiple Queries. A *Snippet* does not have to be a syntactically intact query (for example, it might just be a WHERE clause) but there is no reason why it can't be a complete query (e.g., a reusable sub-query). This feature lets you develop libraries of 'Reusable Query Snippets' to be reused in multiple queries. If a *Snippet* shared by multiple queries needs to change, you can make the change once and all of the queries using the *Snippet* are changed automatically.

1. Under "Design Library" in the lefthand panel select **"QueryPairs"** and then select the 'ZCITY' folder and then the 'ORDERS_FACT' sub-folder.
2. Click the "**Create New QueryPair"** button.
3. Type 'ZCITY – COMMENTS' in the **"QueryPair Name"** field.
4. Click the **Save** button.
5. Type or copy/paste the following under the **"Source"** editor panel:

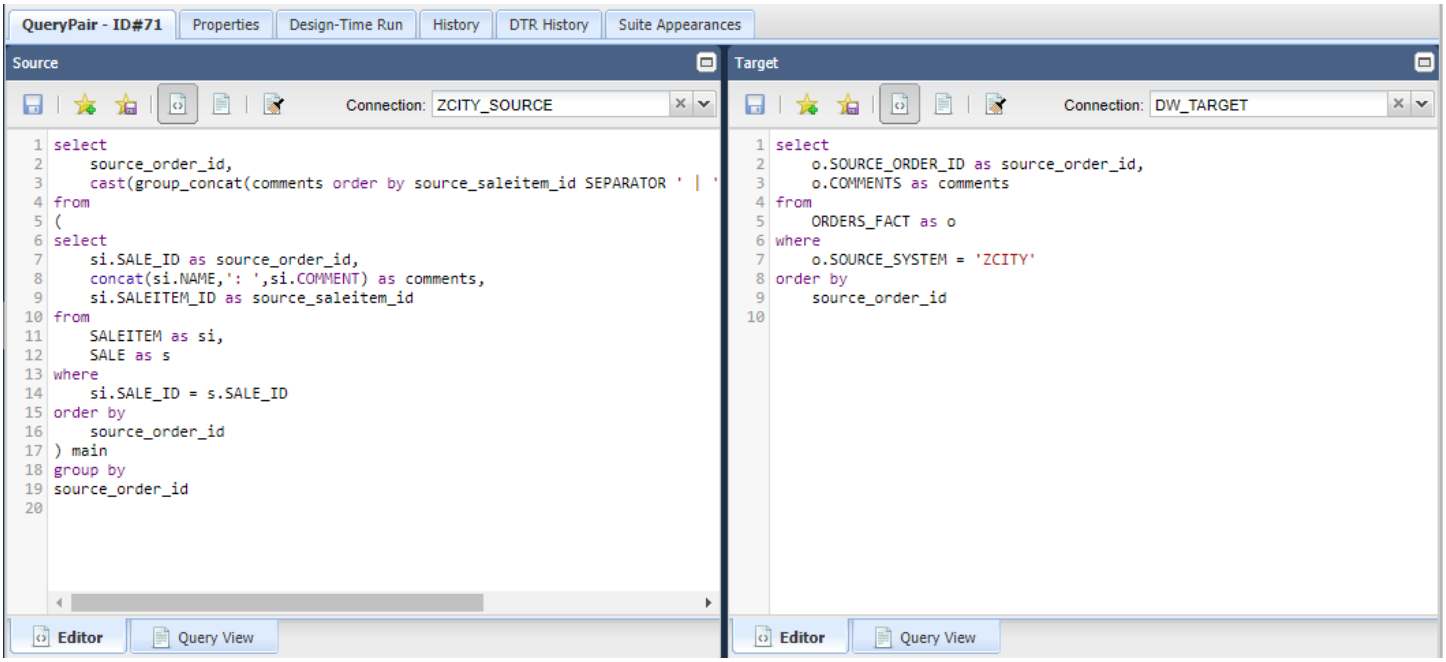
```
select
    source_order_id,
    cast(group_concat(comments order by source_saleitem_id SEPARATOR ' | ') as
char(100)) as comments
from
(
select
    si.SALE_ID as source_order_id,
    concat(si.NAME,': ',si.COMMENT) as comments,
       si.SALEITEM_ID as source_saleitem_id
from
    SALEITEM as si,
    SALE as s
where
       si.SALE_ID = s.SALE_ID
order by
       source_order_id
) main
group by
       source_order_id
```


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6. Type or copy/paste the following under the **"Target"** editor panel:

```
select
    o.SOURCE_ORDER_ID as source_order_id,
    o.COMMENTS as comments
from
       ORDERS_FACT as o
where
    o.SOURCE_SYSTEM = 'ZCITY'
order by
       source_order_id
```

7. In the **"Source"** editor panel select "ZCITY_SOURCE"' from the "'Connection" dropdown menu.
8. In the **"Target"** editor panel select "DW_TARGET" from the "Connection" dropdown menu.
9. Click the **"Save"** button on both the source and target queries.



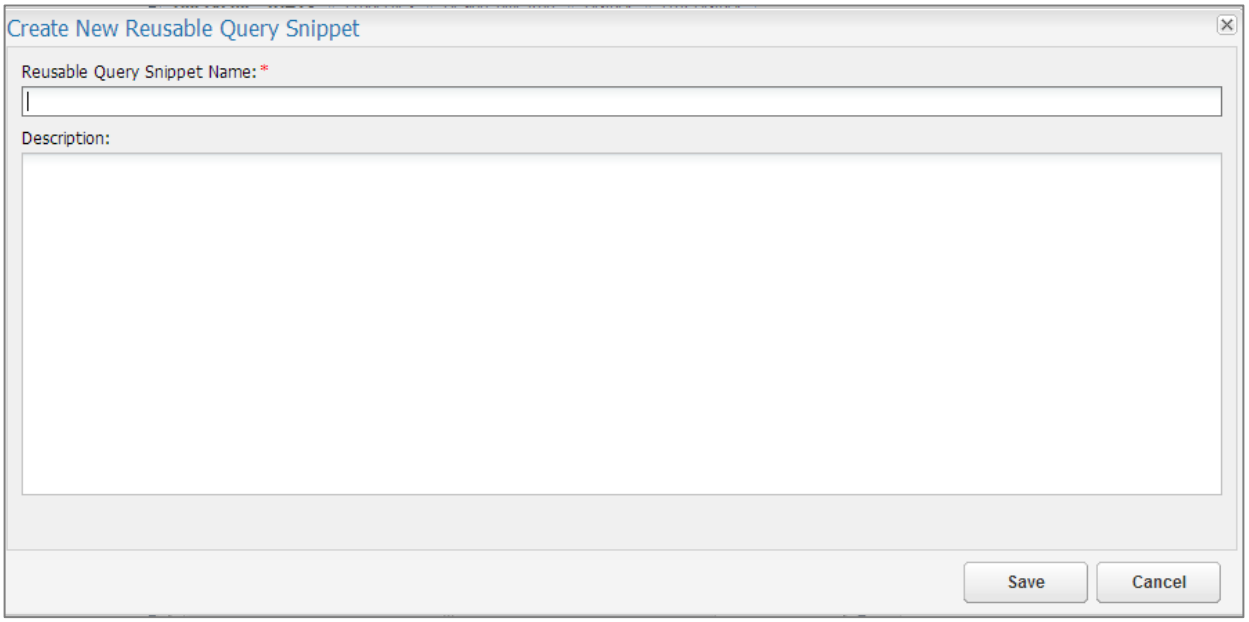
Points of interest:

- The "ZCITY – COMMENTS" *QueryPair* is to validate that the source comments column was loaded into the data warehouse (see tutorial mapping document from the 'Setup Wizard')
  - This *QueryPair* is validating that the COMMENT column from source ZCITY, to the target data warehouse, is loaded with the following logic:
    - Populate the COMMENTS column on the target table ORDERS_FACT by concatenating columns from source ZCITY with the following logic:

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- saleitem.NAME + ' : ' + saleitem.COMMENTS
- For multiple comments separate source comments with ' | '

10. In the **"Target"** editor panel, highlight the following text: 'o.SOURCE_SYSTEM = 'ZCITY'
11. Click the **"Save selected SQL as Reusable Query Snippet"** icon.

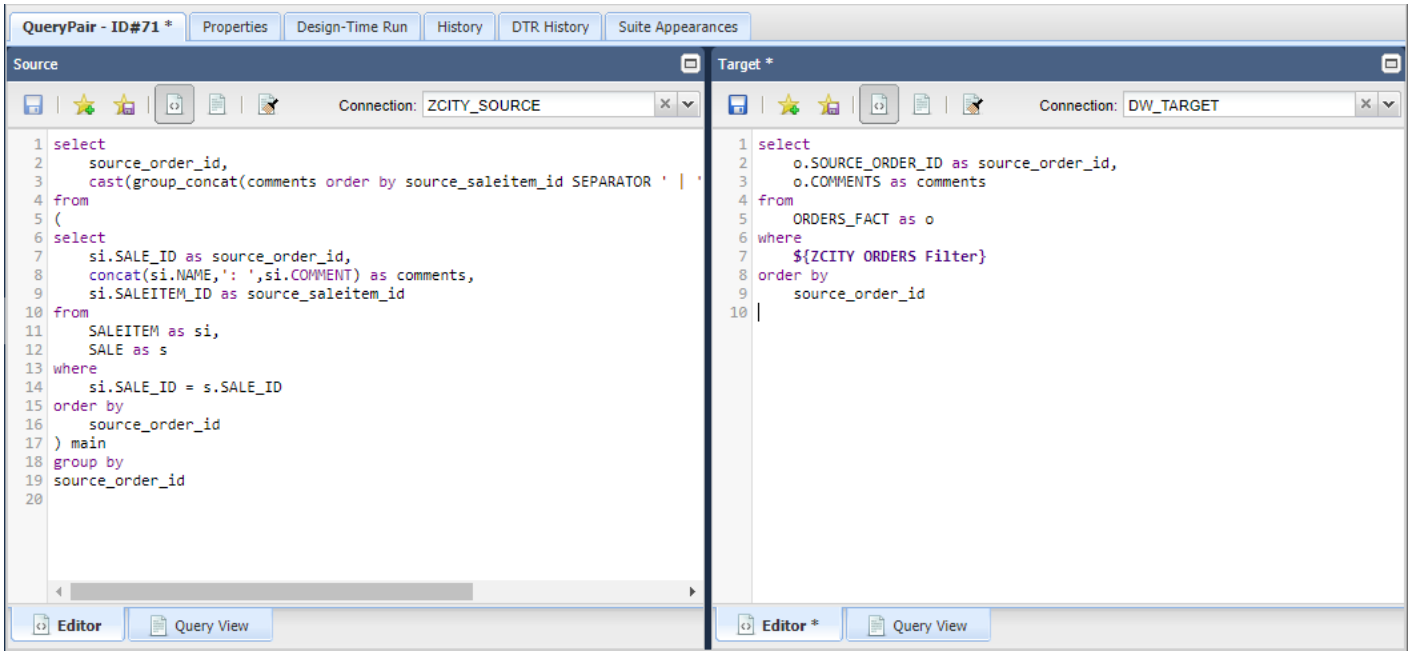


Points of interest:

- The 'Create New Reusable Query Snippet' dialog contains:
  - Reusable Query Snippet Name field – a name for your *Snippet*
  - Description field – a description for your *Snippet*

12. Type 'ZCITY ORDERS Filter' in the **"Reusable Query Snippet Name"** field.
13. Click the **Save** button.
14. Click the **Save** button under the **"Target"** editor panel.

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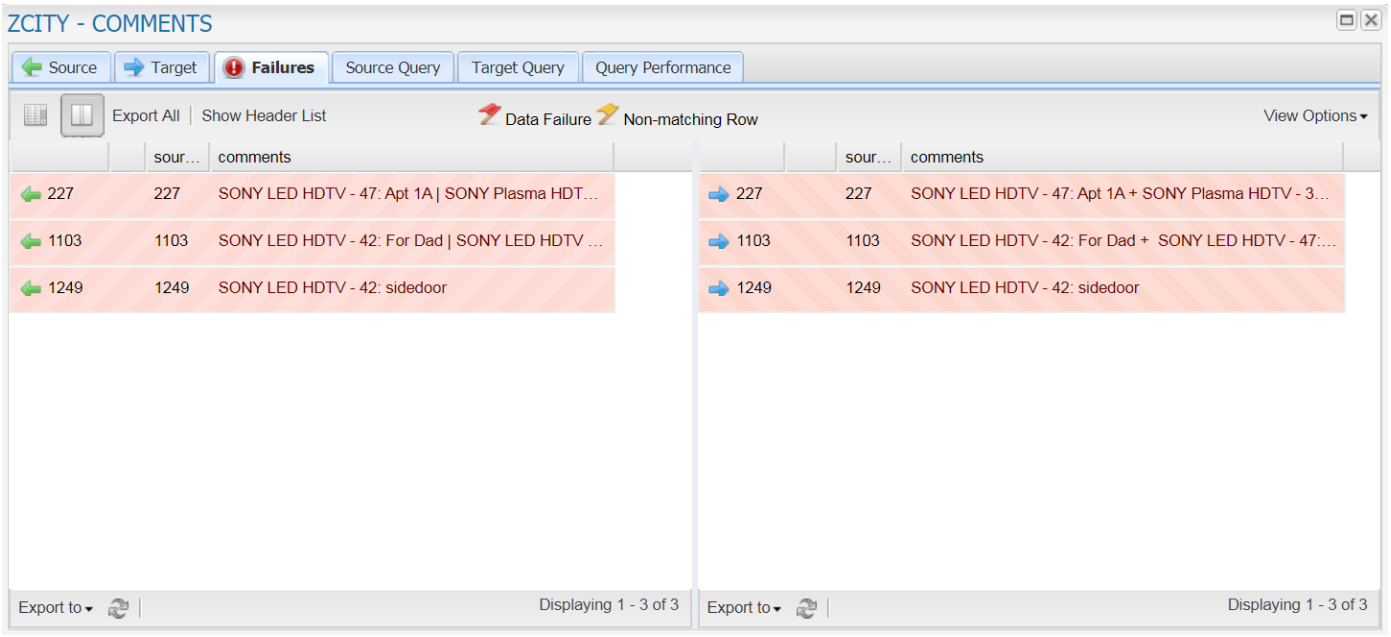


Points of interest:

- Notice that 'o.SOURCE_SYSTEM = 'ZCITY'' has been replaced with '**${ZCITY ORDERS Filter}**', signifying a reusable query snippet
- Hover your mouse over the *Snippet* to resolve, or see, the actual query
    - You can click the "Query View" tab at the bottom of the pane to resolve the *Snippet* as well

15. Execute a Design-Time Run by selecting the **"Design-Time Run"** tab then click the **Run** button.
16. Click the **View Query Results** button.
17. Click the **"ailures"** tab. (expand the comments field to view the entire column)

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Points of interest:

- Notice that on the target side, the development team is not using the correct delimiter in their ETL process
    - The target data warehouse is using '+' as its delimiter between comments, but according to the mapping document, '|' was specified
- Notice line 1249 on the source/target. The data looks the same; however, the target has an extra space after "sidedoor". Use View Options -> Show Whitespace to highlight this extra space. ("**View options"** is located in the upper right of the results dialog window)

18. Close the "ZCITY – COMMENTS" query results window.

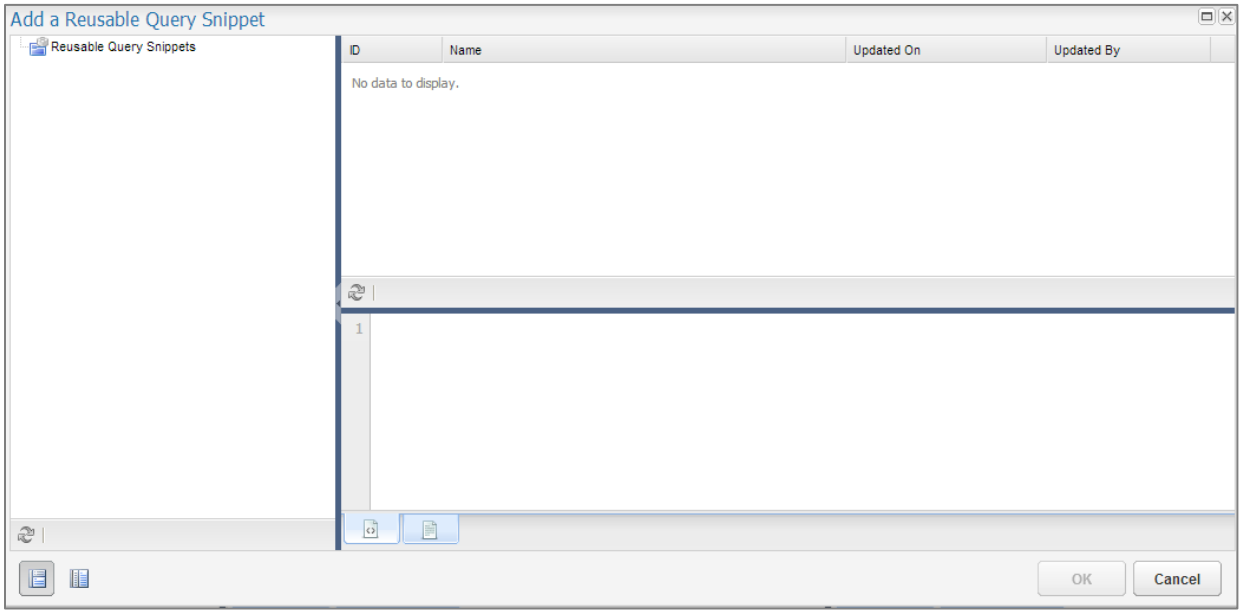
For more information on Reusable *Snippets*, please visit *Design Library* → *Working with Reusable Snippets* in the Knowledge Base

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Add a Reusable Query *Snippet* to a *QueryPair*

Add a *Snippet* to an existing *QueryPair*

1. Under **"Design Library",** select the 'ZCITY' folder and then select the 'ORDERS_FACT' sub-folder.
2. Double click the 'ZCITY - ORDER_DT' *QueryPair*.
3. Under the **"Target"** editor panel highlight the following text: 'o.SOURCE_SYSTEM = 'ZCITY''.
4. Click the "Add Reusable Query Snippet" icon. 

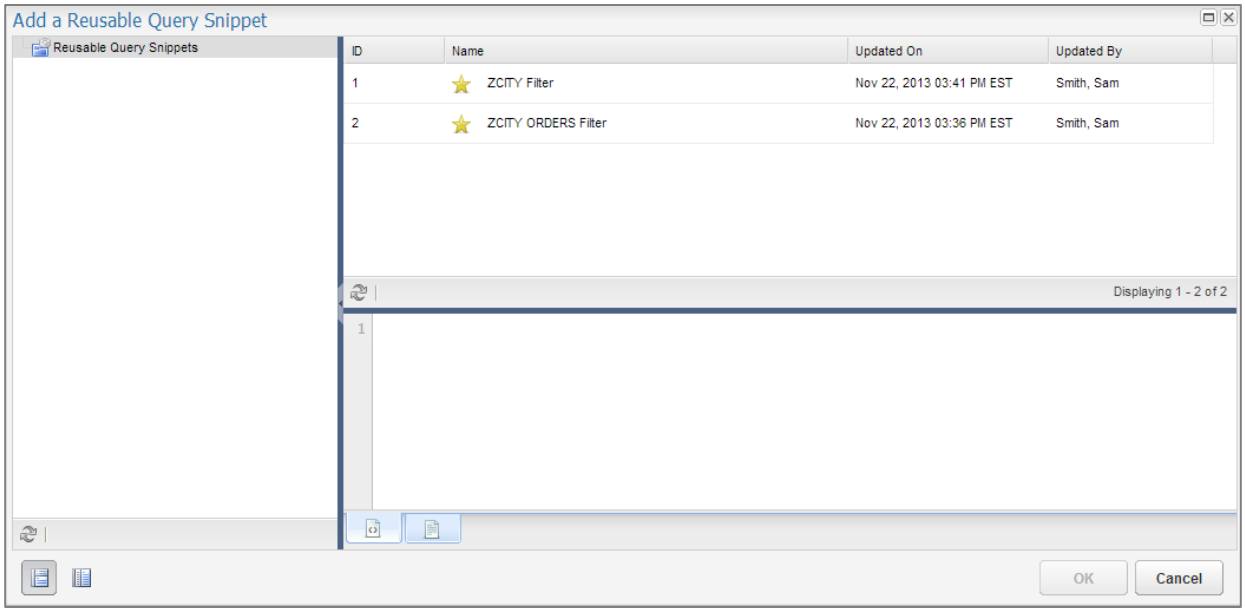


Points of interest:

- The left panel utilizes a folder structure similar to Microsoft Windows Explorer (our *Snippets* are saved in the root folder, but can be grouped by subfolders manually if desired)
- The top right panel will display the 'Reusable Query *Snippets*' contained within the selected folder once a folder is selected
- The bottom right panel displays the SQL of the selected *Snippet*

5. Click the **"Reusable Query Snippets"** folder in the left panel.

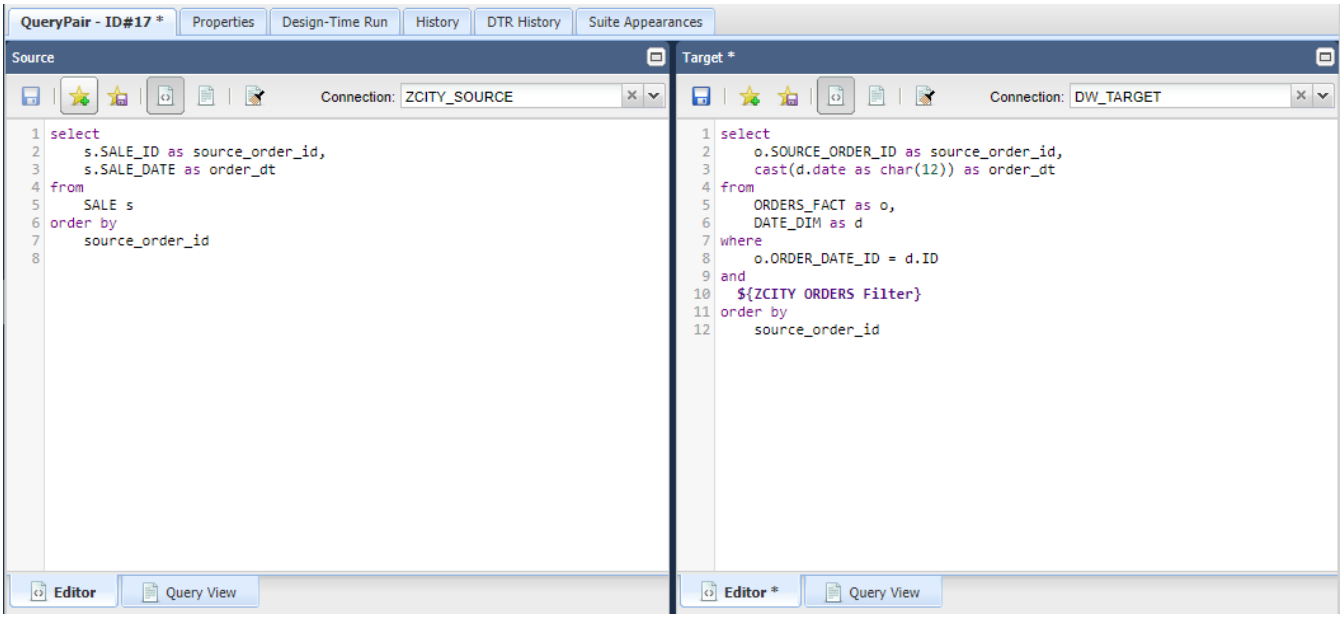
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Points of interest:

- The top right pane displays the 'Reusable Query Snippets', including the one you created earlier

6. Select the "ZCITY ORDERS Filter" snippet and click the **OK** button.



Points of interest:

- The "ZCITY ORDERS – Filter" snippet replaces the highlighted text with the Reusable Query *Snippet*
- Now the *Snippet* is in two locations. If you need to change that part of the query, it can be done in one place instead of two. (What if the same SQL fragment was in hundreds of *QueryPairs*? That would save quite a bit of time.)

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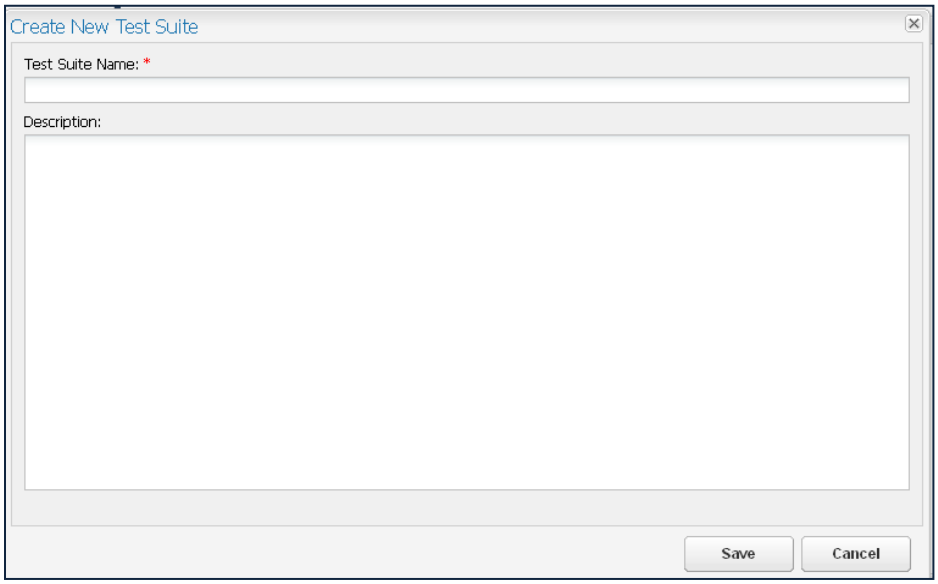
7. Click the **Save** icon on the target query. 

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Create a *Test Suite* and Schedule an Execution *Scenario*

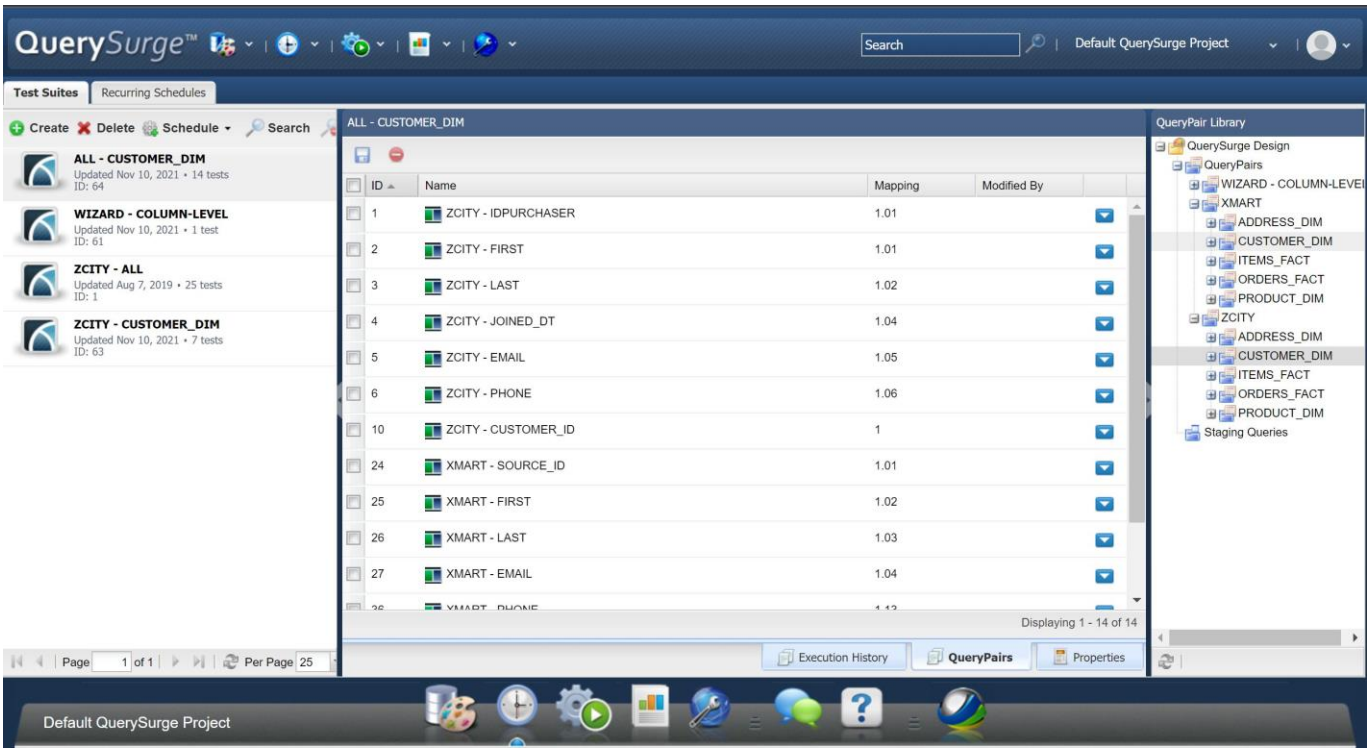
QuerySurge Scenarios are collections of *Test Suites* that are scheduled for execution. *Scenarios* can be run immediately after creation, or scheduled for a future date/time, and are viewable in the 'Run Dashboard'.

1. Click the **"Scheduling"** icon located on the bottom panel. 
2. Click the **Create** button to create a new test Suite.



3. Type 'ALL – CUSTOMER_DIM' in the **"Test Suite Name"** field.
4. Click the **Save** button.
5. Expand the 'XMART' folder in the "QueryPair Library" panel on the right.
6. Drag and drop the 'CUSTOMER_DIM' folder to the 'ALL – CUSTOMER_DIM' test suite panel in the middle.
7. Expand the 'ZCITY' folder in the 'QueryPair Library' panel on the right.
8. Drag and drop the 'CUSTOMER_DIM' folder to the 'ALL – CUSTOMER_DIM' test suite panel in the middle.

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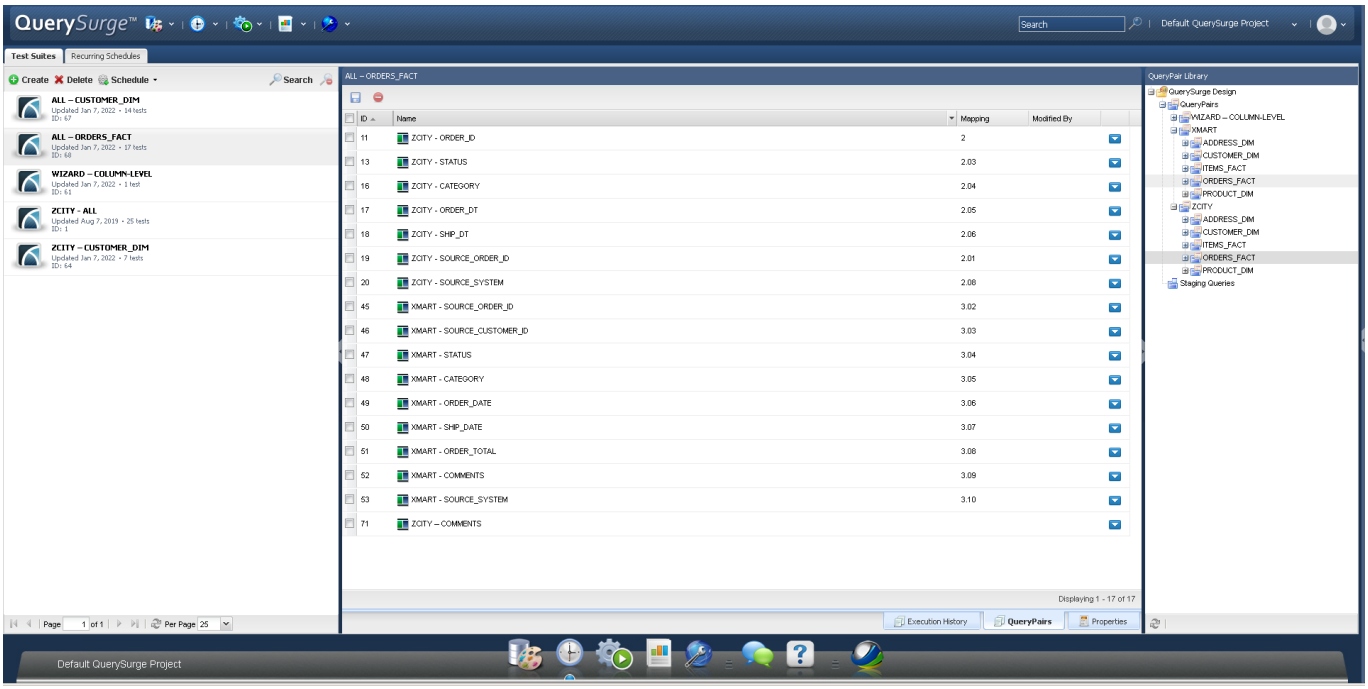


Points of interest:

- The 'ALL – CUSTOMER_DIM' *Test Suite* now contains all *QueryPairs* from the 'CUSTOMER_DIM' folders from both 'XMART' and 'ZCITY' (14 *QueryPairs* in total)
- *Test Suites* can contain a combination of *QueryPairs* from various sources (XMART and ZCITY) and varying connection types (flat file and database, respectively)

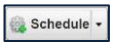
9. Click the **Save** icon in the upper left corner of the middle panel to save your *Test Suite.*
10. Click the **Create** button to create a new test Suite.
11. Enter 'ALL – ORDERS_FACT' in the **"Test Suite Name"** field.
12. Click the **Save** button.
13. Expand the 'XMART' folder in the "QueryPair Library" panel on the right.
14. Drag and drop the 'ORDERS_FACT' folder to the 'ALL – ORDERS_FACT' test suite panel in the middle.
15. Expand the 'ZCITY' folder in the "QueryPair Library" panel on the right.
16. Drag and drop the 'ORDERS_FACT' folder to the "ALL – ORDERS_FACT" test suite panel in the middle.

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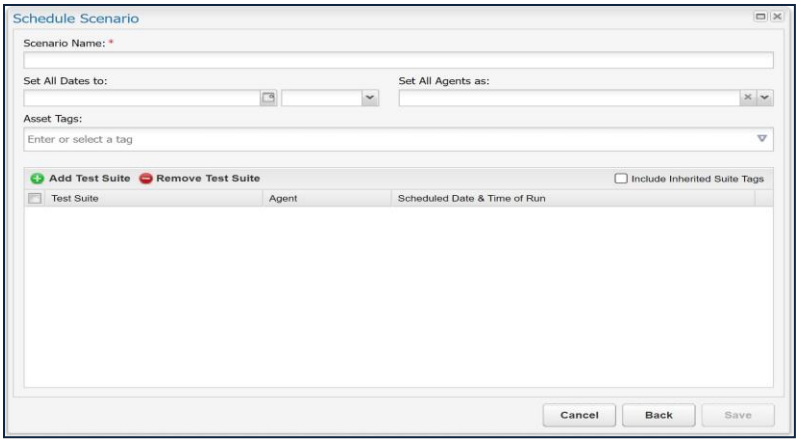
Points of interest:

- The 'ALL – ORDERS_FACT' *Test Suite* now contains all *QueryPairs* from the 'ORDERS_FACT' folders from both 'XMART' and 'ZCITY' (17 *QueryPairs*)

17. Click **Save** in the upper left corner of the middle panel to save your *Test Suite.* 
18. Click the **Schedule** button on the right panel. 

19. Choose the **One-time** radio button and click **Next**.

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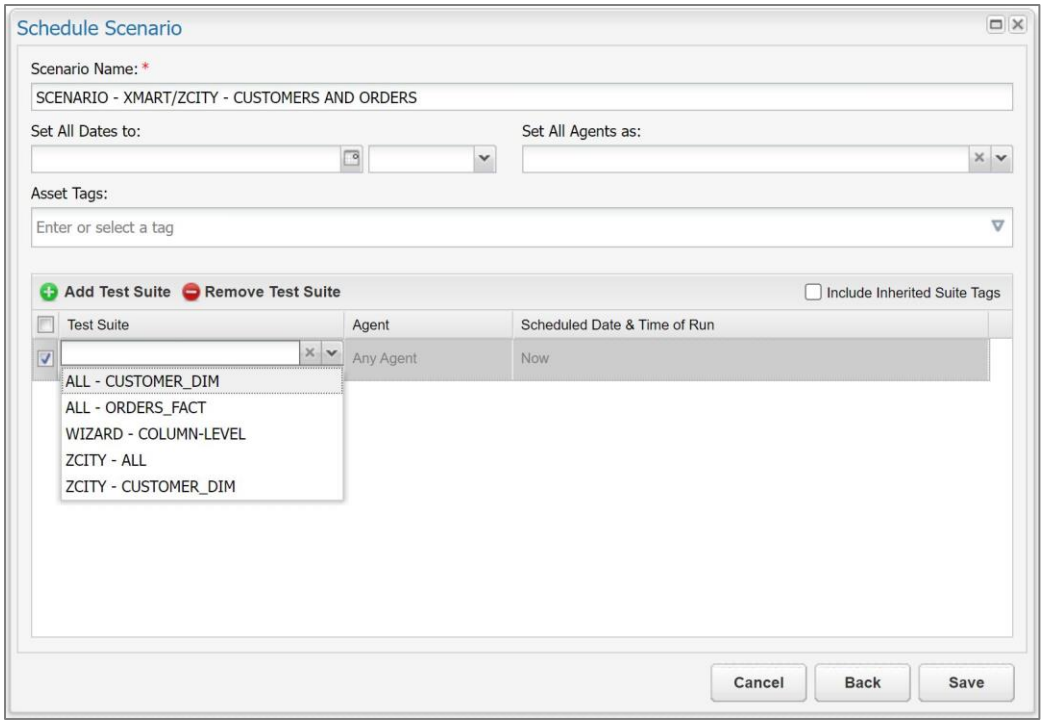
Points of interest:

- The 'Schedule Scenario' window contains:
  - Scenario Name: field – the name of the *Scenario*
  - Set All Dates to: field – sets all *Test Suites* to run on a particular date and time
  - Set All Agents as field – sets all *Test Suites* to run on a particular *Agent*
  - Asset Tags: field – optional tag descriptors for this *Scenario*
  - A *Test Suite* schedule area with columns containing:
    - *Test Suite* – *Test Suites* for this *Scenario*
    - *Agent* – allows you to select a particular *Agent*
    - Scheduled Date & Time of Run – for selecting a specific date and time for a run
      - Useful for executing tests, e.g., during non-working hours

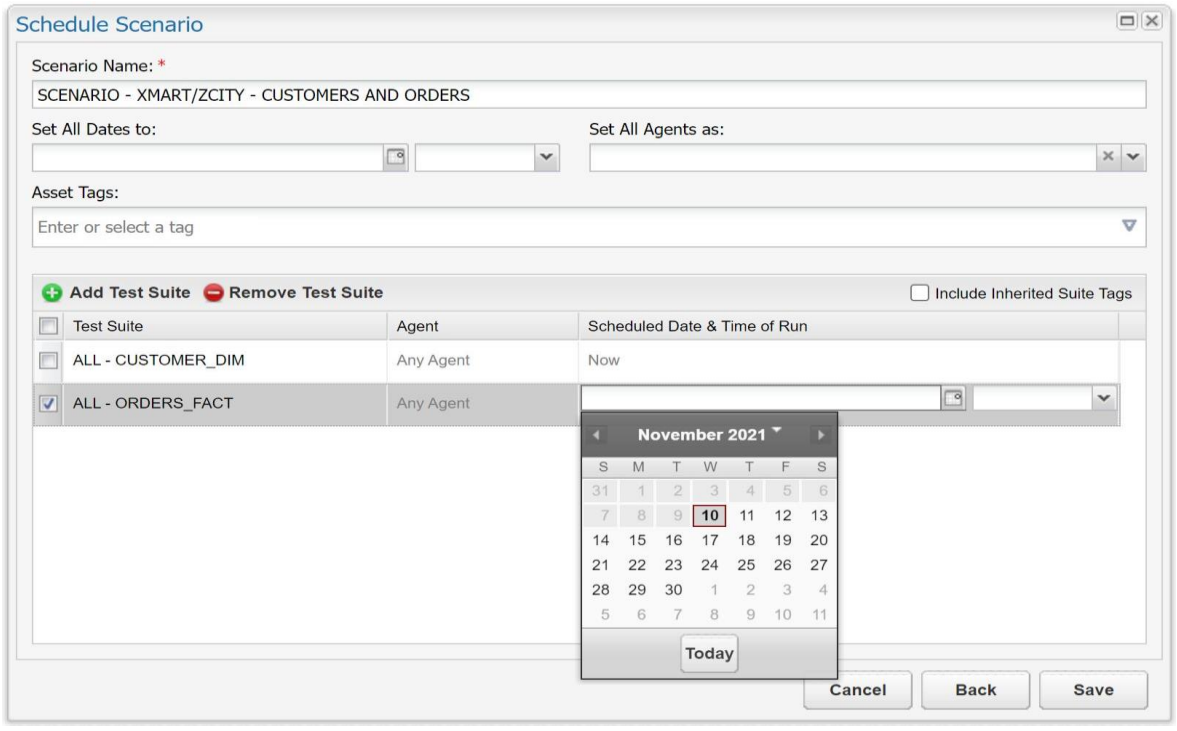
20. Enter "SCENARIO – XMART/ZCITY – CUSTOMERS AND ORDERS" into the "Scenario Name" field
21. Click the **Add Test Suite** button. 

22. In the "Test Suite" column, click the drop down list and select the 'ALL – CUSTOMER_DIM' *Test Suite*

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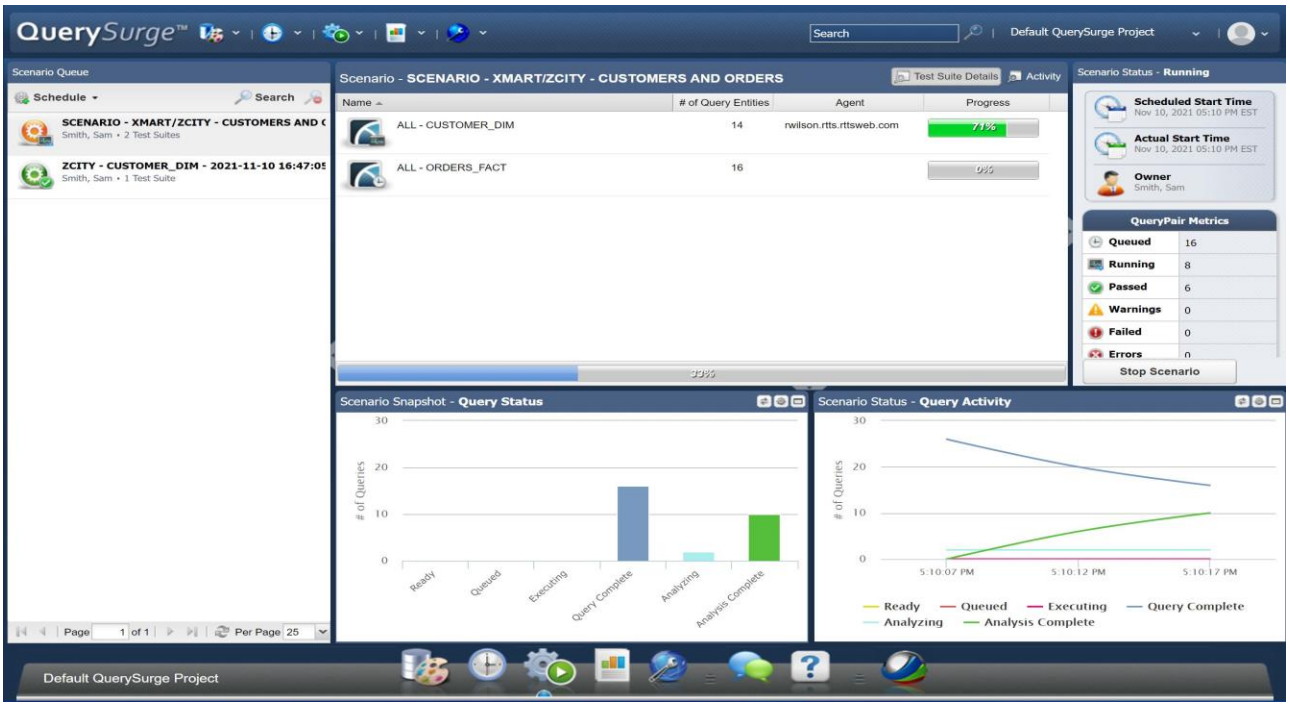


23. Click the "**Add Test Suite"** button and add a second *Test Suite* by selecting the 'ALL – ORDERS_FACT' *Test Suite*
24. Click in the **"Scheduled Date & Time of Run"** field for the 'ALL – ORDERS_FACT' *Test Suite* to enable the selection of a date and time for the run. Then click the "Calendar" icon 



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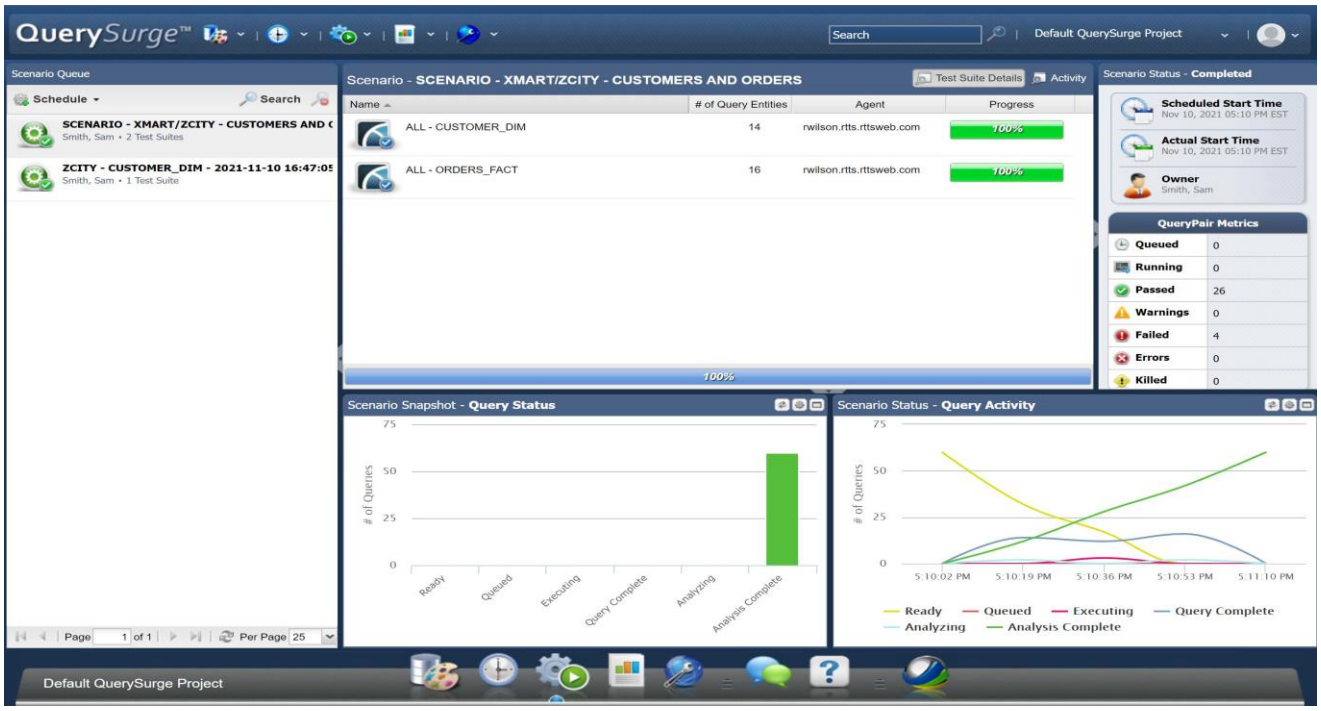
25. Click the **Today** button for the date portion, and then set the time for the nearest future time from now (your current time), and click the **Save** button
26. Click the **"Run Dashboard"** icon located on the bottom pane. (Your system may have toggled you over to the run dashboard automatically) 
27. Click the **Refresh** at the bottom of the 'Scenario Queue' pane (left pane) 
28. Click the "SCENARIO – XMART/ZCITY – CUSTOMERS AND ORDERS" *Scenario* in the "Scenario Queue" pane (this may take a few seconds to refresh and appear). Observe that the *Scenario* just created is now executing



Points of interest:

- The 'ALL – CUSTOMER_DIM' *Test Suite* executes immediately while the 'ALL – ORDERS_FACT' *Test Suite* remains idle since it was set to execute to the nearest time in the future
- Even when the "ALL – CUSTOMER_DIM" *Test Suite* completes its execution, the "ALL – ORDERS_FACT" *Test Suite* will remain idle until its set execution time is reached

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Points of interest:

- After the time has elapsed, the 'ALL – ORDERS_FACT' *Test Suite* will execute automatically

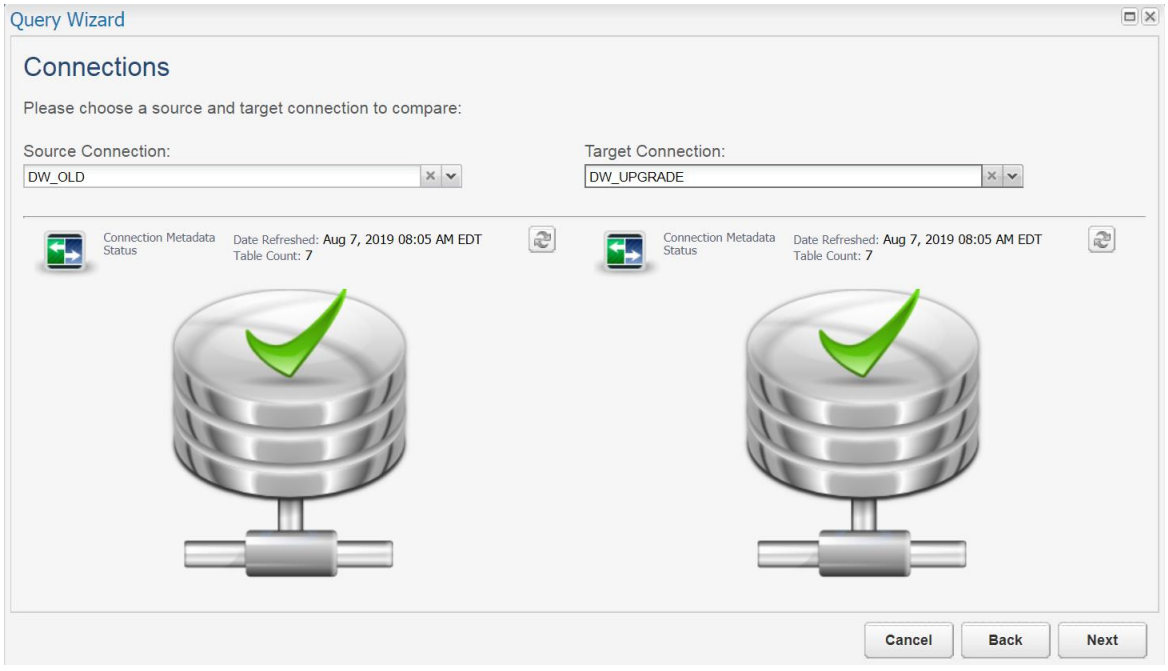
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Create *QueryPairs* Using the *Query Wizard* for *Table-Level Comparison*

The *Query Wizard* is a tool that allows you to generate *QueryPairs* automatically. This is great for those with limited or no SQL experience, as well as for experts who are looking for a fast and easy way to create *QueryPairs* for data migrations, database upgrades, and other database comparisons that do not have transformations. The *Query Wizard* can generate *QueryPairs* for *Row Count Comparison, Table-Level Comparison, and Column-Level Comparison* validations. A folder can be created in the Wizard that contains all of the *QueryPairs* generated. These *QueryPairs* can also be added to a new *Test Suite* so they can be easily run together.

Note: This part of the tutorial illustrates the use of the *Query Wizard* for validating a database upgrade. In this case, the data warehouse was upgraded to a new version 'Connection DW_UPGRADE' from an older version 'Connection DW_OLD'

1. In the "**Design Menu"** dropdown menu on the top toolbar, select **"Launch Query Wizard".** 
2. Click **Next** under the *Query Wizard* welcome screen.
3. Select 'DW_OLD' from the **"Source Connection"** drop down menu.
4. Select ''DW_UPGRADE' from the **"Target Connection"** drop down menu.

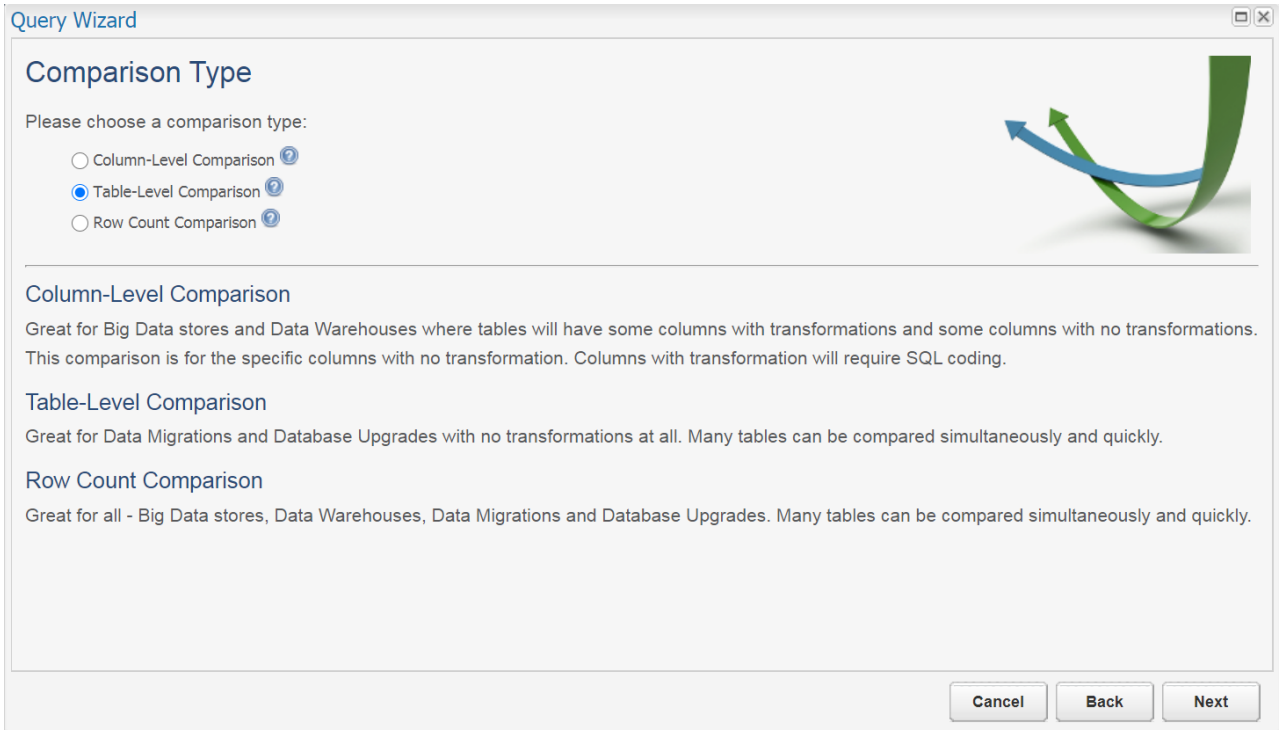


Points of interest:

- Metadata (or data about our data) pre-populates for the selected databases

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5. Click the **Next** button
6. Select the **"Table-Level Comparison"** radio button in the **"Comparison Type"** section of the *Query Wizard* window

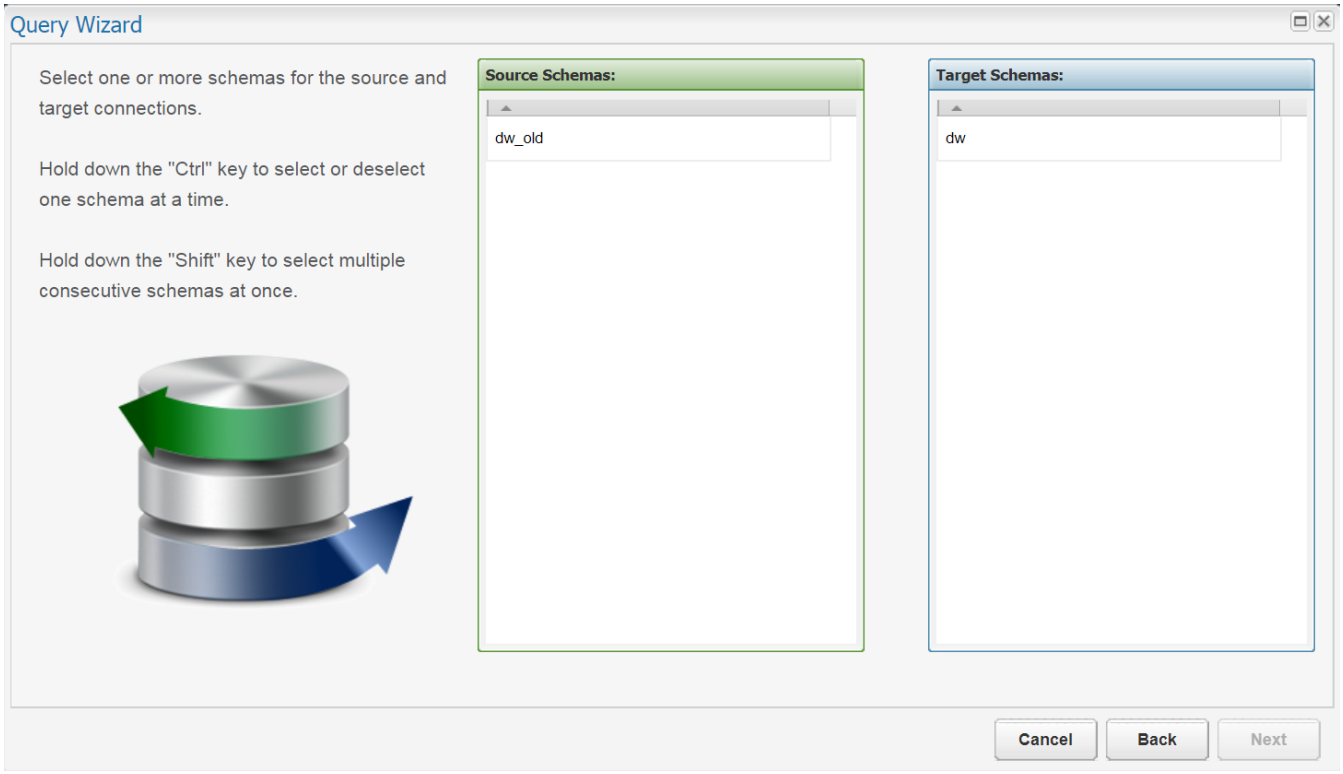


Points of interest:

- *Column-Level Comparison* validates data points in the selected columns and allows for the addition of Filtering (WHERE Clause) and Sorting (ORDER BY)
- *Table-Level Comparison* validates each data point in the source table to its corresponding data point in the target table
- *Row Count Comparison* validates the number of rows in each of the selected tables

7. Click the **Next** button.

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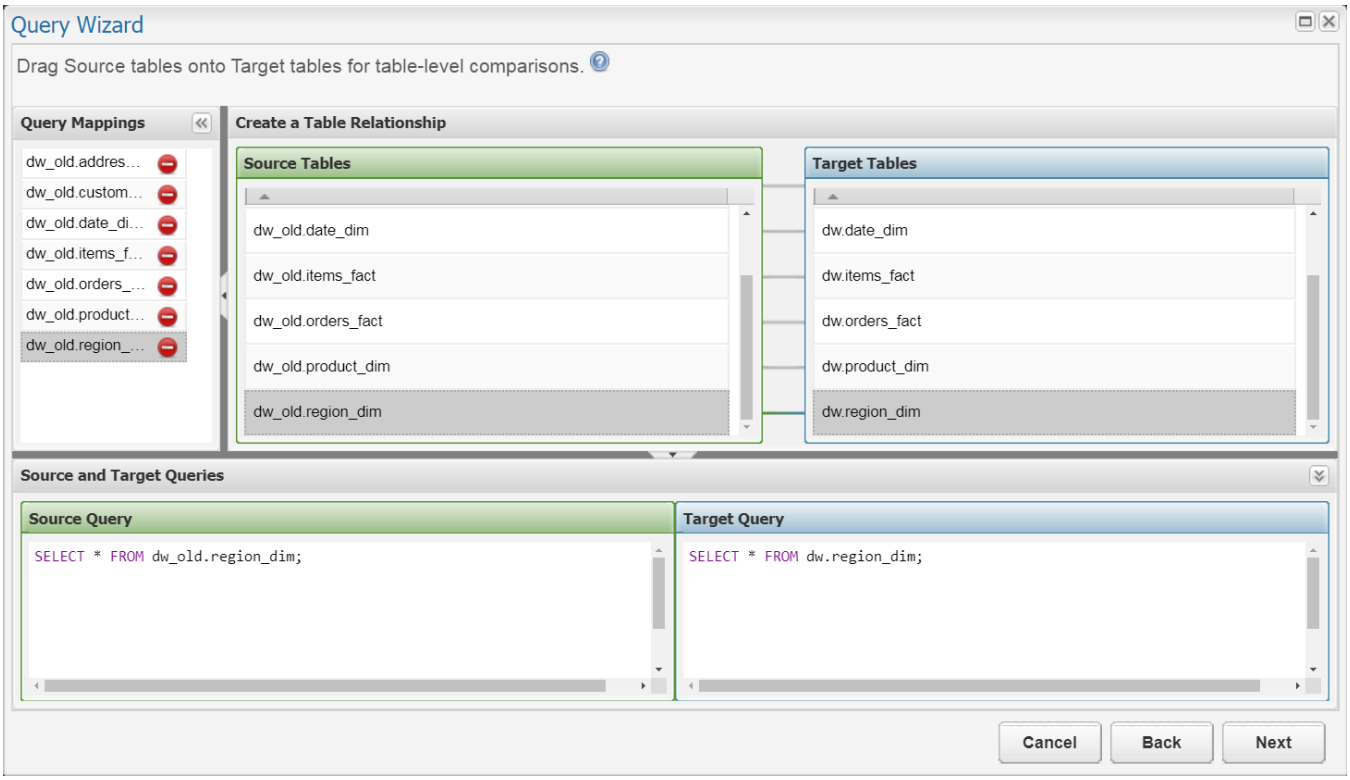


Points of interest:

- Schemas available for selection are retrieved from the source and target databases that you previously selected
- One, many, or all schemas can be selected

8. Select 'dw_old' (the location for the data warehouse data before the upgrade) for the source schema and 'dw' (the location for the data warehouse data after the upgrade) for the target schema.
9. Click the **Next** button.
10. Drag each individual source table and drop it onto its corresponding target table. Scroll down in the 'Source Tables' and 'Target Tables' windows if needed.

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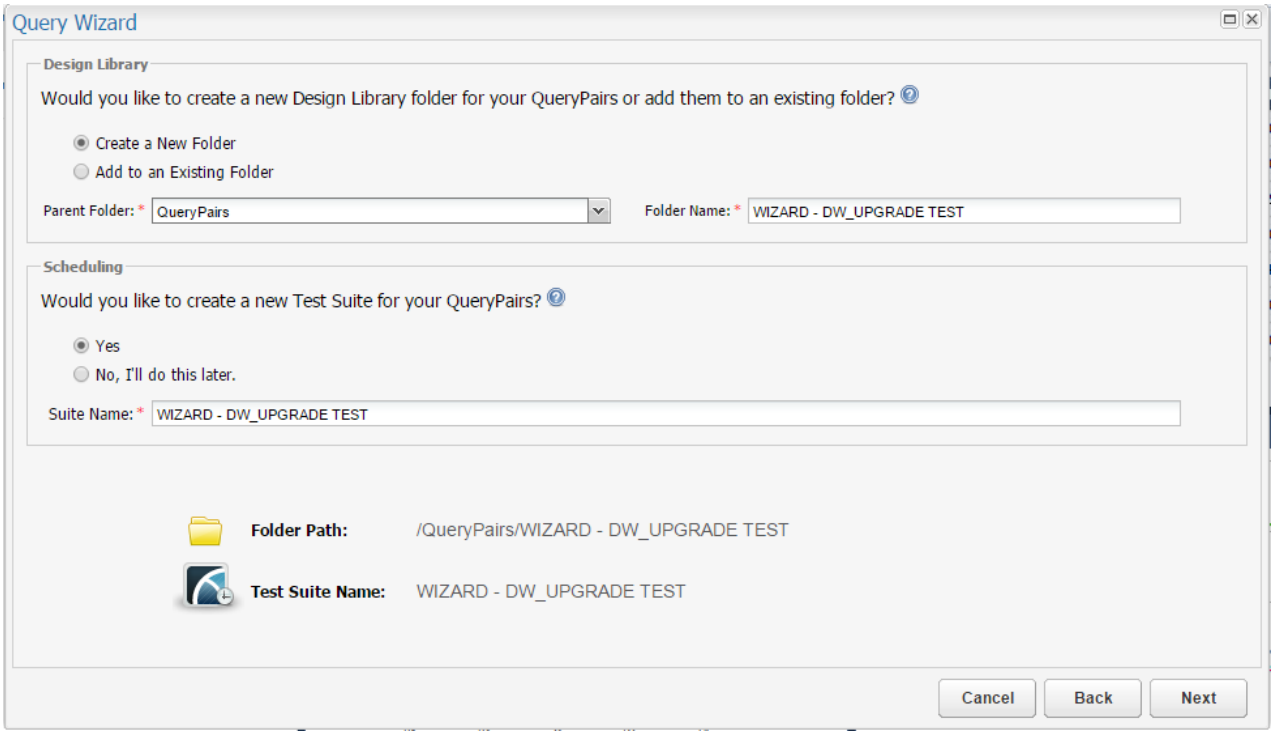


Points of interest:

- As each individual source table is dropped onto its corresponding target table, its *Query Mapping* appears in the left-hand column and can be renamed if desired
- Mapped tables show a line connecting the source and target tables with the selected *Query Mapping* line highlighted in green
- Auto generated SQL appears in the 'Source Query' and 'Target Query' panes at the bottom of the *Query Wizard* window

11. Click the **Next** button.
12. Select the **"Create a New Folder"** radio button in the "Design Library" section of the *Query Wizard* window.
13. In the **"Parent Folder"** drop down menu, select the root "**QueryPairs**" folder.
14. Enter 'WIZARD – DW_UPGRADE TEST' in the **"Folder Name"** field.
15. Select the **"Yes"** radio button in the "Scheduling" section of the *Query Wizard* window.
16. Enter "WIZARD – DW_UPGRADE TEST" in the "Suite Name" field.

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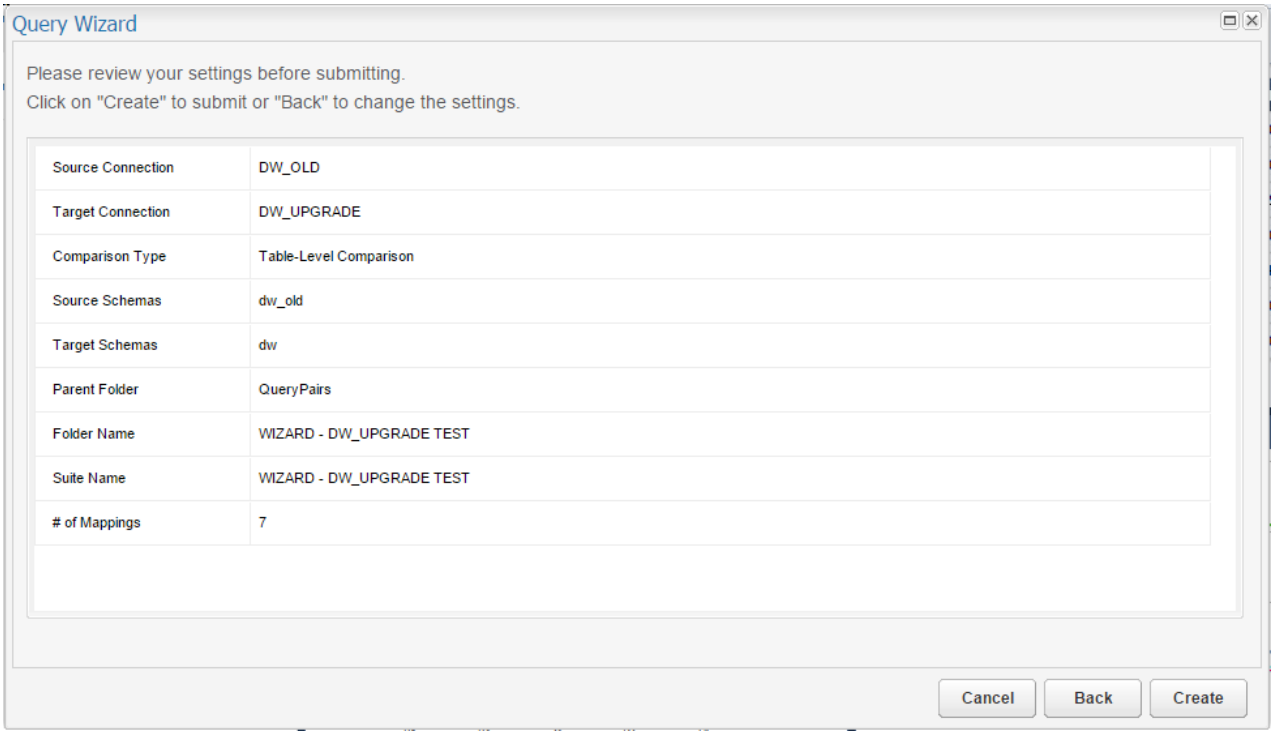


Points of interest:

- The *Query Wizard* will automatically create the test assets:
  - A new folder in the 'Design Library' with the name entered in the 'Folder Name' field
  - All *QueryPair(s)* for testing the tables selected
  - A new *Test Suite* in the 'Scheduling' module with all of the generated *QueryPairs*

17. Click the **Next** button.

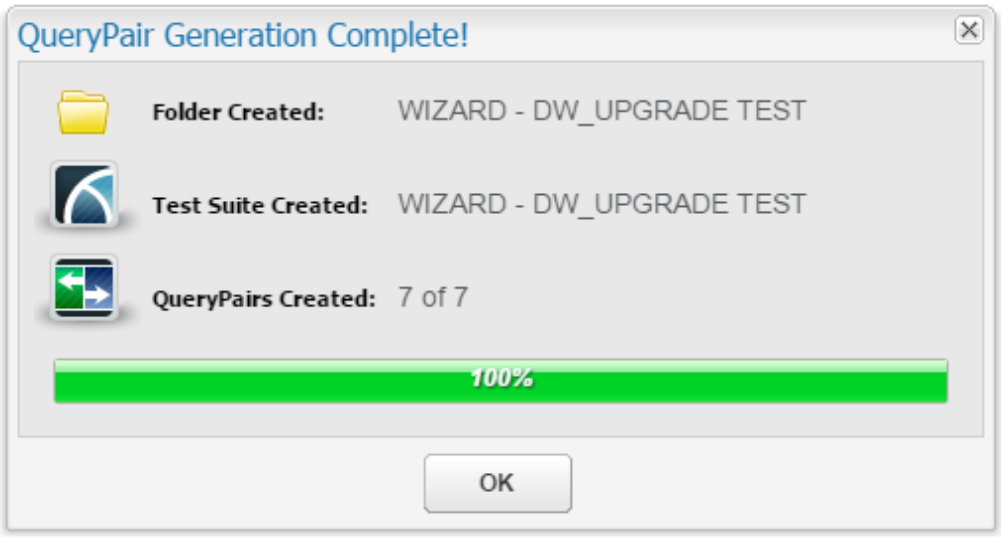
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Points of interest:

- A review screen appears with the details for your *Query Wizard* execution

18. Click the **Create** button.

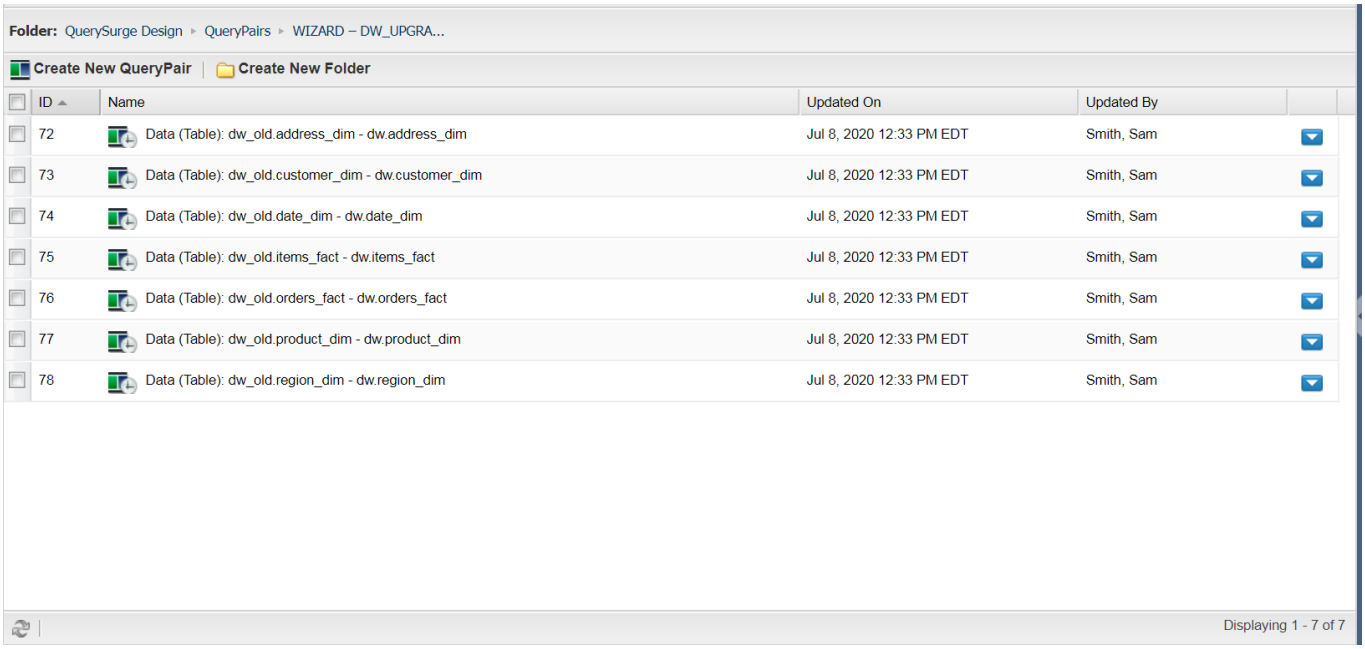


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Points of interest:

- A test folder named 'WIZARD – DW_UPGRADE TEST' was created in the 'Design Library' module
- A *Test Suite* named 'WIZARD – DW_UPGRADE TEST' was created in the 'Scheduling' module
- Seven *QueryPairs* were created (one for each of the source and target tables mapped)

19. Click the **OK** button.
20. In the **"Design Menu"** dropdown menu on the top toolbar, select **"Library Explorer".**
21. In the **"Design Library**" tree (left panel), navigate to the folder 'QuerySurge Design -> QueryPairs -> WIZARD – DW_UPGRADE TEST'.

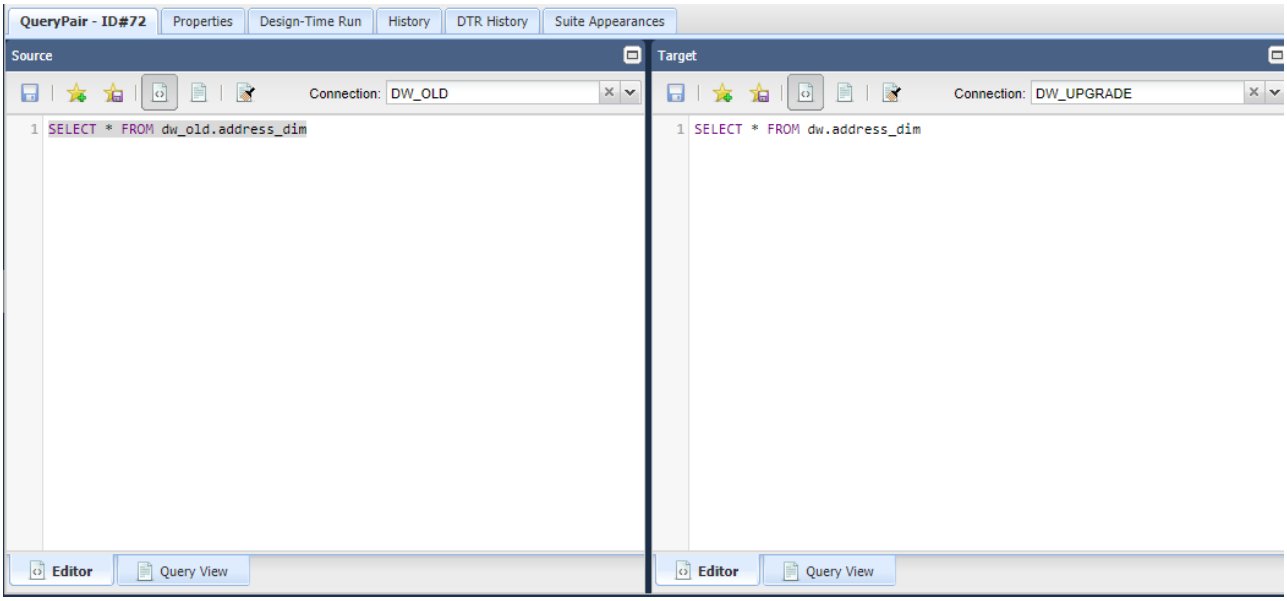


Points of interest:

- The name of the folder is the name you entered into the 'Folder Name' field in the *Query Wizard*
- The name of a *QueryPair* is in the following format:
  - Validation Type: source schema.source table - target schema.target table

19. Double click the 'Data (Table): dw_old.address_dim - dw.address_dim' *QueryPair* and notice the SQL for source and target queries that was automatically generated.

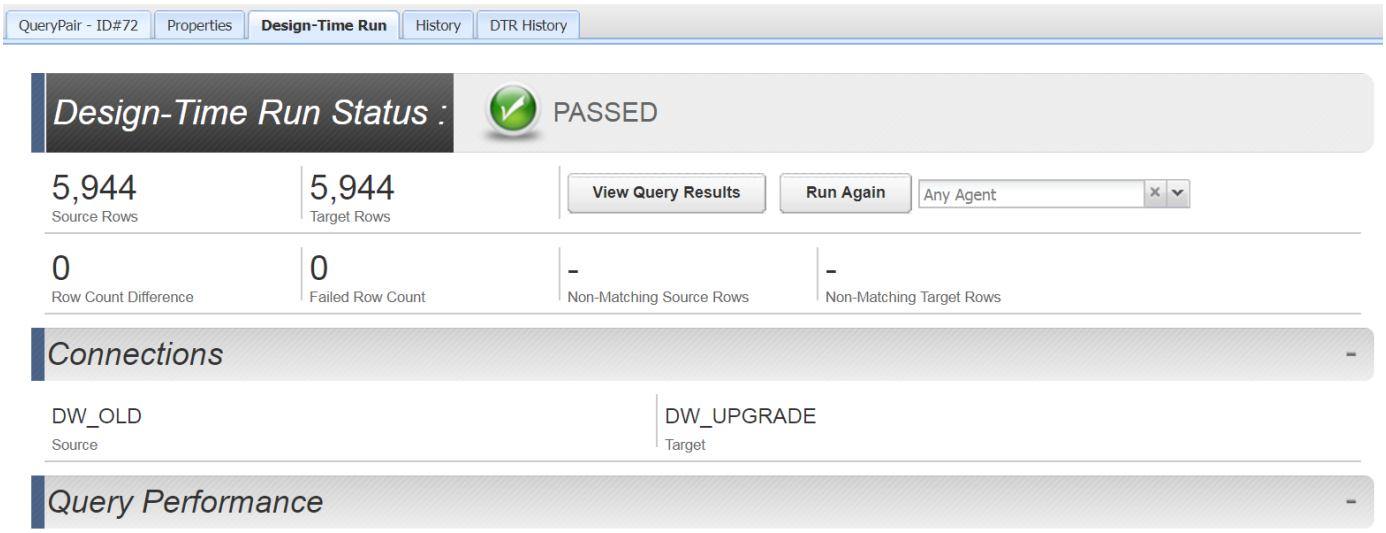
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Points of interest:

- The *QueryPair* is selecting all columns from the source connection and target connection
- During execution, all columns will be compared between source and target

20. Execute a 'Design-Time Run' by clicking the **"Design-Time Run"** tab and clicking the **"Run"** button.

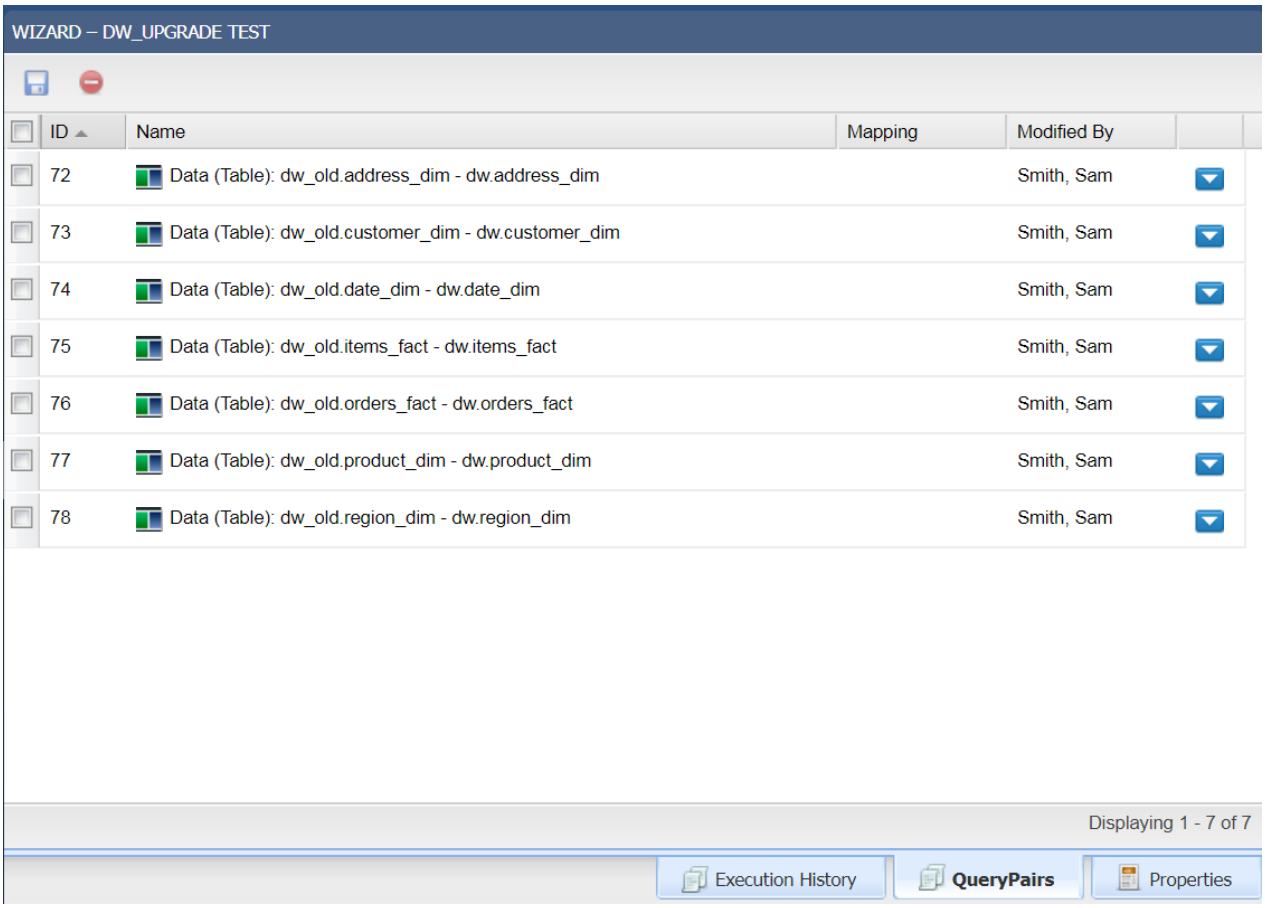


Points of interest:

- The *QueryPair* compared 5,944 source and target rows, which passed
- Feel free to view the results

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21. Click the **"Scheduling"** icon located on the bottom panel. 
22. Select the 'WIZARD – DW_UPGRADE TEST' *Test Suite* in the left pane.

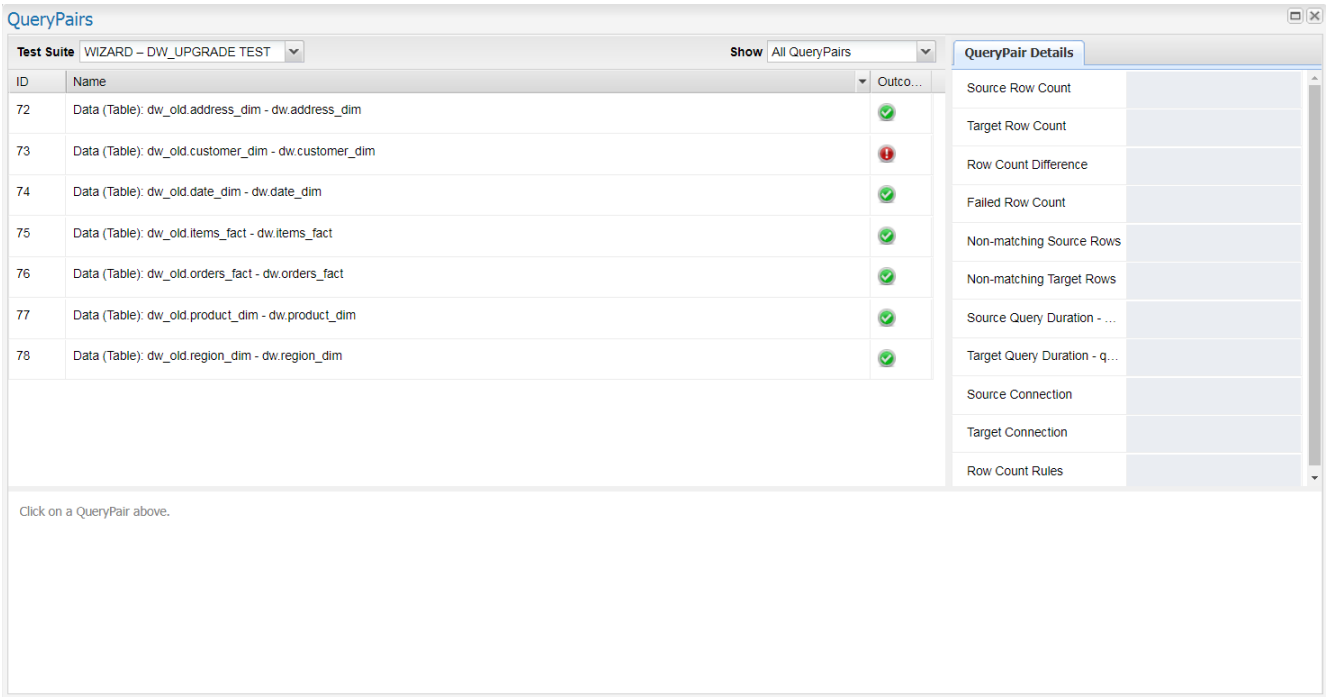


Points of interest:

- The name of the *Test Suite* is the name entered into the 'Name' field during the *Query Wizard*
- All the *QueryPair(s)* are included in the *Test Suite* upon generation

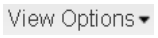
23. Right click the 'WIZARD – DW_UPGRADE TEST' *Test Suite* and select the **"Run Now"** option.
24. If a "Run Scenario" dialog is displayed, click the **OK** button.
25. The "Run Dashboard" is displayed. Double click the 'WIZARD – DW_UPGRADE TEST' row in the middle panel to review the results.

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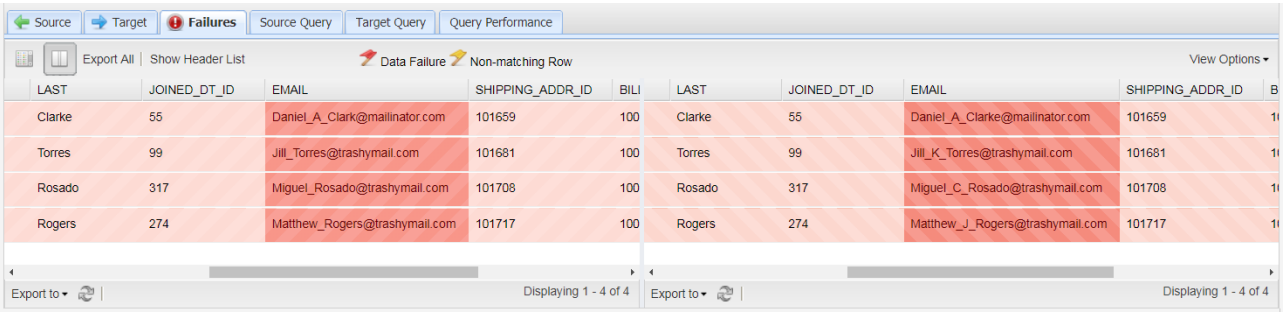


Points of interest:

- One *QueryPair* of the seven total failed

26. Click the 'failed' *QueryPair.*
27. Select the **"Failures"** tab.
28. Click **"View Options"** dropdown in the upper right of the lower panel. 

29. Select **"Show Inferred Data Mismatches"** radio button – Since no keys have been specified, using **Inferred Data Mismatches** provides some additional insights into the data causing failures.



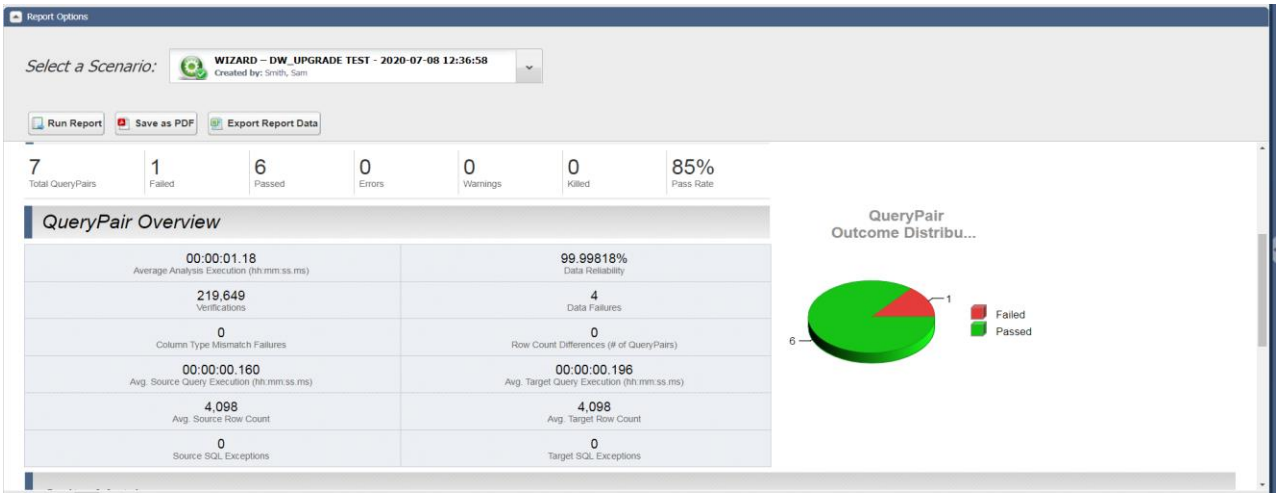
Points of interest:

- Four email addresses do not match from source to target
- Investigation is needed to determine the reason for the data differences that occurred during the upgrade

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30. Close the **"QueryPairs"** query results window.
31. Click the **"Data Intelligence Reporting"** icon located on the bottom panel. 
32. Select the **"Scenario Detail"** report from the left panel.
33. Click the **"Open Report"** button.
34. Select the 'WIZARD – DW_UPGRADE TEST'. (a date time stamp will be appended to the end of the name in the 'Select a Scenario' dropdown list)

    Click the **Run Report** button.



Points of interest:

- Notice the number of individual field verifications that QuerySurge was able to accomplish, just under a quarter million, in less than 9 seconds on our tutorial instance (completion time will vary based on your hardware)

For more information on Reports, please visit *Reporting* in the Knowledge Base

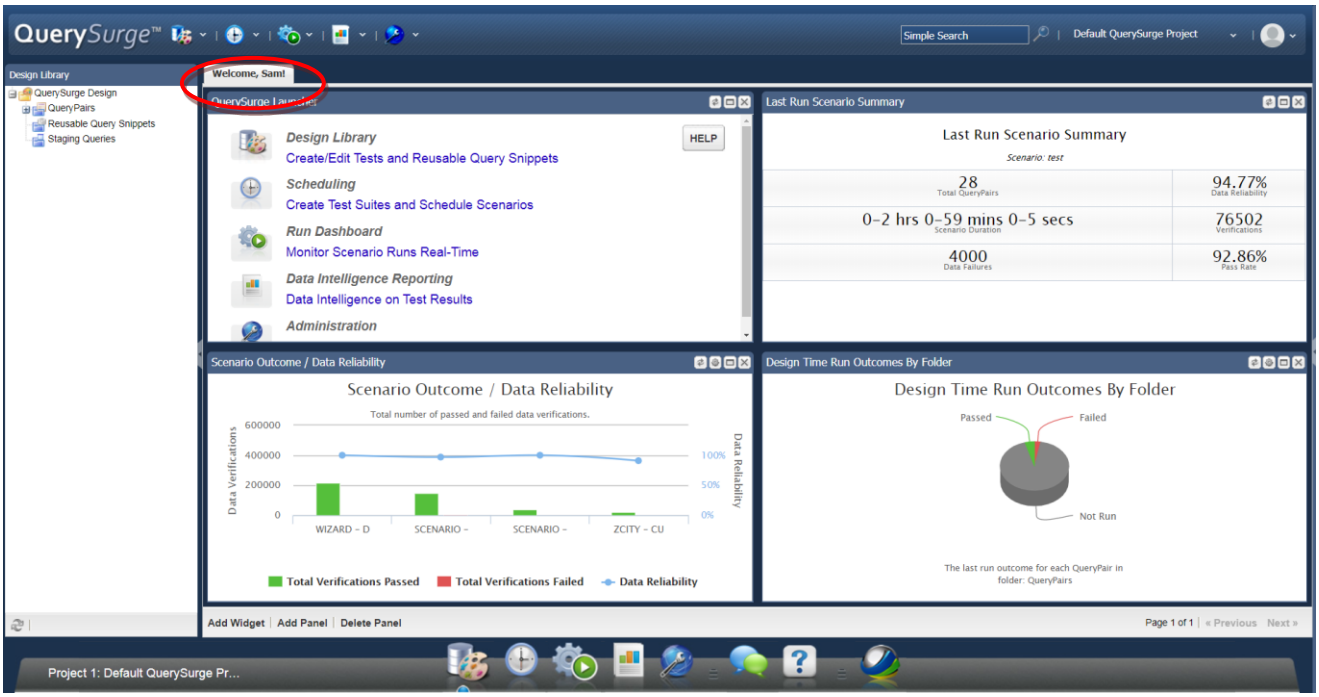
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Review of Data Health Dashboard

The Data Health Dashboard gives you a real-time view into your project progress at all levels, from *QueryPair* development to execution and results. Data *Widgets* are configurable, so you can see the slice of your project that you want to see, in the way you want to see it.

1. Click the **"Design Library"** icon located on the bottom panel. 
2. Click the **"Welcome"** tab if it's not already in view.

Note: If the available widgets panel is not displayed, then click the 'Add Widget' Button

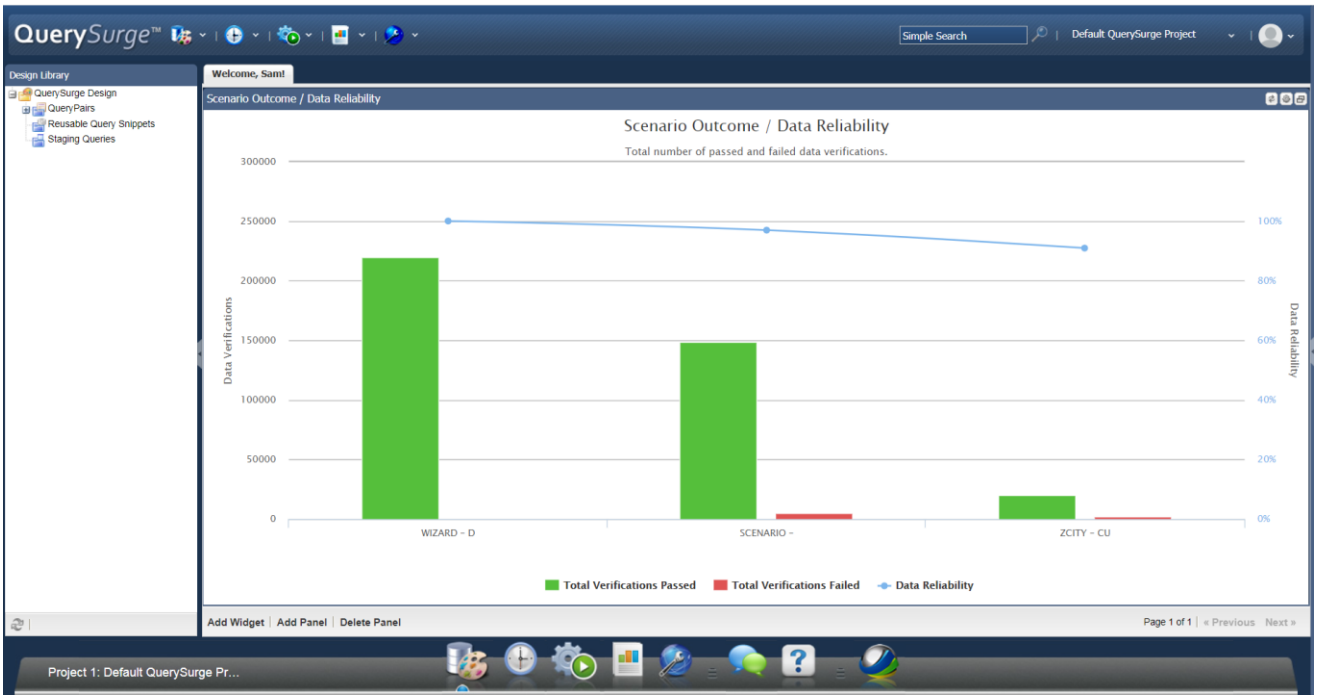


Points of interest:

- The "Welcome" tab defaults to one panel containing four *Widgets* (center panel)
- Click the 'Add Widget' button for a list of additional 'Available Widgets' (right panel)
- Panels can be configured by dragging and dropping *Widgets* as desired

3. Click the **Maximize** button on the "Scenario Outcome / Data Reliability" *Widget* for a closer view. 

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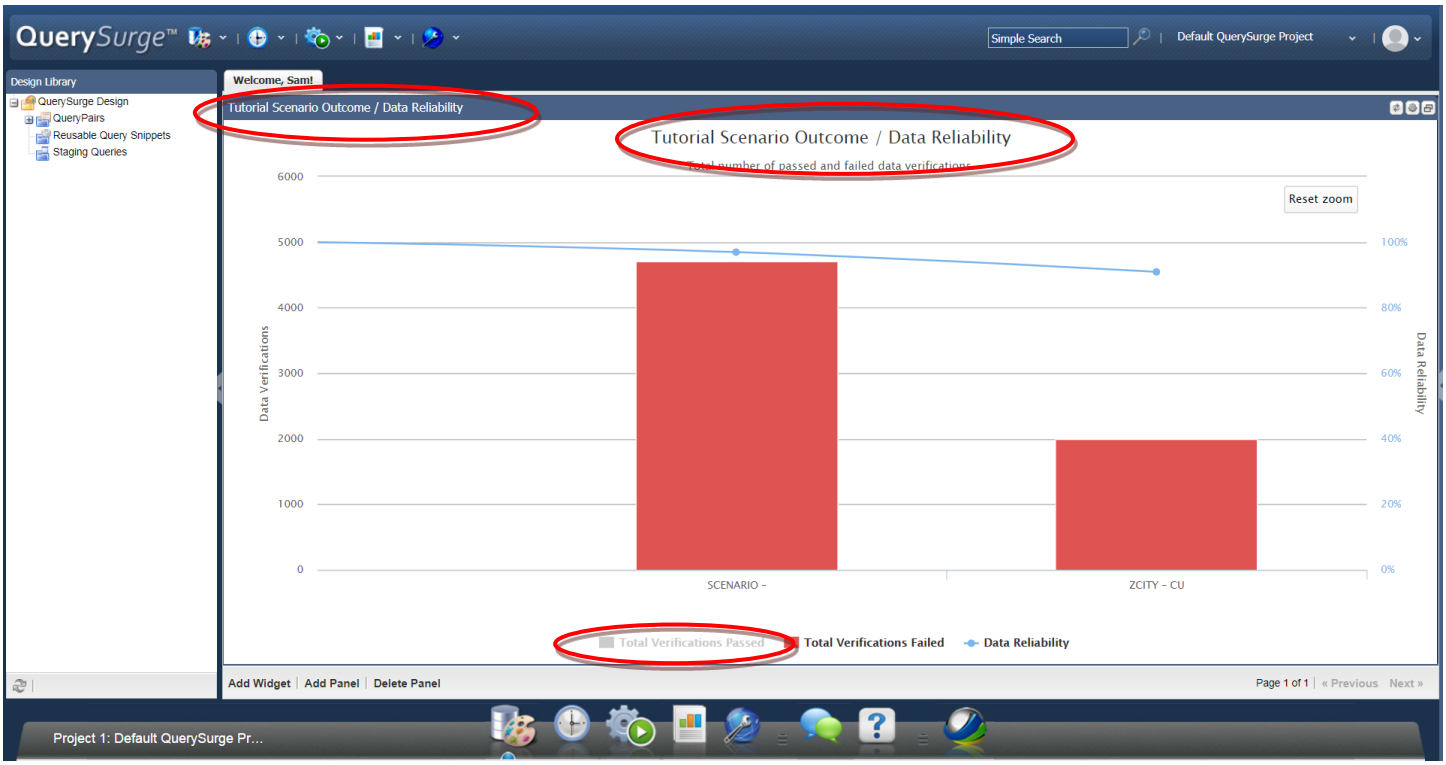


Points of interest:

- Pass/Fail Verifications of your recent *Scenario* Runs displayed as a bar graph of Pass/Fail Rate
- Data Reliability per *Scenario* metric displayed as a blue line
- The vertical axis for the bar graph is on the left, while the vertical axis for the blue line is on the right

4. Click the 'Scenario Outcome / Data Reliability' *Widget* title to make it editable, type 'Tutorial' at the beginning of the existing title and hit enter.
5. Turn off the 'Total Verifications Passed' data in the *Widget* by clicking on the legend at the bottom.
6. Zoom in on the data by clicking on the "SCENARIO" bar in the graph and dragging your mouse across to the left. (Zoom occurs upon the release of the mouse button)

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Points of interest:

- Title changes are reflected above the graph and in the title bar
- 'Total Verifications Passed' in no longer reflected in the graph, you can add the data back by clicking on the greyed-out legend
- The graph is zoomed in on the bars that were highlighted in the previous step, and a 'Reset zoom' button now appears to revert. Reset zoom
- You can point many of the *Widgets* at specific data by clicking on the 'View Settings' icon
- You can have multiple panels of *Widgets* by clicking on the 'Add Panel' button

For more information on Widgets, please visit *Reporting* → *Project Widgets* in the Knowledge Base

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Summary

You have just finished the QuerySurge tutorial. This is what you completed:

- Created a *QueryPair* using the *Column-Level Comparison Query Wizard*
- Reviewed existing and created new *QueryPairs*
- Created Reusable Query *Snippets*
- Executed Design-Time Runs
- Investigated 'Design-Time Run' failures
- Scheduled and executed a *Scenario*
- Generated and reviewed reports
- Generated QueryPairs using the *Table-Level Comparison Query Wizard*
- Reviewed *Widgets*

There are additional examples within the sample data and *QueryPairs*, many of them which are more complex than the exercises you completed in this tutorial. Now that you have a solid foundation on using QuerySurge, try them out. You can access the mapping and data model document from the "Setup Wizard."

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Deleting the Tutorial Data

This section is for removing the tutorial data from QuerySurge.

Warning: Deleting the tutorial data will permanently remove data. Please read this section carefully before proceeding!

The following entities will be permanently deleted (original sample data):

- The ZCITY folder and all *QueryPairs* contained within
- The XMART folder and all *QueryPairs* contained within
- The *Snippet* named "ZCITY – Filter"
- The *Test Suite* named "ZCITY – ALL"

The following entities are not deleted: (can be deleted manually)

- *QueryPairs* created with the *Query Wizard*
- *Snippets* created during the tutorial
- *Test Suites* created during tutorial
- *Scenario* execution data and results

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Appendix

Documentation

- System Requirements Guide – hardware and software needed to run QuerySurge minimally and optimally
- Installation Guides
  - for Windows (Single Machine)
  - for Windows (Multi Machine)
  - for Linux (Single Machine)
  - for Linux (Multi Machine)
- Configuration Guides for connecting to a broad range of technologies.

QuickStart User Guide – Everything you need to know to get started with QuerySurge

Data Warehouse Testing

Comprehensive testing of a data warehouse at every point throughout the ETL (extract, transform, and load) process is becoming increasingly important, as more data is being collected and used for strategic decision-making.

Data warehouse projects are undertaken because of mergers and acquisitions, compliance and regulations, data consolidation, and the increased reliance on data-driven decision making (using Business Intelligence tools, etc.).

Any way you slice it, the data warehouse/business intelligence (BI) platform is complex and presents many data quality and testing problems to overcome. Some of the main challenges of data warehouse testing are:

- **Data Completeness.** Verifying that all data has been loaded from the sources to the target data warehouse
- **Data Transformation.** Ensuring that all data has been transformed correctly during the ETL process
- **Data Quality.** Ensuring that the ETL process correctly rejects, corrects, ignores, substitutes default values, and reports invalid data
- **Regression Testing.** Ensuring existing functionality remains intact each time a new release of code is completed

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QuerySurge has five different modules:



Design Library

Use the QuerySurge "Design Library" tools to create collections of powerful tests across your data warehouse architecture.

- ❑ Create test *QueryPairs* between any two points (Source, Staging, Data Warehouse, Data Mart) in your architecture
- ❑ See query history for all of your queries
- ❑ Run *QueryPair* executions as you build queries to ensure they produce the required results
- ❑ Make your design flexible with Reusable Query *Snippets* - *Snippet* libraries of query fragments that you can use to modularize your queries, helping to speed up the process of bulk *QueryPair* updates
- ❑ Run the 'Query Wizard', which provides information for (1) row count compares, (2) full table compares, and (3) column compares without writing any SQL code



Scheduling

Schedule your testing by time and location for maximum productivity. Schedule your tests for the specific time and day or run tests when an event, like the ETL process, completes.

- ❑ Mix-and-match test *QueryPairs* in QuerySurge *Test Suites* to meet specific project execution goals
- ❑ Build groups of QuerySurge *Test Suites* to test specific mappings, ETL logic or Data Warehouse Sources
- ❑ Build QuerySurge *Scenarios* for scheduling your execution runs at specific dates and times
- ❑ Use the execution API for event-based scheduling



Run Dashboard

Visualize and track the real-time progress of your running Scenarios on the QuerySurge 'Run Dashboard'. A graphical display helps you to follow the flow of your queries, and you can drill-down to see test details and data failures.

- ❑ View query execution progress live via graphical run status displays
- ❑ Drill-down into data as QuerySurge executes to examine results as they become available during execution
- ❑ See real-time statistics for each *QueryPair* executed and for the *Scenario* execution as a whole
- ❑ Alert your team about the status of execution via custom email notifications
- ❑ Export detailed results in Excel, CSV, or XML formats to share with team members and other project stakeholders



Data Intelligence Reporting

Get a comprehensive audit and share detailed results with others. Use QuerySurge reports to share both high-level and detailed views of your testing with team members, managers, and business stakeholders.

- ❑ Choose from a wide selection of configurable reports within the QuerySurge 'Reporting Center'
- ❑ Built-in reports range from high-level summary reports to lower-level, detailed reports that contain a complete audit trail of test modifications

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- ❑ Configure reports for specific date ranges, asset types, or specific executions, to get the view into your results that you need
- ❑ Export your reports as Excel or PDF files to share within your organization, or to archive for future audit needs



Administration

The administration module provides access to the control features of your QuerySurge installation. Here, you have the ability to create and manage user profiles, database connections, Agent configuration, data archiving and automated email notification options.

- ❑ Create and manage QuerySurge users and *Agents*
- ❑ Manage QuerySurge data storage with the included storage tools
- ❑ Create and manage connections to source and target data stores
- ❑ View QuerySurge server information, including configuration settings and your application licensing details
- ❑ Create automated email notifications to send results to your team at the end of a test run

The QuerySurge Testing Process

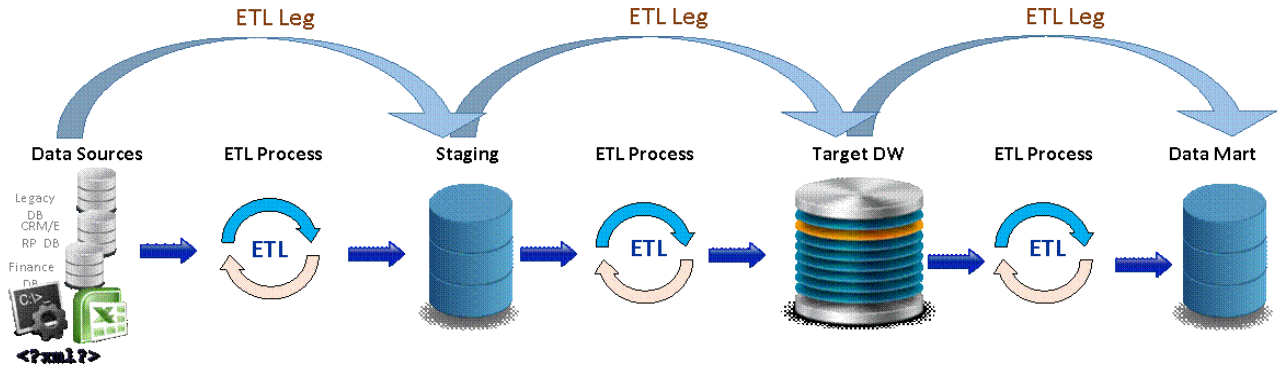
Testing and validating the ETL process is the key component to the success of a data warehouse implementation. Bad data caused by defects in the ETL process can cause data problems in reporting that can result in poor strategic decision making.
Example: A large fast-food company depends on BI reports to determine how much raw beef to order on a monthly basis, by sales region and time of year. If these reports are not correct, then the company could order an incorrect amount, which could cost the company millions of dollars in either lost revenue or wasted product.

Developers utilize ETL tools to accomplish the extracting, transformation and loading of data from sources into target systems.

Test Points and ETL Legs

- An "ETL Leg" refers to a single ETL process that moves/transforms data between two discrete points in the data warehouse architecture
- A full ETL process may have multiple legs
- Test points and single ETL leg: the verification is between the source and the target for that leg
  Example: An operational source database (source test point) is extracted, transformed, and loaded into a data warehouse (target test point). Testing is conducted across this ETL leg

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- Test points and multiple ETL legs: The multi-leg approach is to validate multiple legs of the ETL process in a single test, 'jumping over' ETL legs in the process. A typical multi-leg test is to validate the entire ETL process from data sources to final data endpoint utilizing test points only at the data sources and at the final data endpoint

Data Mapping Document

A source-to-target map is a "data mapping" document and is the most important artifact when building or testing a data warehouse. Data mapping documents are often created in Excel spreadsheets or Word documents. The document acts as a central listing of the "functional" requirements. Testers use the mapping document to verify that the data has been extracted from the source databases, data stores and files and into the target data warehouse and data marts correctly. The following information is contained within the mapping document:

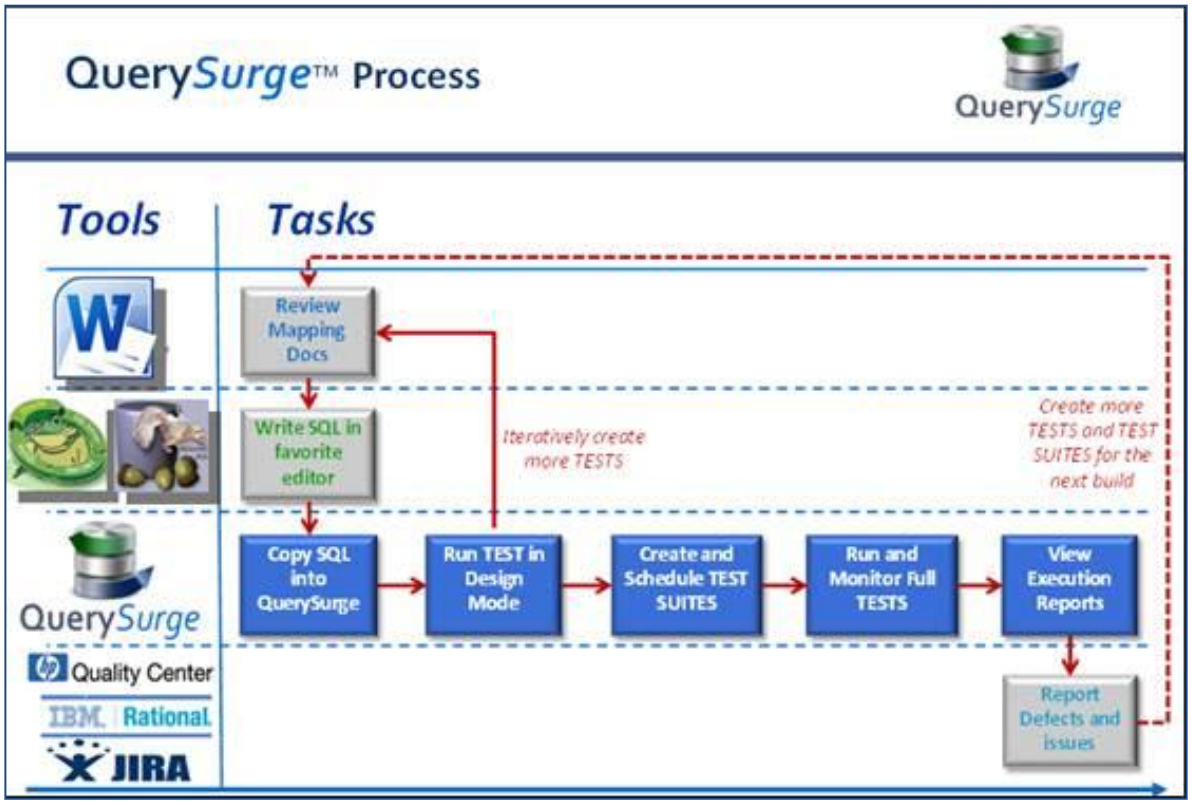
- Source database Information
  - Source table
  - Source column
- Target database Information
  - Target table
  - Target column
- Data transformation logic
- Optional requirements

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QuerySurge: The Process in a Nutshell

- Review the mapping document
- Determine the optimum percent of coverage or amount of data that is required to be tested, based upon time and resources
- If not testing 100% of the data, determine the exact sampling of data needed
- Create test cases that exercise the requirements of the mapping document
- Create pairs of SQL queries (*QueryPairs*) – one aimed at the source database or file, one at the target data warehouse or data mart
- Bundle groups of *QueryPairs (Test Suites)* into test *Scenarios*
- Schedule *Scenarios* to run (a) immediately, (b) at a certain day and time, or (c) automatically after an event
- Analyze and drill down into your results and identify bad data and data defects with our robust reporting engine.
- Report defects in your defect tracking tool (i.e., HP Quality Center, IBM Rational Quality Manager, Jira, Bugzilla, etc.)
- Have reports sent automatically via email to team members

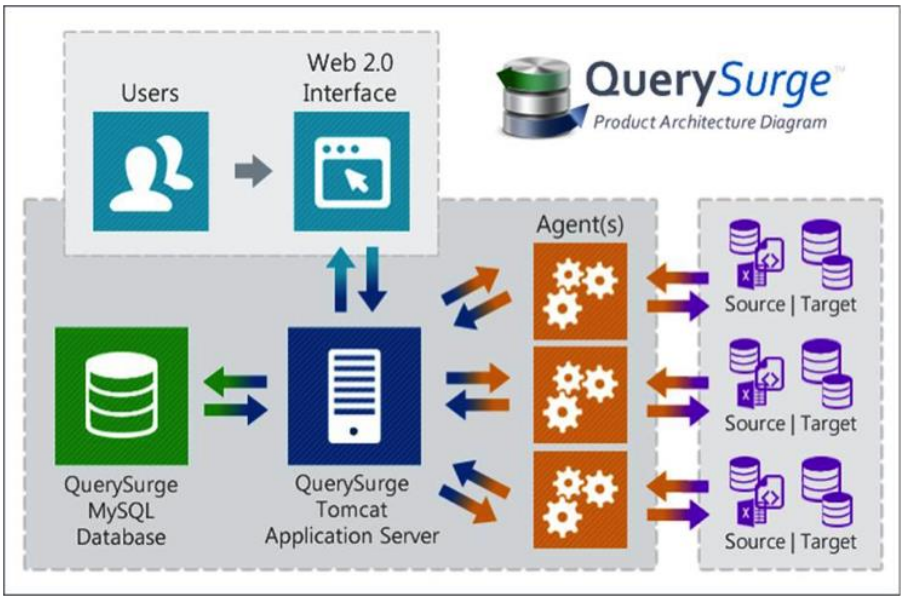
The QuerySurge Architecture



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About the Query*Surge*™ Architecture

QuerySurge is a locally installed, browser-based testing tool. Supporting all current browsers (Chrome, Firefox, IE, Safari, etc.), QuerySurge is perfect for teams that are geographically distributed. QuerySurge extracts the data-under-test to its own, separate infrastructure, which eliminates processing overhead on the source and target database servers in your data warehouse architecture. The QuerySurge architecture is comprised of an application server, database server and Agents.



QuerySurge Application Server and Database

QuerySurge utilizes a Tomcat application server and a MySQL database. The database is bundled with and embedded within the product.

QuerySurge *Agents*

QuerySurge *Agents* are the components of the architecture that execute queries against source and target data sources, returning the results to the QuerySurge database. The Agents execute SQL queries, validating each piece of data throughout the ETL process. This exposes all data mismatch failures, row count differences, and column type mismatch failures, affording you the ability to test to 100% of your data quickly.

Although the Agents issue queries to both the source and target databases, they do not reside on the physical source or target database boxes. QuerySurge Agents can be deployed on the same box as the QuerySurge *Application Server* and QuerySurge *Database Server* in a single box install or on any "satellite" boxes in the environment (often, these are test lab boxes or available desktop boxes). The QuerySurge Agent, when it receives a bundle of queries, can run multiple queries simultaneously (in parallel).

One of the features QuerySurge gives you is the ability to raise your 'testing throughput'. The idea is to be able to fire off *QueryPairs* in bulk against your sources and targets so that you can execute at the highest level your environment can support. The QuerySurge Agent lets you do this, because you can deploy multiple Agents in your environment – on 'satellite' boxes surrounding your QuerySurge server (note that each Agent can run multiple query threads as well). QuerySurge allows you to install up to 10 Agents.

More *Agents* = more queries = more throughput

How many Agents are appropriate for your environment? The answer is – you find out by experimentation. Once you have built an initial test library, start with 2 or 3 Agents, and see how your sources and targets behave. Add additional Agents in a subsequent cycle, again monitoring the source and target behavior. As you continue to add Agents, the loads on sources and targets will grow with query volume – and source/target response times will start to grow as well. Once

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you have identified the level where response times have started to increase, back down to the previous level. This level is roughly the maximum throughput your environment can support.

Database / Data Warehouse Support

In principle, QuerySurge can support any JDBC-compliant data source. QuerySurge currently ships with Oracle, MySQL, Teradata, Aster, SQL Server, DB2, Informix, Netezza, Sybase, in-memory database, flat file support, and supports many other data sources.

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About RTTS

RTTS was founded in 1996, and has cultivated partnerships with the world's leading test tool vendors, including IBM, Microsoft, and HP. We are headquartered in New York City and our satellite locations are in Philadelphia, Atlanta, and Phoenix. Many of our consulting and education services are also offered through the cloud, so that no matter where you are, RTTS will ensure application functionality, performance, scalability, and security for your organization.

RTTS is the premier professional services organization that specializes in providing software quality for critical business applications. We offer the most comprehensive suite of quality assurance services. We've helped 400+ organizations drive positive results from their software development projects.

For more information, please visit www.rttsweb.com

About QuerySurge™

QuerySurge is the only automated software tool built specifically for ETL testing. It can verify as much as 100% of all data from source systems, through the ETL process, to the target data warehouse and data marts. QuerySurge has increased test coverage and reduced test cycle time for numerous Fortune 500 organizations, helping them to mitigate risk and meet business requirements. For more information, please visit RTTS' team of test experts developed QuerySurge (www.QuerySurge.com) to address the unique testing needs in the data warehousing and data migration area. It has been implemented on projects ranging from large data warehousing and ETL processes to data migrations, database upgrades, integration testing, data load testing and system patch testing.

For more information, please visit www.querysurge.com

Click here to contact us for more information

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